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1

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.

2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.

3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

Fri May 9 13:11:49 2014

N56 MLB: CARRIER BUILD

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N/A

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N/A

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JUMPER

N56 RADIO MLB

05/07/2014

N56 BOM CALLOUTS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-0517	1	SCH,MLB,N56	SCH	CRITICAL	?
820-3675	1	PCBF,MLB,N56	PCB	CRITICAL	?
825-6838	1	EEEE FOR 639-4924 16GB	EEEE_FKV8	CRITICAL	EEEE_16G
825-6838	1	EEEE FOR 639-6157 32GB	EEEE_FRR9	CRITICAL	EEEE_32G
825-6838	1	EEEE FOR 639-6158 64GB	EEEE_FRRC	CRITICAL	EEEE_64G
825-6838	1	EEEE FOR 639-00150 128GB	EEEE_G05J	CRITICAL	EEEE_128G
825-6838	1	EEEE FOR 639-00196 16GB NON-TDD-LTE	EEEE_G166	CRITICAL	EEEE_16G_DTD
825-6838	1	EEEE FOR 639-00194 32GB NON-TDD-LTE	EEEE_G163	CRITICAL	EEEE_32G_DTD
825-6838	1	EEEE FOR 639-00195 64GB NON-TDD-LTE	EEEE_G165	CRITICAL	EEEE_64G_DTD
825-6838	1	EEEE FOR 639-00197 128GB NON-TDD-LTE	EEEE_G164	CRITICAL	EEEE_128G_DTD

NAND BOM OPTIONS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S0992	1	NAND,1Y8M,16GB8,MLC,PPN1.5,64G,SHLGA60	U0604	CRITICAL	NAND_16G
335S0993	1	NAND,1Y8M,32GB8,MLC,PPN1.5,64G,SHLGA60	U0604	CRITICAL	NAND_32G
335S1000	1	NAND,19NM,64GB8,MLC,PPN1.5	U0604	CRITICAL	NAND_64G
335S00010	1	NAND,19NM,128GB8,TLC,PPN1.5	U0604	CRITICAL	NAND_128G
138S00003	1	CAP,XSR,150UF,20V,6.3V,0.65MM,HEMTL,0402	C0614,C0633	CRITICAL	NOSTUFF
138S0867	1	CAP,XSR,100UF,20V,6.3V,0.65MM,HEMTL,0402	C0610,C0611,C0613,C0626	CRITICAL	NAND_16G
138S0867	1	CAP,XSR,100UF,20V,6.3V,0.65MM,HEMTL,0402	C0610,C0611,C0613,C0626,C0614,C0633	CRITICAL	NAND_32G
138S0867	1	CAP,XSR,100UF,20V,6.3V,0.65MM,HEMTL,0402	C0610,C0611,C0613,C0626,C0614,C0633	CRITICAL	NAND_64G
138S00003	1	CAP,XSR,150UF,20V,6.3V,0.65MM,HEMTL,0402	C0610,C0611,C0613,C0626,C0614,C0633	CRITICAL	NAND_128G

INVENSENSE ACCEL/GYRO

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S00017	1	IC,CARBON,MPU-6700-12,LGA16	U2203	CRITICAL	INVSNS_GYRO
132S0395	1	CAP,CER,0.1UF,10V,6.3V,X6S,0201	C2211	CRITICAL	INVSNS_GYRO

BOSCH ACCEL/GYRO

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S00028	1	IC,CARBON,BMI162BC,LGA16	U2203	CRITICAL	BOSCH_GYRO
132S0395	1	CAP,CER,0.1UF,10V,6.3V,X6S,0201	C2211	CRITICAL	BOSCH_GYRO

ST ACCEL/GYRO

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S00029	1	IC,CARBON,AP6DS2AA,LGA16	U2203	CRITICAL	ST_GYRO
132S0391	1	CAP,CER,0.01UF,10V,25V,X5R,0201	C2211	CRITICAL	ST_GYRO

ALTERNATE BOM OPTIONS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S1844	152S1836	ALTERNATE	L1604	TY ALT INDUCTOR
152S1802	152S1849	ALTERNATE	L1519	CYNTEC ALT INDUCTOR
152S1842	152S1849	ALTERNATE	L1519	TY ALT INDUCTOR
197S0392	197S0369	ALTERNATE	Y1200	ESPON ALT XTAL
197S0399	197S0369	ALTERNATE	Y1200	NDK ALT XTAL
338S1285	338S1202	ALTERNATE	U1601	L21 B0
152S2034	152S2033	ALTERNATE	L1605,L1519,L1514	CYNTEC 1.0UH 1.2MM
152S2034	152S2033	ALTERNATE	L1605	CYNTEC 1.0UH 1.2MM
152S00004	152S2049	ALTERNATE	SH2503,SH2504,SH2505	CYNTEC 0.47UH 1.2MM
343S0688	343S0638	ALTERNATE	U2401	CUMULUS C1, FAB14
155S00012	155S00009	ALTERNATE	L1519,L1510,L1514	MURATA, 650HM CHOKE
118S0764	118S0717	ALTERNATE	R1309	3.92KOHM, 01005
155S0773	155S0453	ALTERNATE	FL0801	TY,1200HM,FERRITE
155S0885	155S0610	ALTERNATE	FL0802,FL1803	TY,1500HM,FERRITE
377S0168	377S0140	ALTERNATE	DZ1113	AMOTECH,6.8V,100PF
138S0648	138S0652	ALTERNATE	C1018	TY,4.7UF,0402
138S0657	138S0702	ALTERNATE	SH2503,SH2504,SH2505	SAMSUNG,4.3UF,0610
138S00005	138S00003	ALTERNATE	C0605	TY,15UF,HRZTL,0402
152S2031	152S1800	ALTERNATE	L1401	MURATA,1UH,2520
152S1840	152S1801	ALTERNATE	L1215	TY,1UH,2016
335S00013	335S0894	ALTERNATE	U0301,U_REF,BP	ST 8K EEPROM
339S0247	339S0246	ALTERNATE	U0201	FIJI 1GB HYNIX B0
339S00006	339S0246	ALTERNATE	U0201	FIJI,B1,E
339S00007	339S0246	ALTERNATE	U0201	FIJI,B1,H
339S00008	339S0246	ALTERNATE	U0201	FIJI,B1,S

138S0917 WILL NOT BE USED AT THE CARRIER BUILD AND WILL BE REPLACED BY 138S0831 AT DVT.

138S0831	138S0917	ALTERNATE	C0204	MURATA,2.2UF,0201
138S00019	138S0917	ALTERNATE	C0204	KYOCERA,2.2UF,0201

ALTERNATE NAND BOM OPTIONS

335S1038	335S0992	ALTERNATE	U0604	NAND HYNIX 16GB
335S0994	335S1000	ALTERNATE	U0604	NAND TOSHIBA 64GB
335S1040	335S1000	ALTERNATE	U0604	NAND HYNIX 64GB
335S00009	335S1000	ALTERNATE	U0604	NAND SANDISK 64GB TLC
335S00014	335S1000	ALTERNATE	U0604	NAND TOSHIBA 64GB TLC
335S00015	335S00010	ALTERNATE	U0604	NAND TOSHIBA 128GB

BOM DEVIATIONS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
604-00416	1	ASSY,SHIELD,LOWER BACK,1P,N56	SH2501	CRITICAL	COMMON
604-8159	1	ASSY,SHIELD,LOWER FRONT,N56	SH2502	CRITICAL	COMMON
806-00401	1	SHIELD,BMI,UPPER BACK,EXTENDED FOOT,N56	SH2503	CRITICAL	COMMON
604-8158	1	ASSY,SHIELD,UPPER FRONT,N56	SH2504	CRITICAL	COMMON
604-00244	1	SURASSY,SHIELD,SA	SH2505	CRITICAL	COMMON

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SCH 051-0517

BRD 820-3675

MCO 056-6398

BOM 639-4924 (16GB)

BOM 639-00196 (16GB,DTD)

BOM 639-6157 (32GB)

BOM 639-00194 (32GB,DTD)

BOM 639-6158 (64GB)

BOM 639-00195 (64GB,DTD)

BOM 639-00150 (128GB)

BOM 639-00197 (128GB,DTD)

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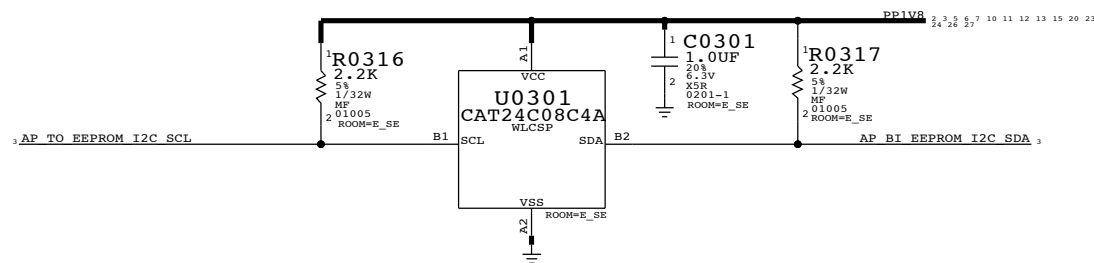
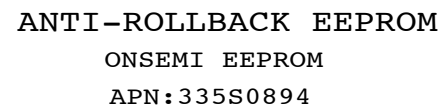
D



B

B

A



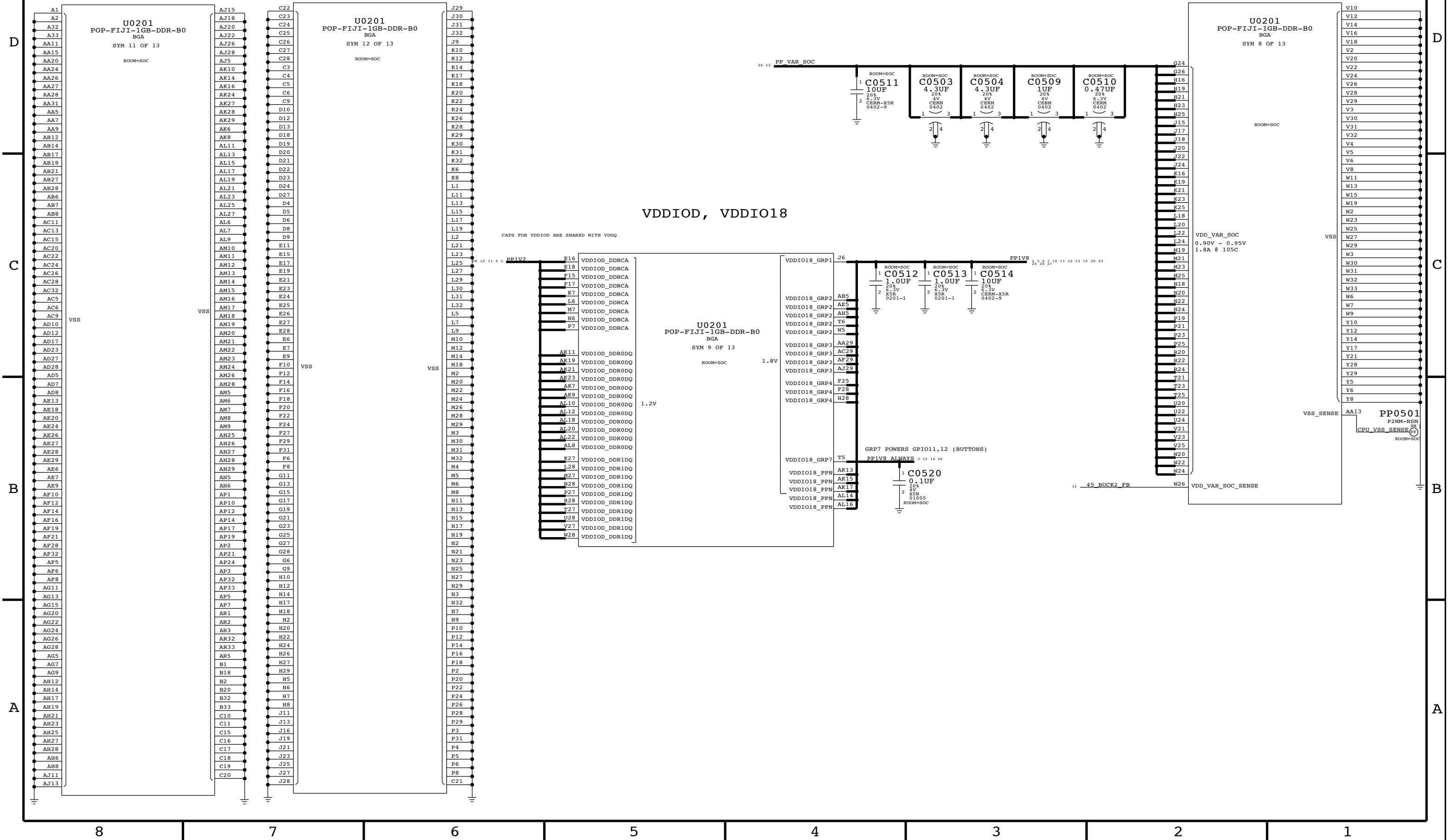




FIJI: VDDIOD, VDDIO18, VDD_VAR_SOC

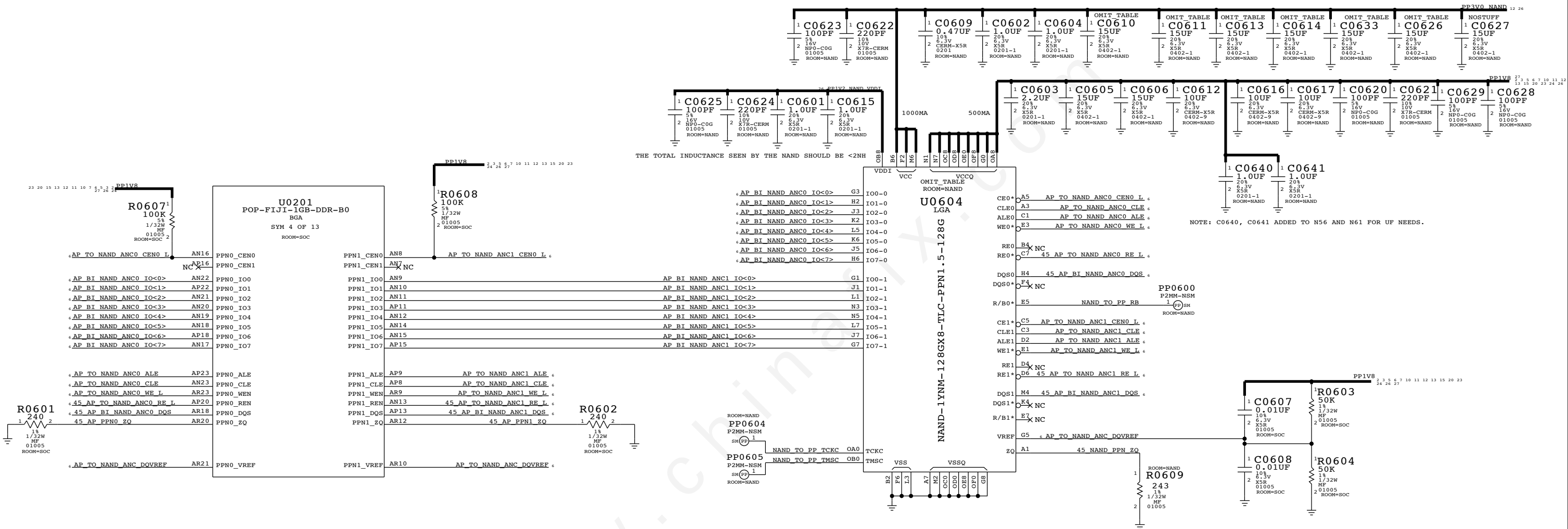
JUST A FEW GNDS

VDD_SRAM, VDD_SOC



FIJI: NAND + 12X17 NAND PKG

SUPPORT FOR PPN1.5 (1.8V IO) ONLY





C

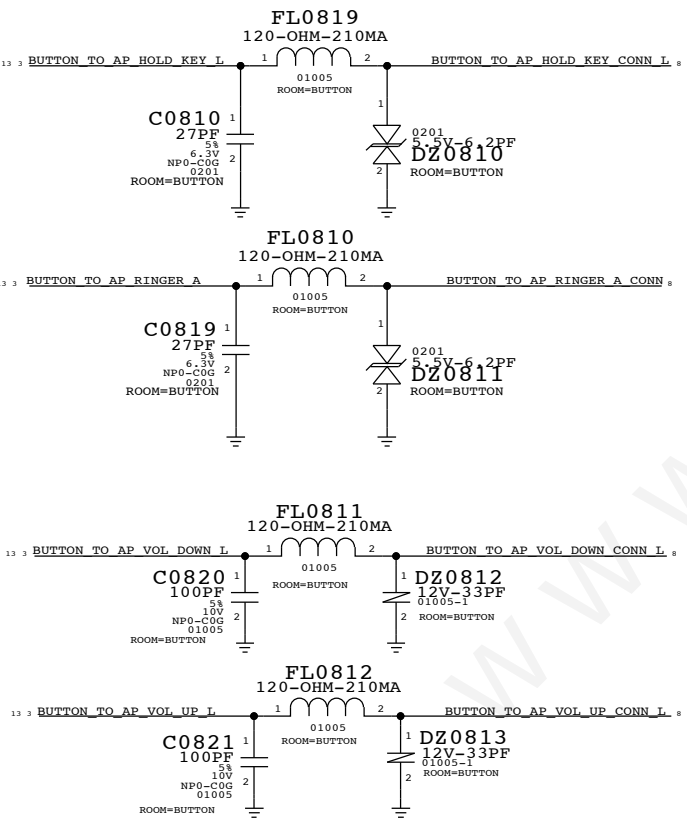
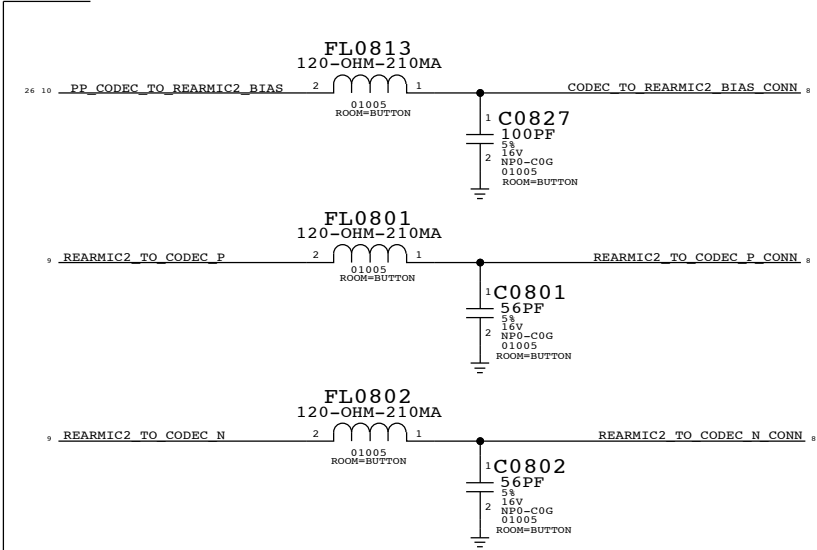




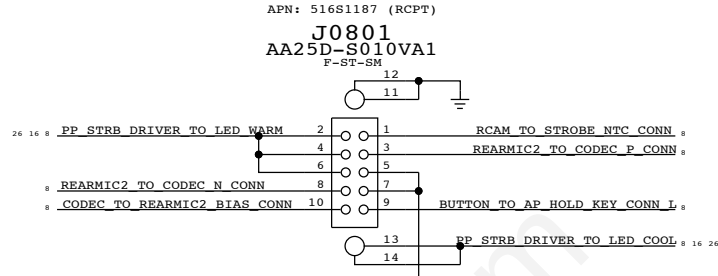
BUTTON FLEX (BUTTONS, ANC REF MIC, STROBE, STROBE_NTC, WIFI FLEX PAC)

MIC2 (ANC REF MIC):
MIC2/4 BIAS,
MIC2_P,_N

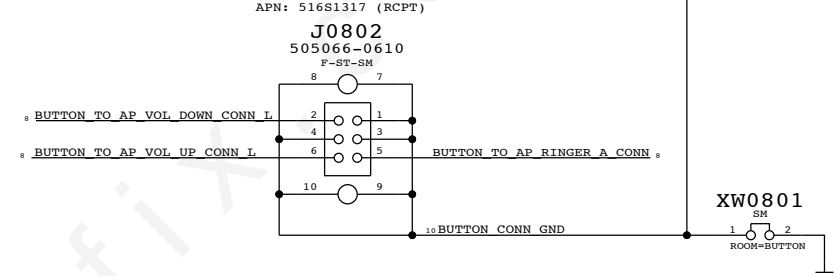
BUTTONS:
RINGER, HOLD,
VOL_UP/DOWN,



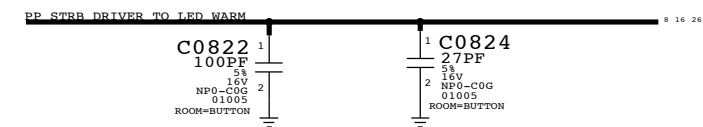
RIGHT BUTTON FLEX B2B



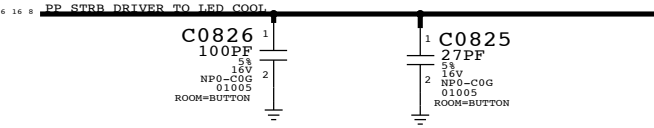
LEFT BUTTON FLEX B2B



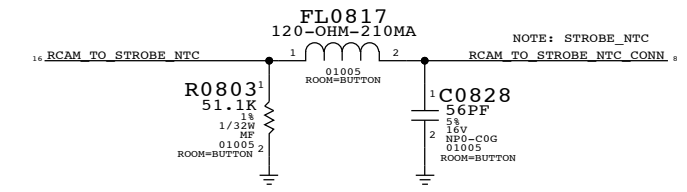
STROBE:
LED WARM



STROBE:
LED COOL



STROBE:
NTC

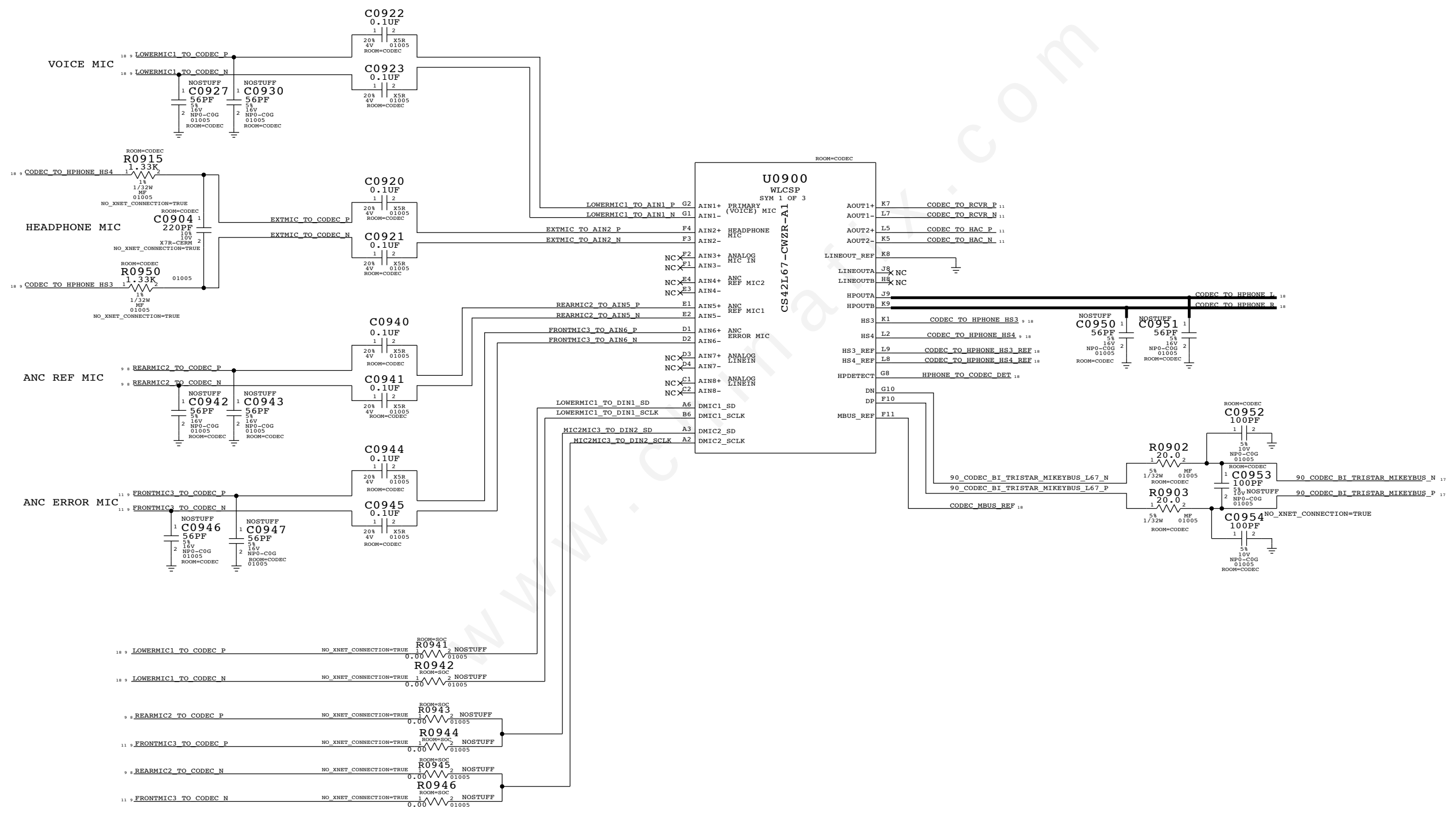




L67 AUDIO CODEC

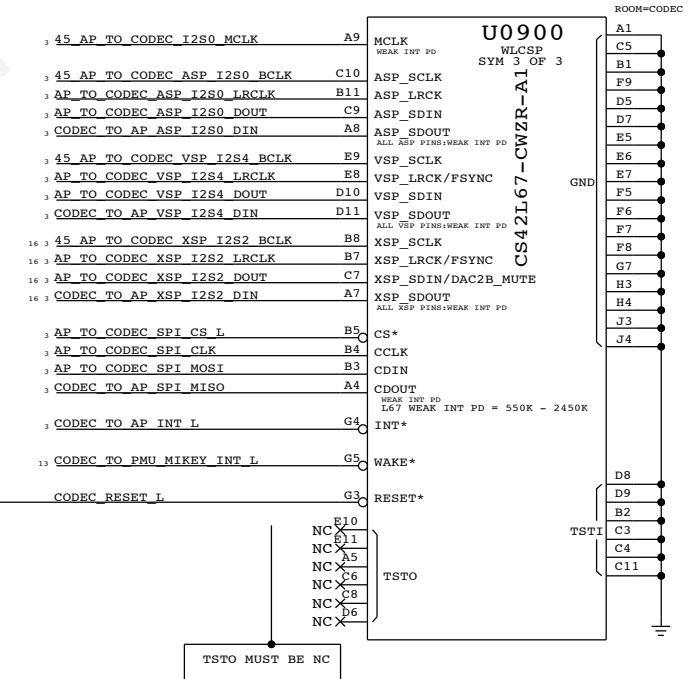
AUDIO I/O

(ANALOG MIC IN, DIG MIC IN, HPOUT, LINEOUT, RECEIVER OUT, MIKEYBUS)



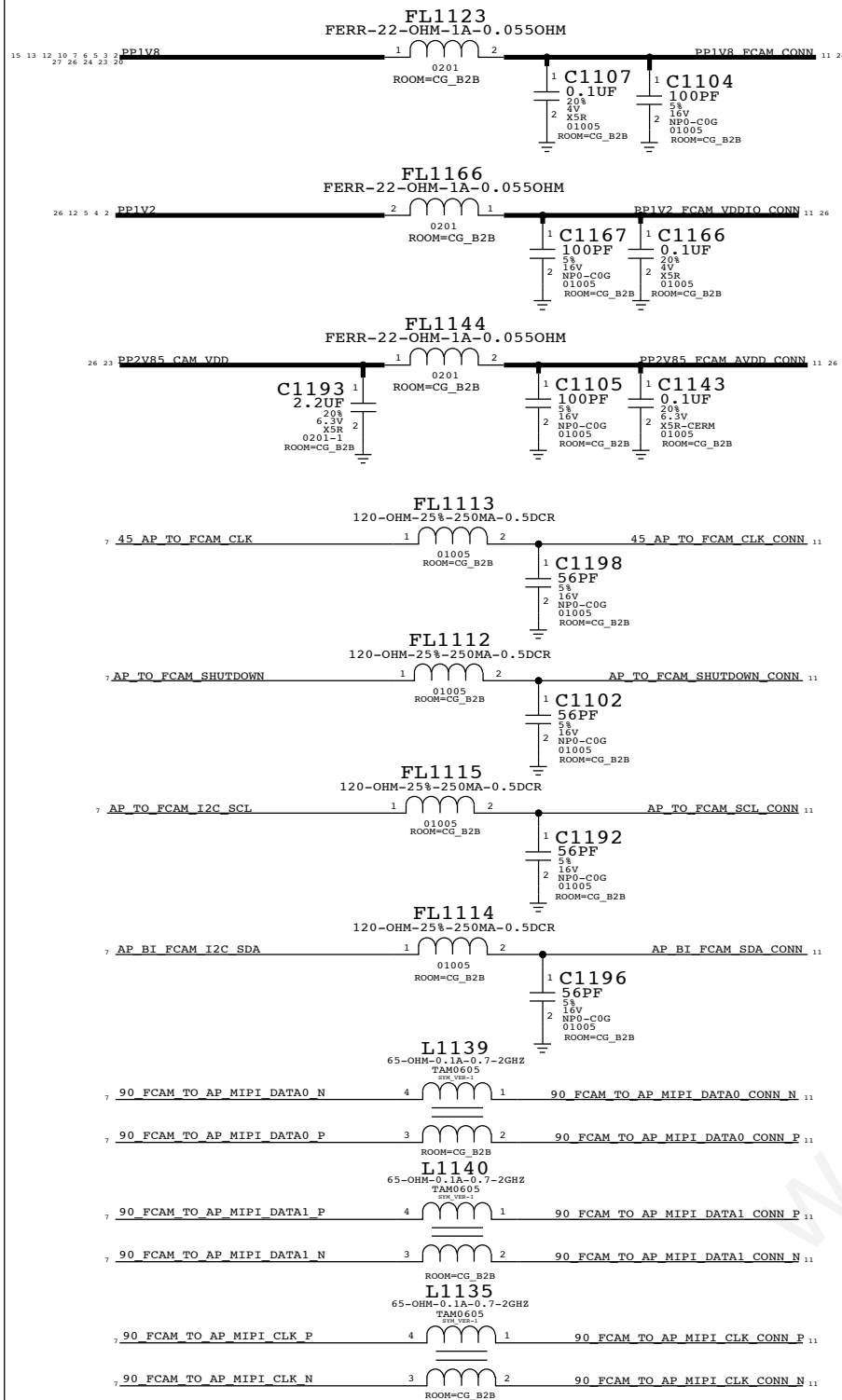


DIGITAL SYSTEM I/O

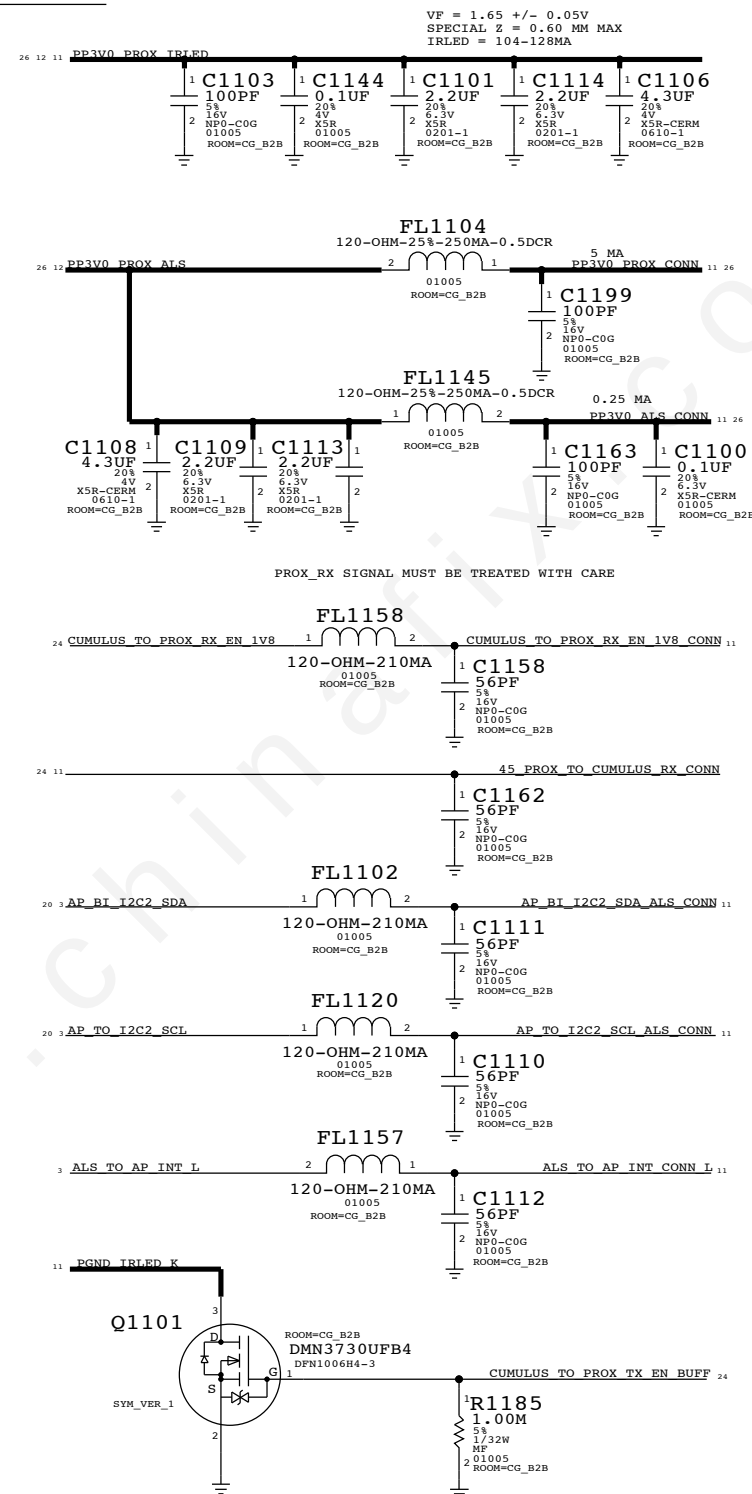
[illegible]

FRONT CAM FLEX B2B

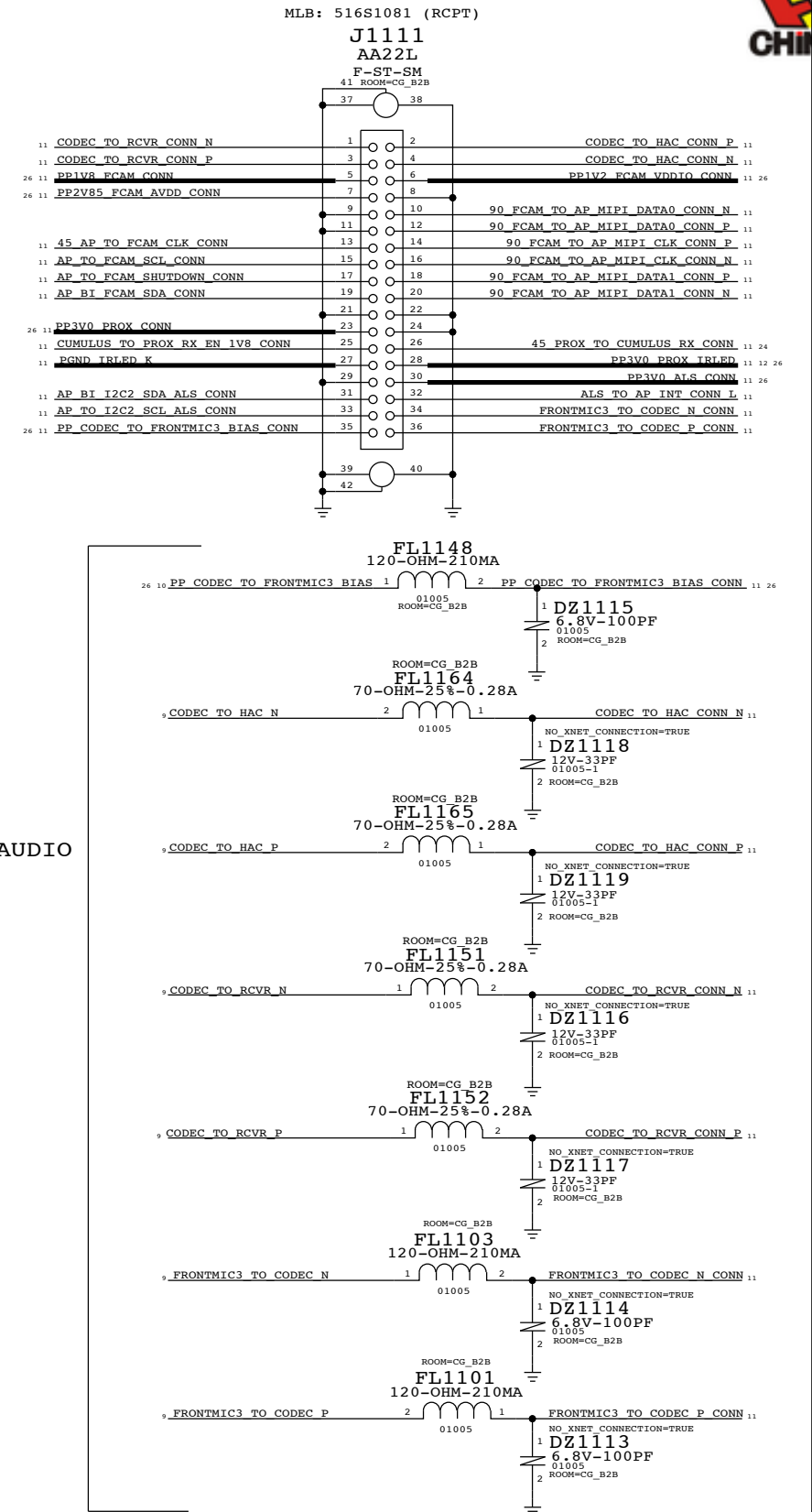
(FCAM, PROX, ALS, RECEIVER, ANC ERROR MIC)



ALS,
PROX



AUDIO





C

B

A



C

B

A



ADI PMU

(AMUX, GPIO, BUTTONS, ADC, THERMISTORS, SYSTEM I/F, GND)

D

D

C

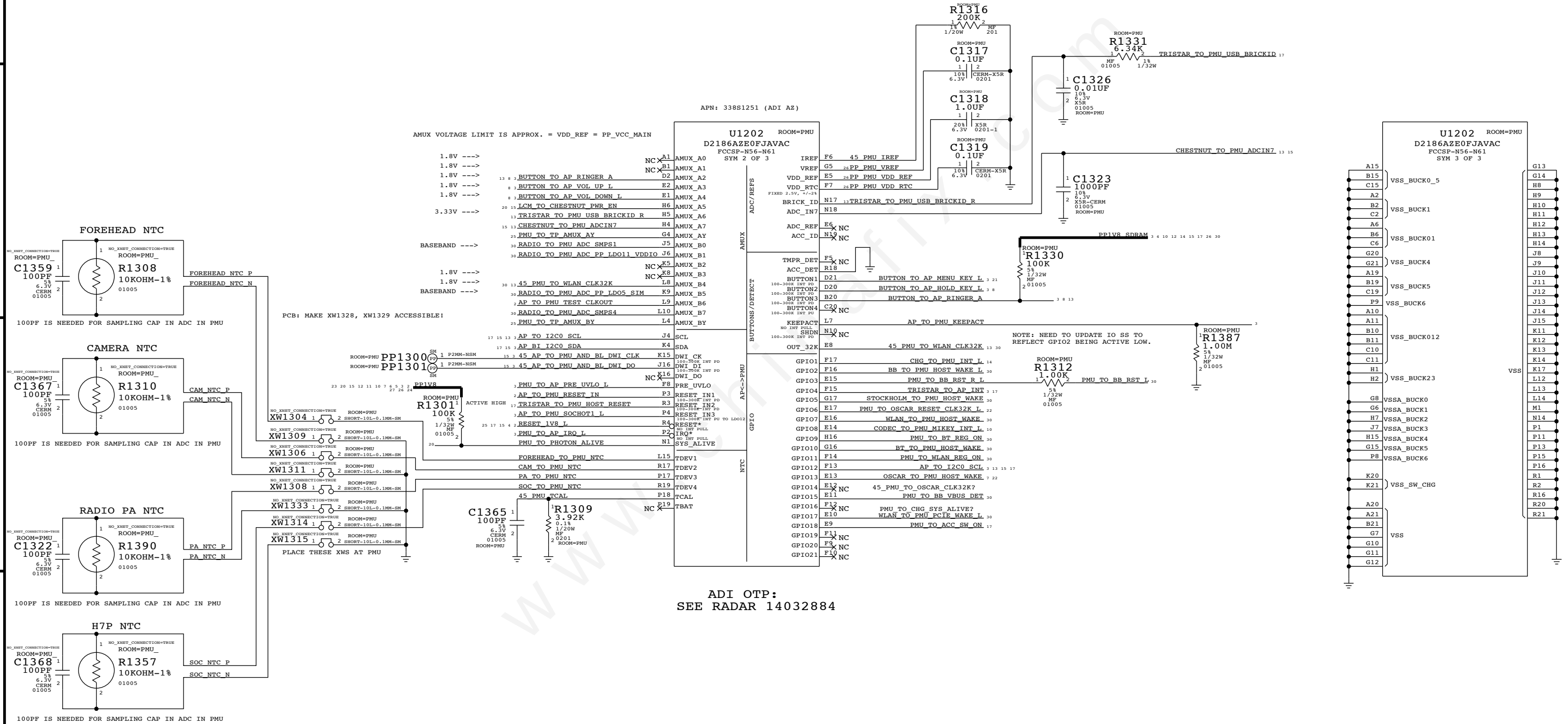
C

B

B

A

A

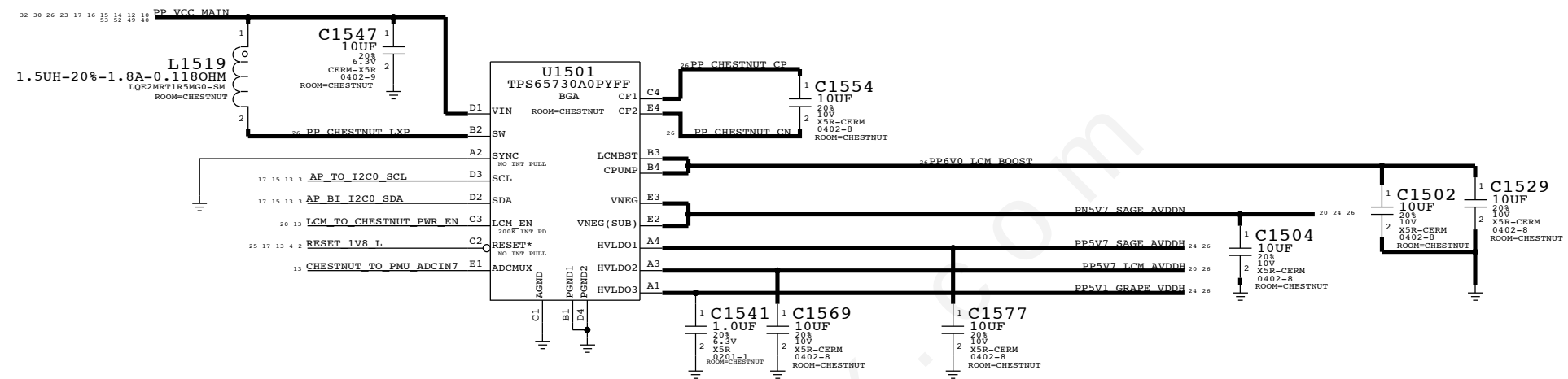




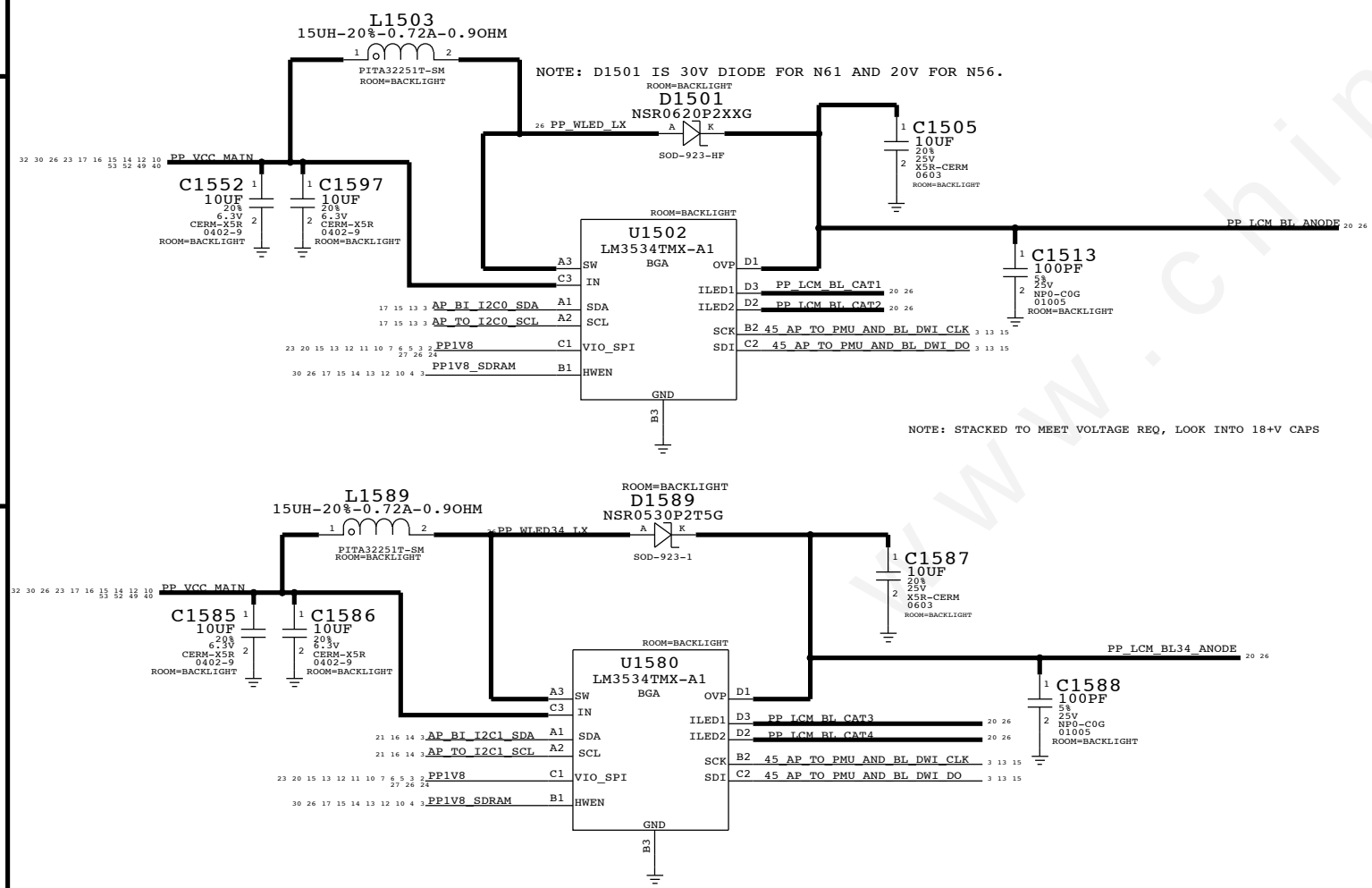


CHESTNUT, BACKLIGHT DRIVER, MESA BOOST

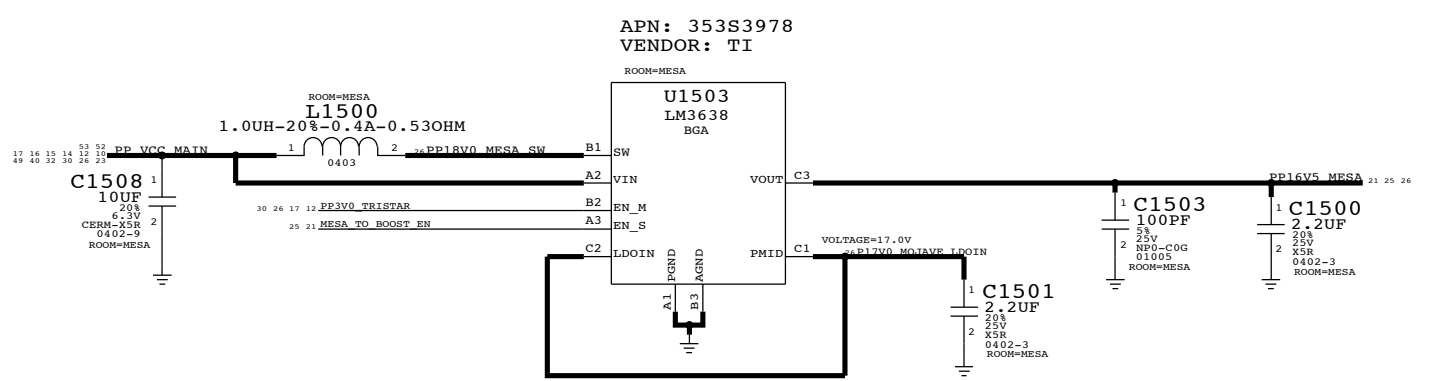
DISPLAY PMU (TI CHESTNUT, 338S1149)



BACKLIGHT DRIVERS

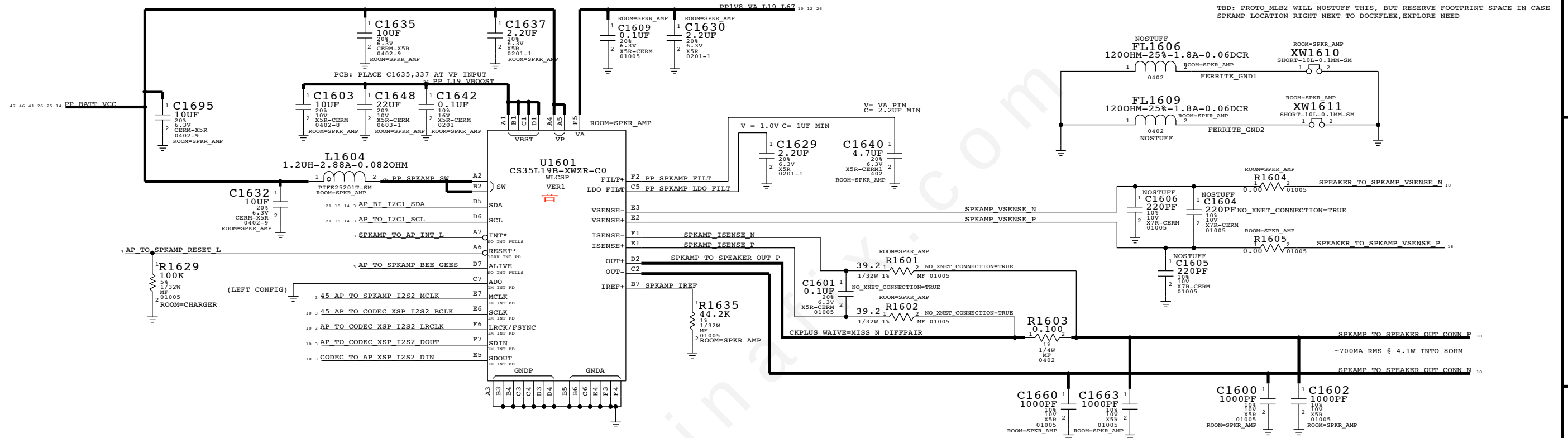


MESA BOOST



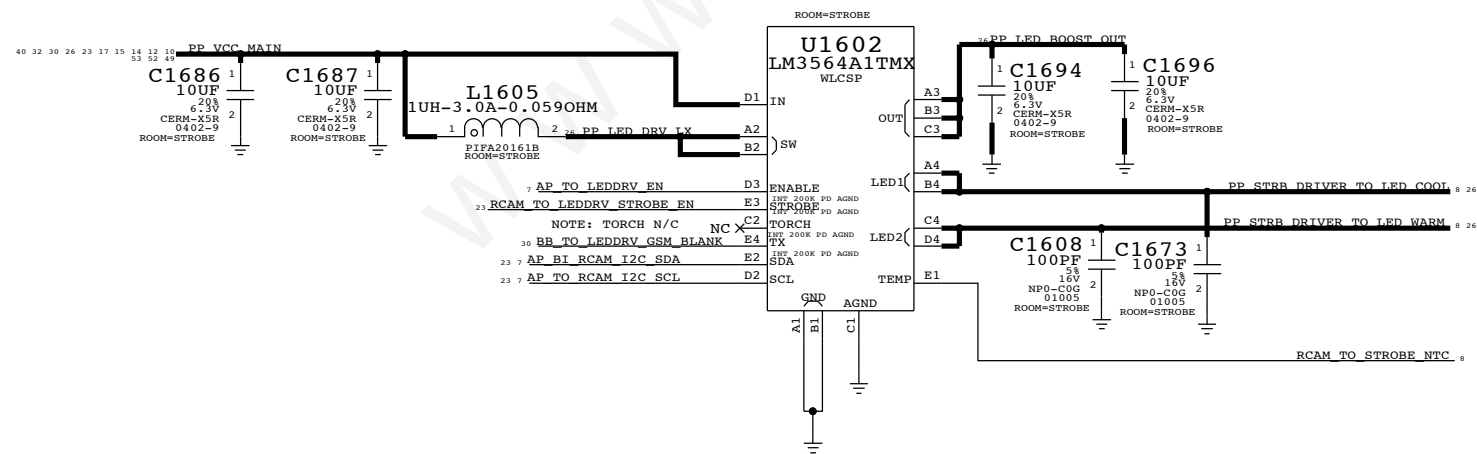


I2C ADDRESS: 1000000X



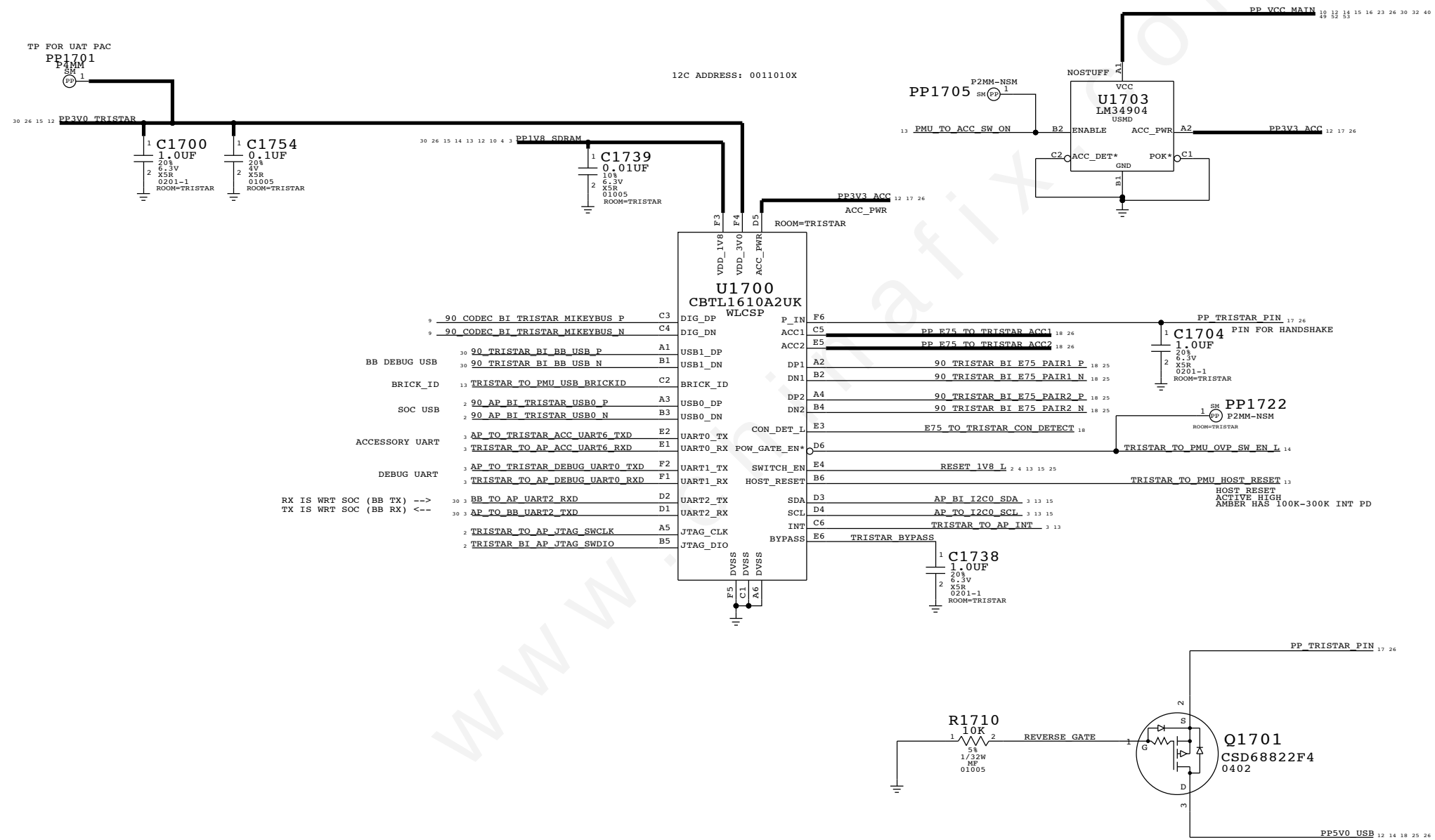
STROBE DRIVER

TI: APN 353S3899



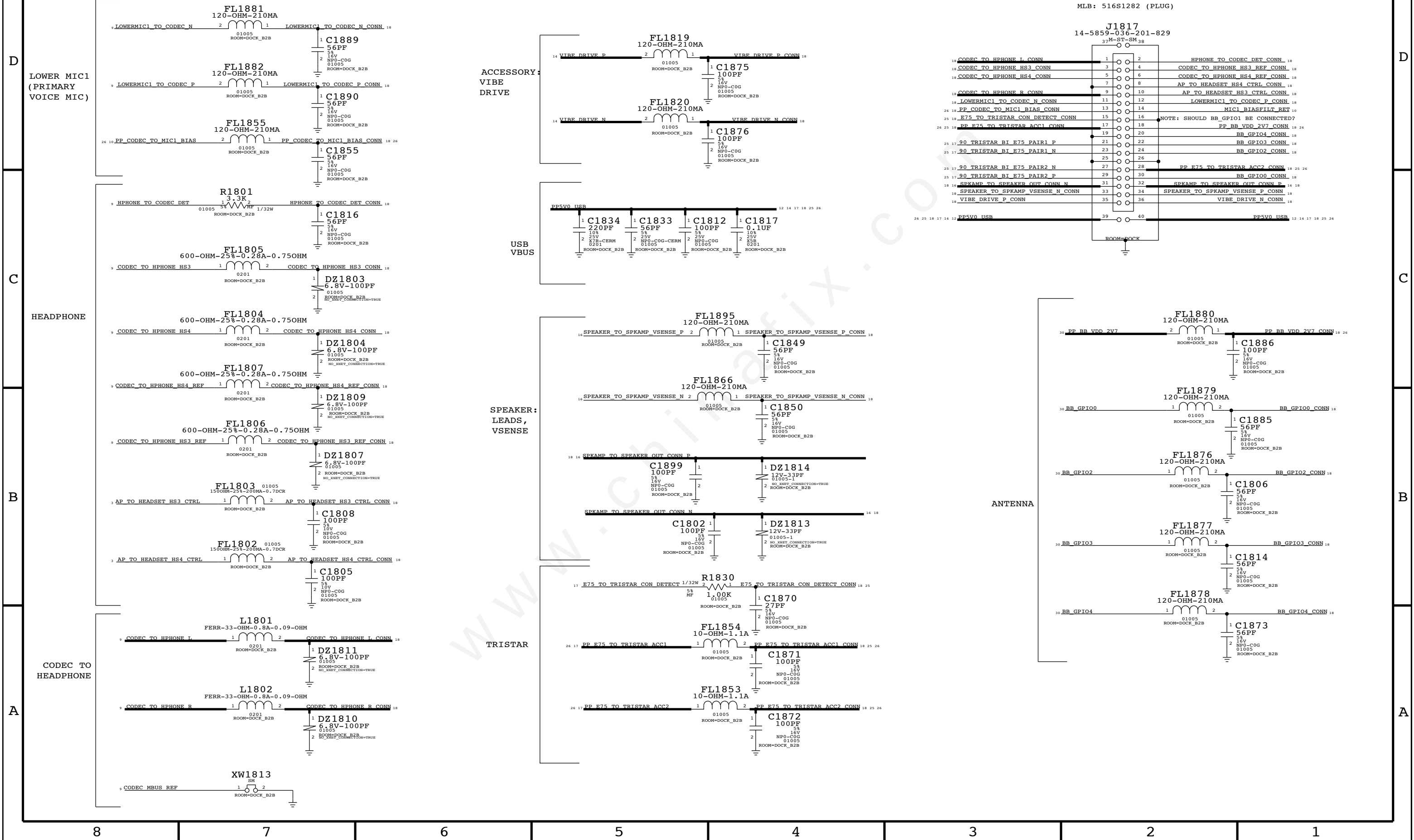


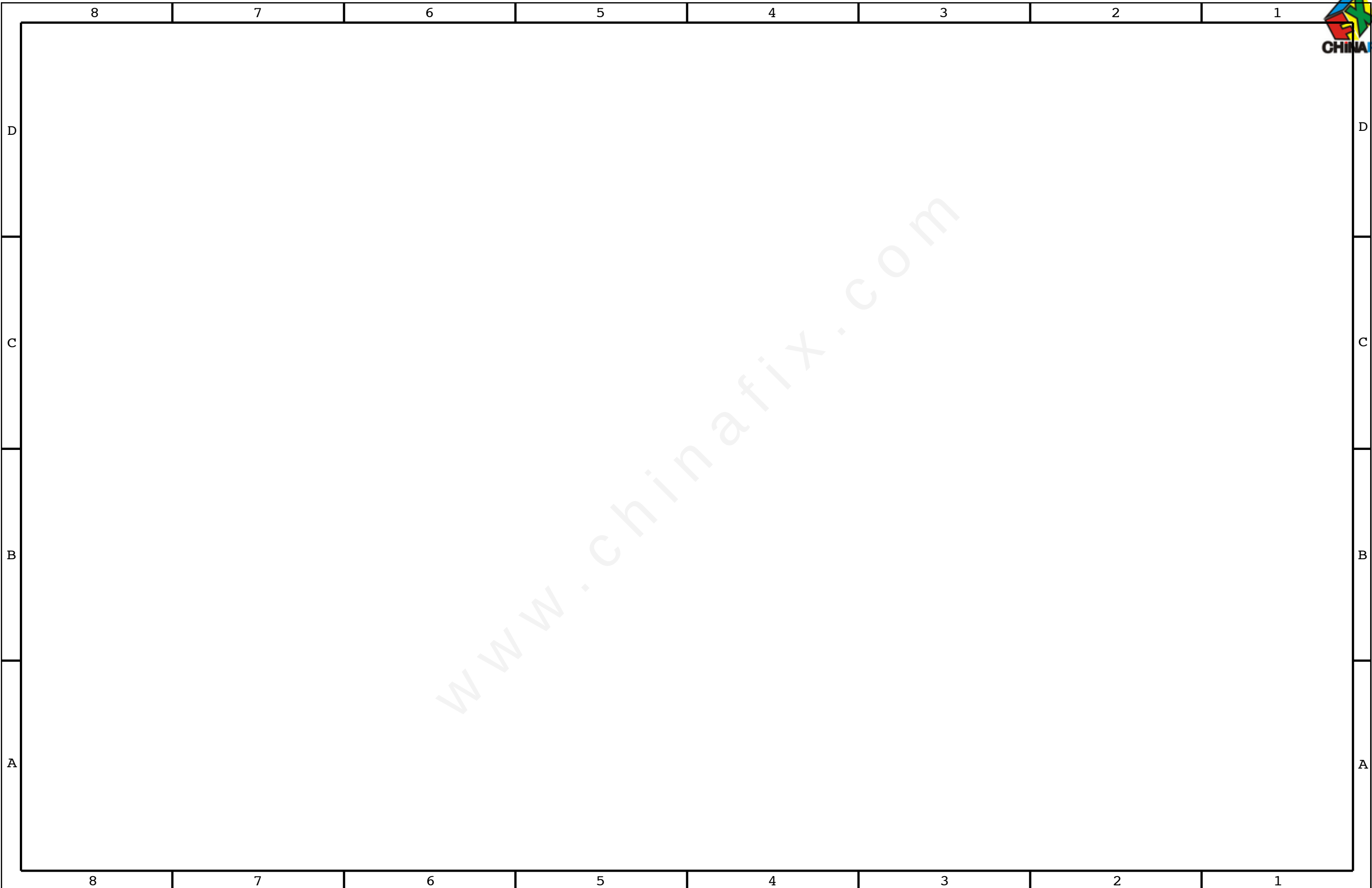
TRISTAR2





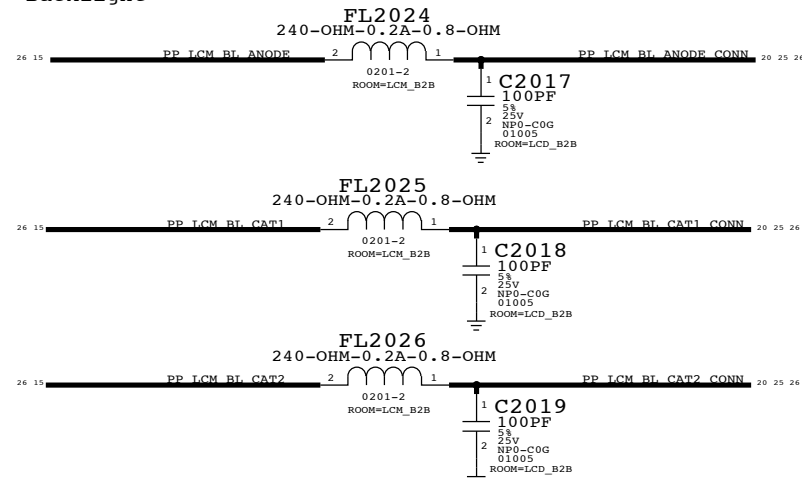
DOCKFLEX B2B (USB VBUS, SPEAKER, ANTENNA LAT SW CTRL, MIC1 (PRIMARY MIC), ACC DET/ID/PWR, E75 DIFFPAIRS)



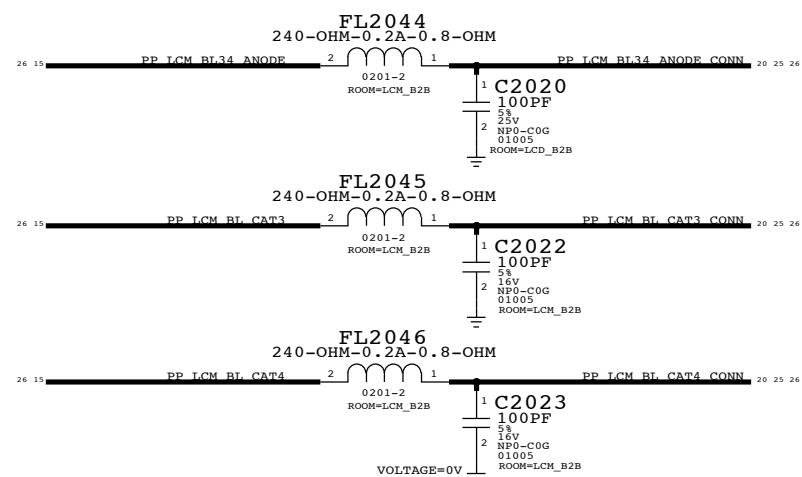


LCD B2B

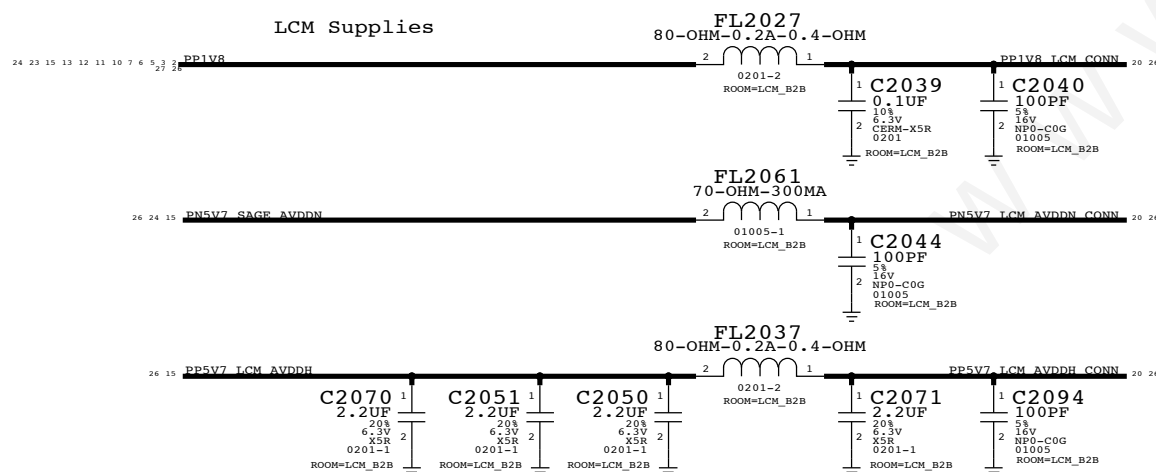
Backlight



BACKLIGHT 2 (N56 ONLY)



LCM Supplies

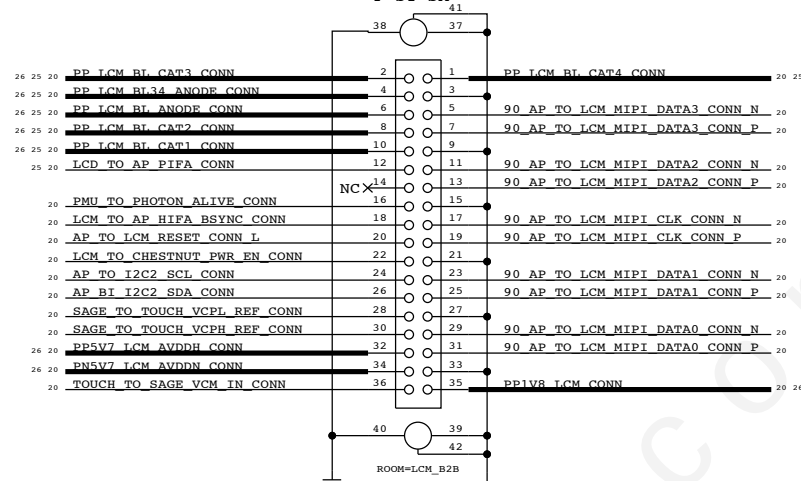


MLB: 516S1185 (RCPT)
(516S1184 ON FLEX)

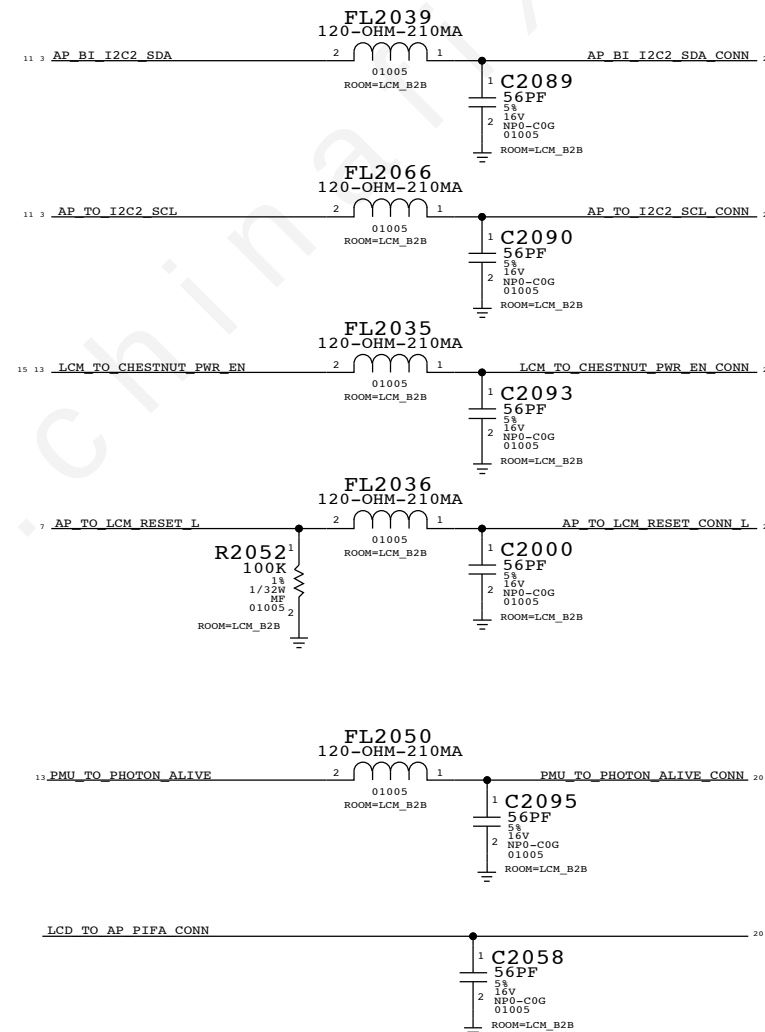
J2019

20-5857-036-001-829

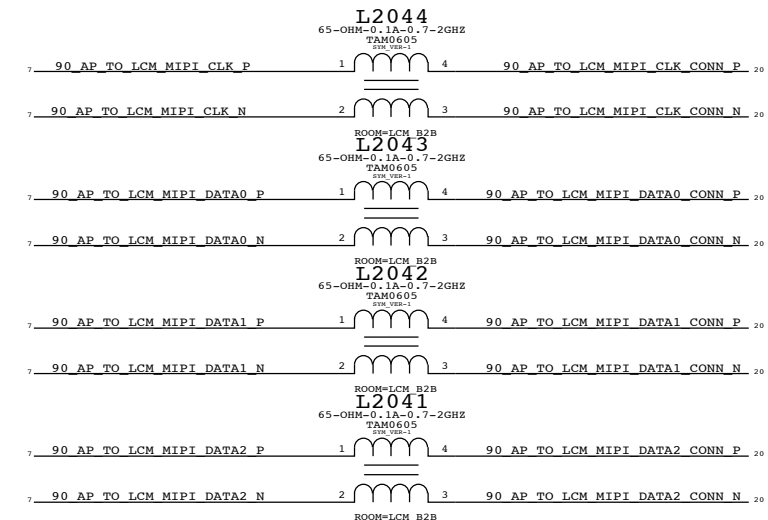
F-ST-SM



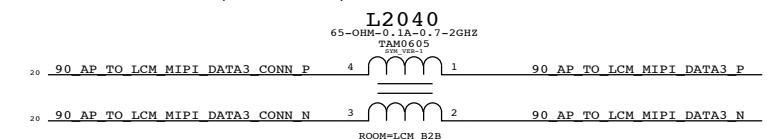
Digital Interfaces



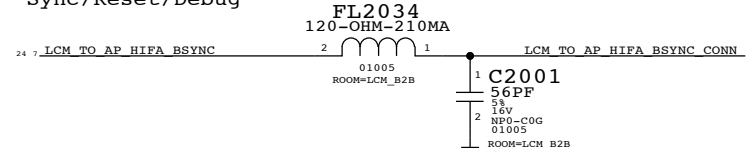
MIPI Common Mode Chokes



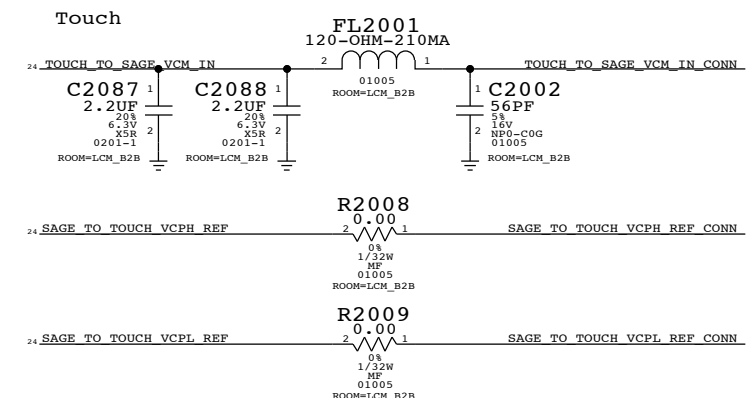
MIPI LANE 3 (N56 ONLY)



Sync/Reset/Debug

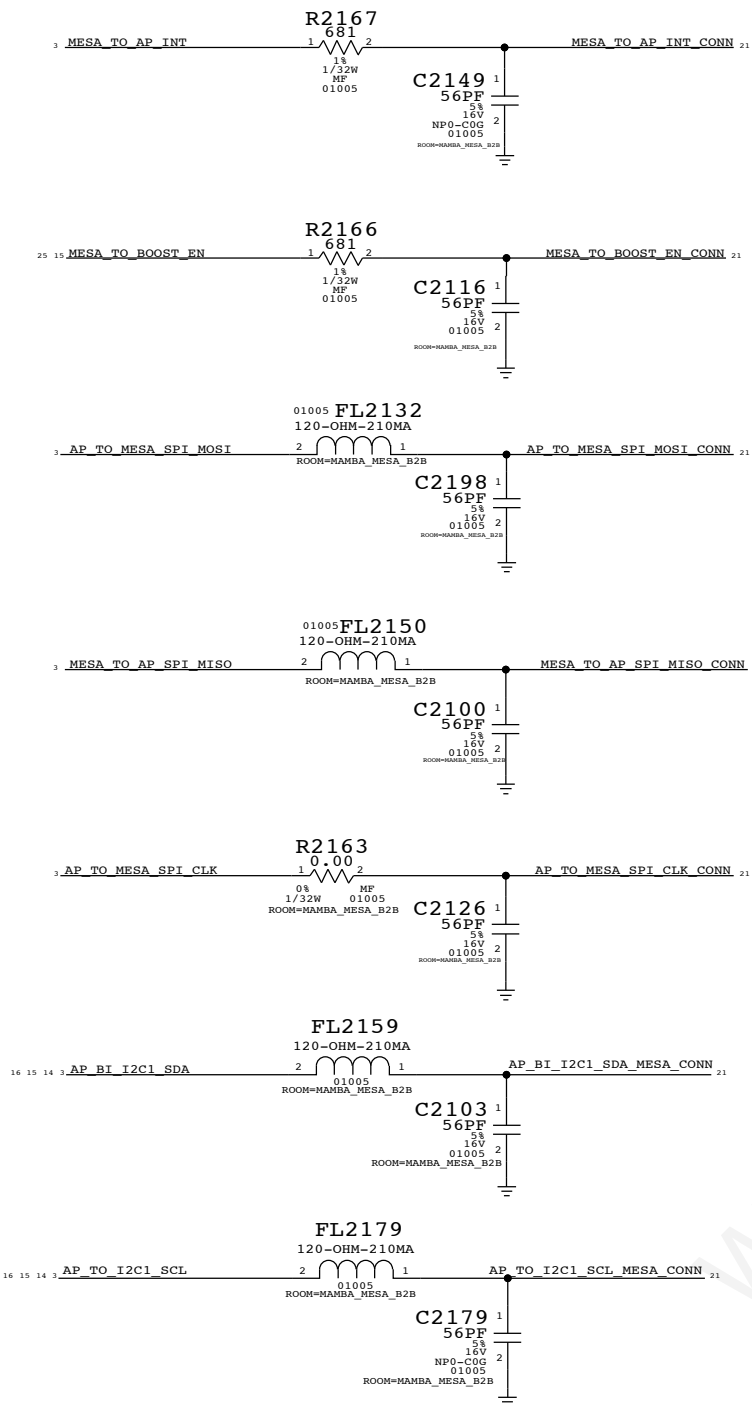


Touch



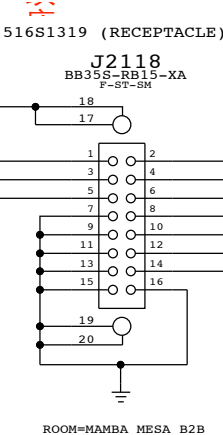
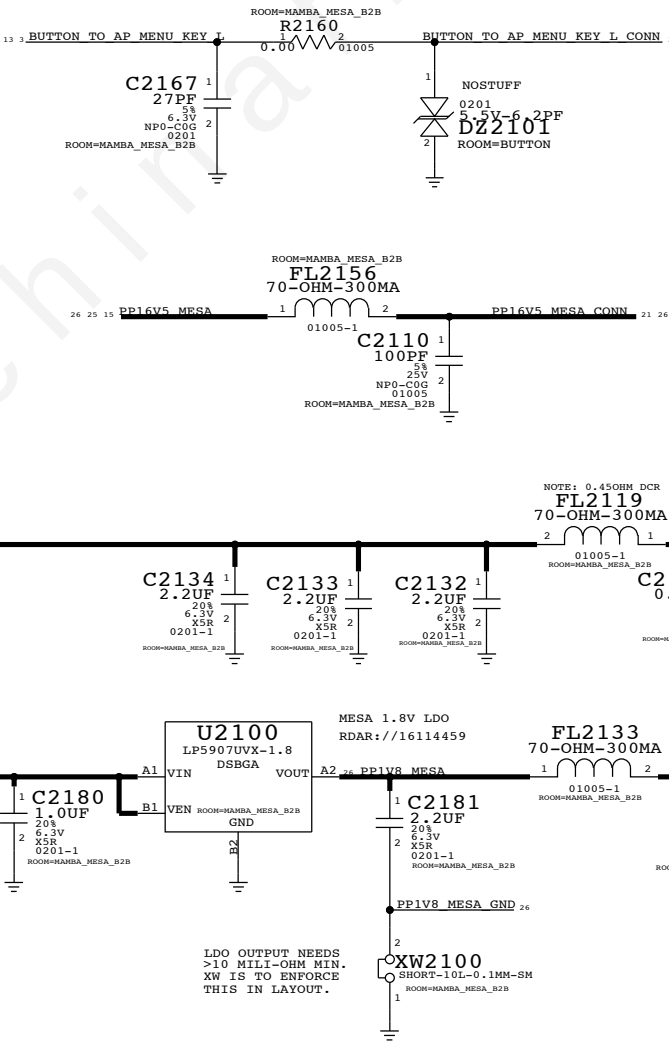
MESA CONNECTOR

MESA:
INT, EN, SPI, I2C



MENU
BUTTON

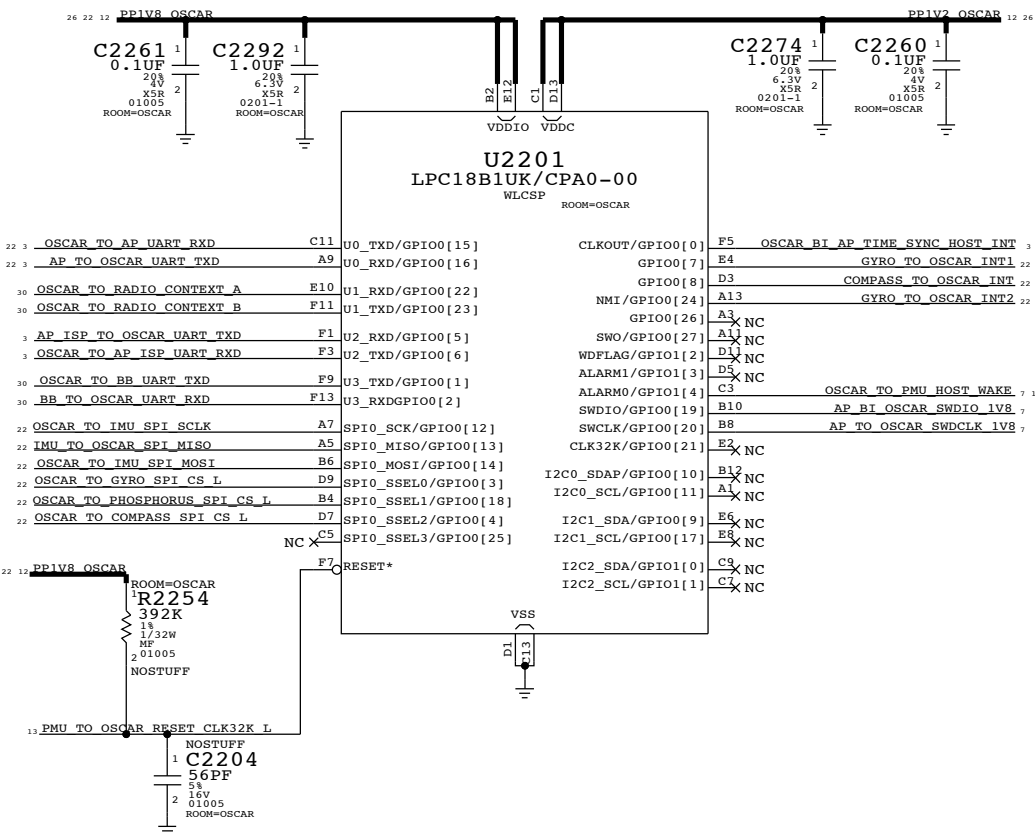
MESA:
POWER



LDO OUTPUT NEEDS
>10 MILLI-OHM MIN.
XW IS TO ENFORCE
THIS IN LAYOUT.

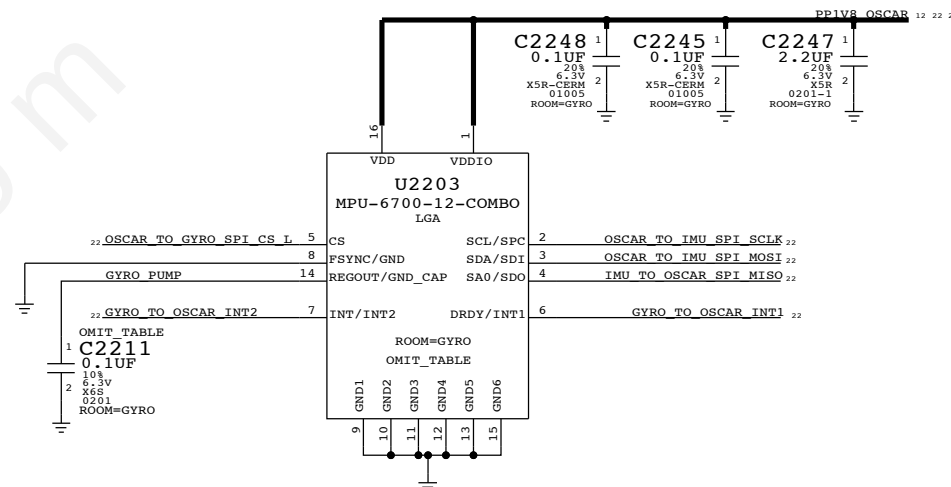
OSCAR + SENSORS

OSCAR VDDIO = 1.8V ALWAYS ON (NEED TO MAKE HOST & RUN PLL)
OSCAR CORE = 1.2V ALWAYS ON (NEED TO RUN IN S2RAM)



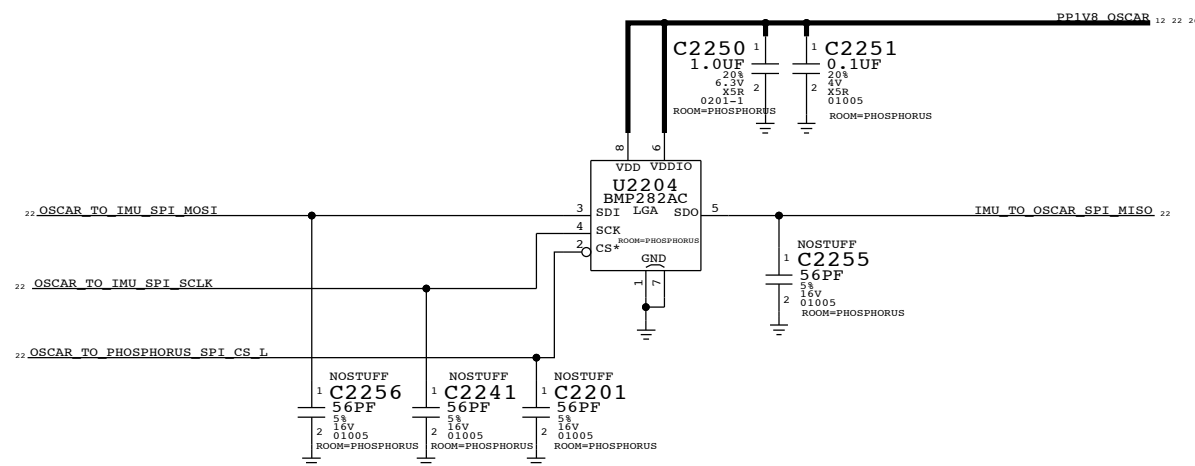
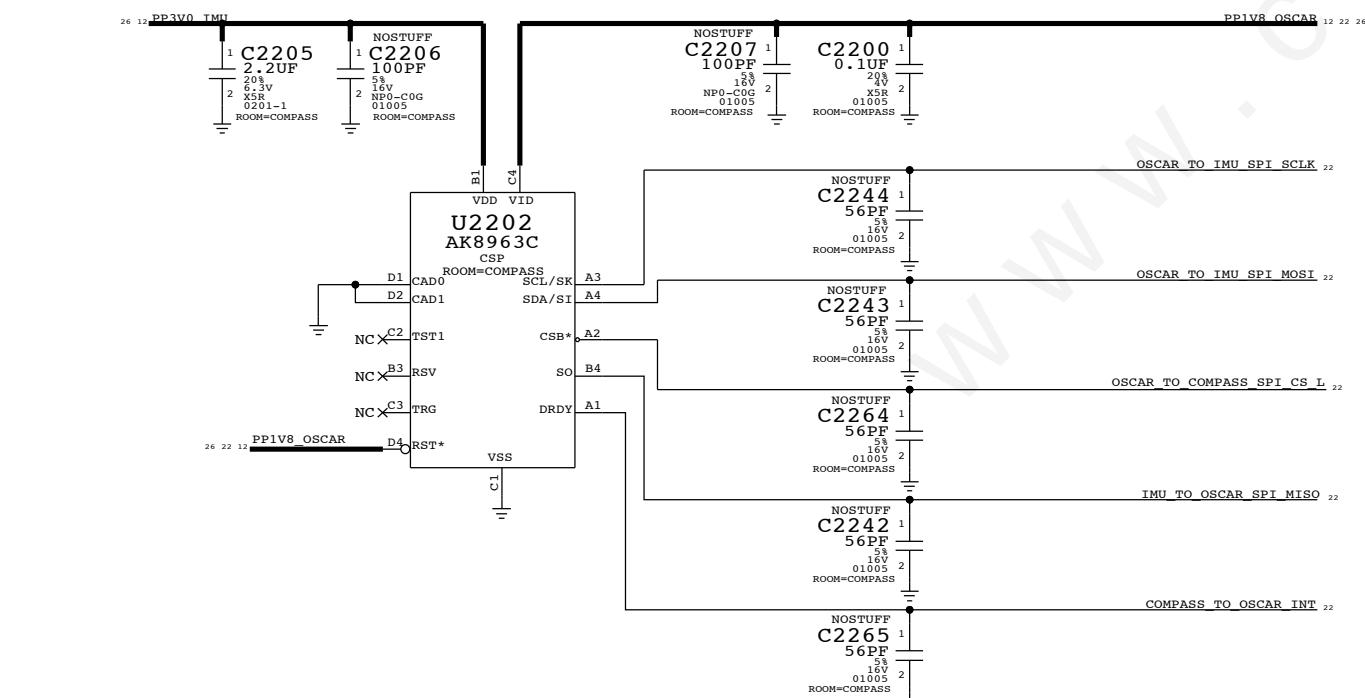
CARBON (ACCEL GYRO COMBO)

INVENSENSE, APN 338S00017, C2211=0.1UF (132S0395)
BOSCH, APN 338S00028, C2211=0.1UF (132S0395)
ST, APN 338S00029, C2211=0.01UF,25V (132S0391)



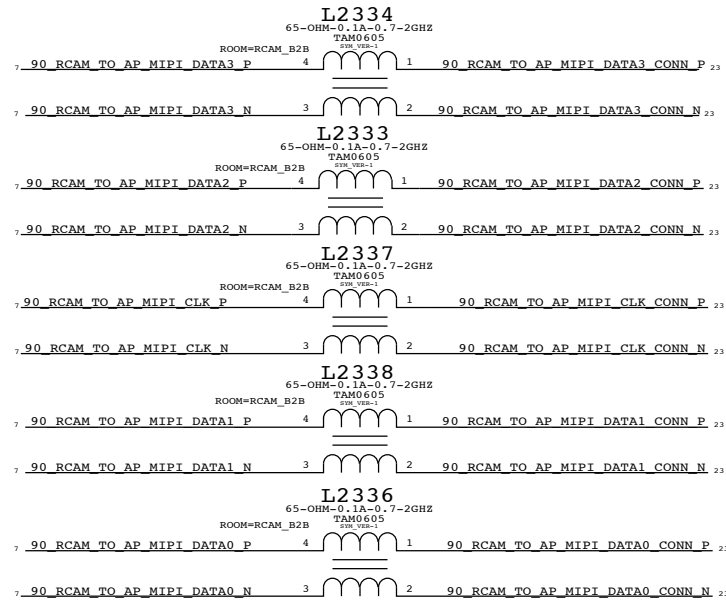
THIS IS OUTSIDE OF SHIELD IN
TO THE RIGHT OF THE NAND
PHOSPHORUS

THIS PART OUTSIDE OF SHIELD ON THE PENINSULA
COMPASS

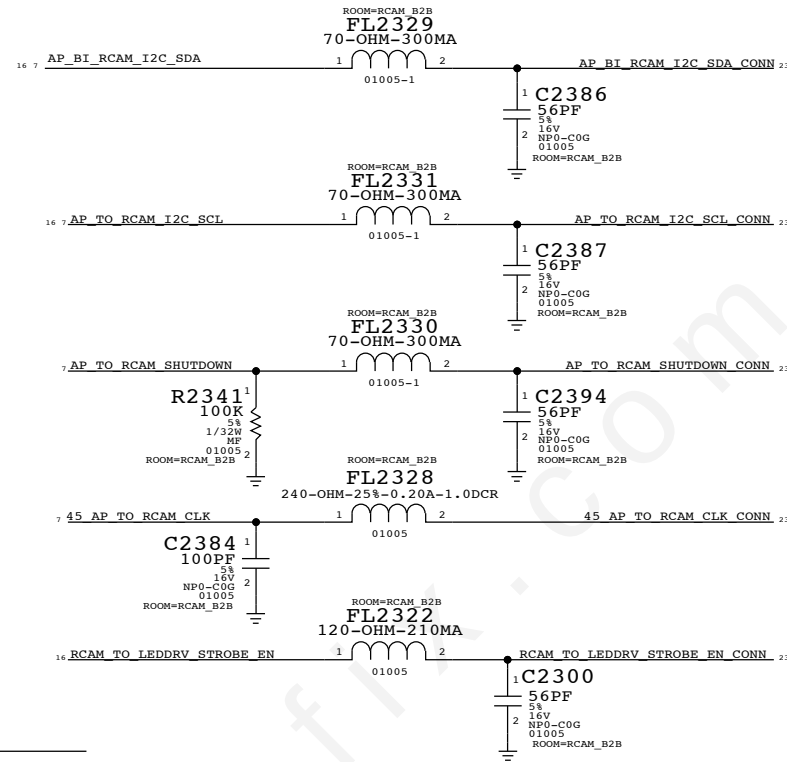


RCAM B2B (REAR CAMERA CONNECTOR)

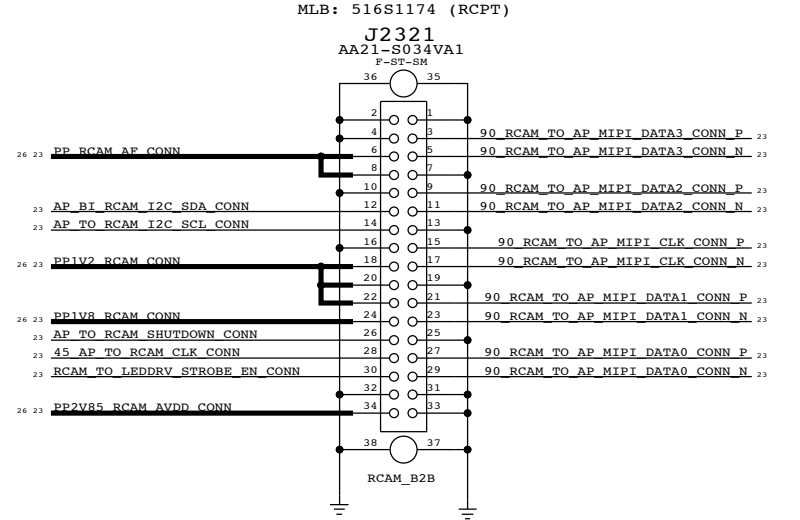
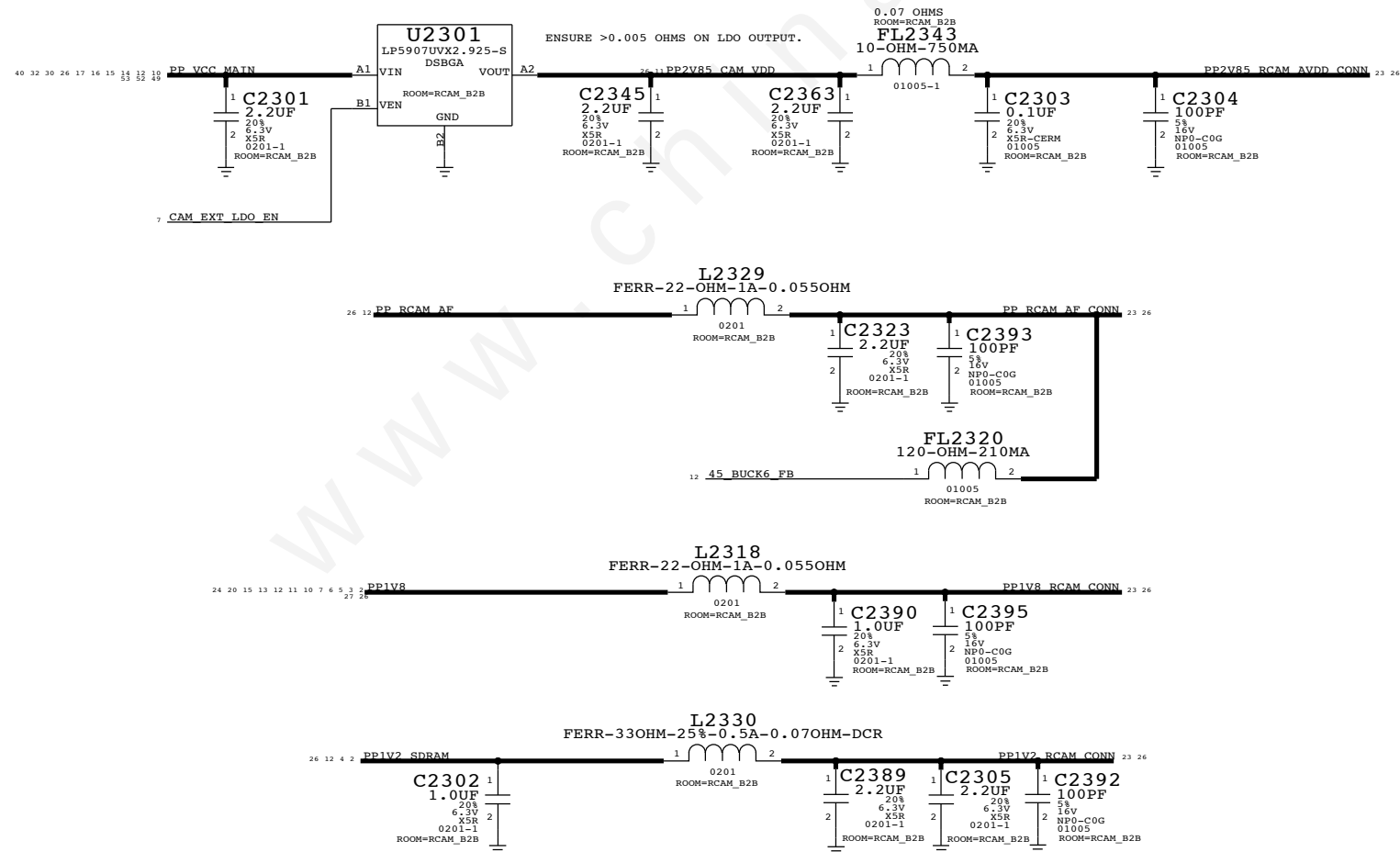
RCAM:
4-LANE MIPI



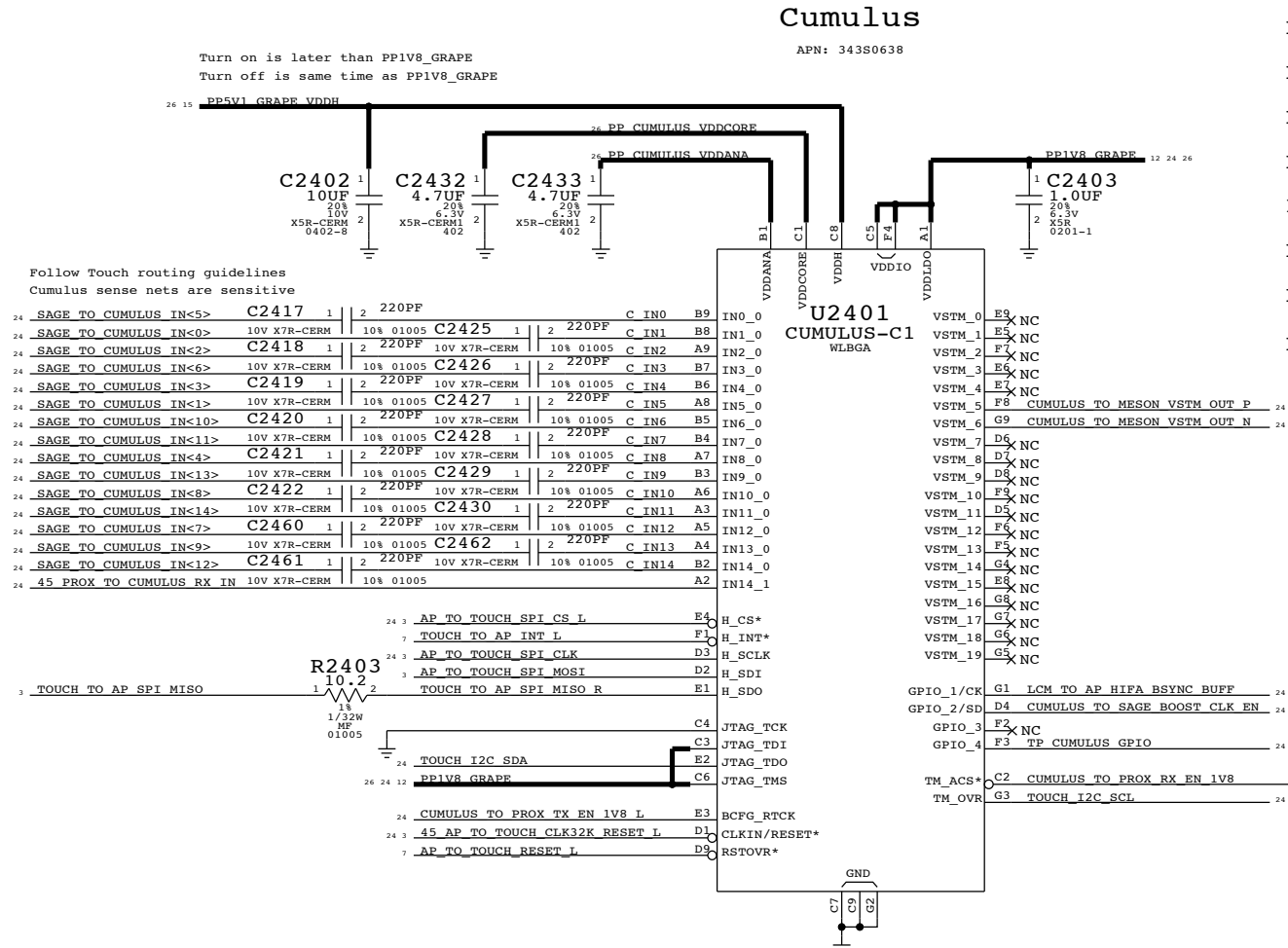
RCAM:
DIGITAL I/F
(I2C, CTRL, CLK)



RCAM:
POWER:
(1.8V DOVDD)
(2.9V AVDD)
(1.2V DVDD)
(2V AF)

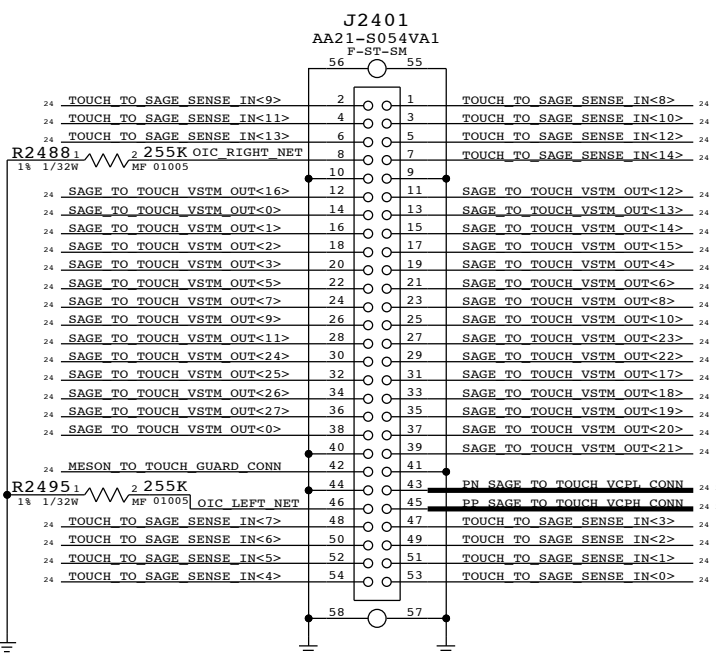


Touch (B2B, Driver ICs)



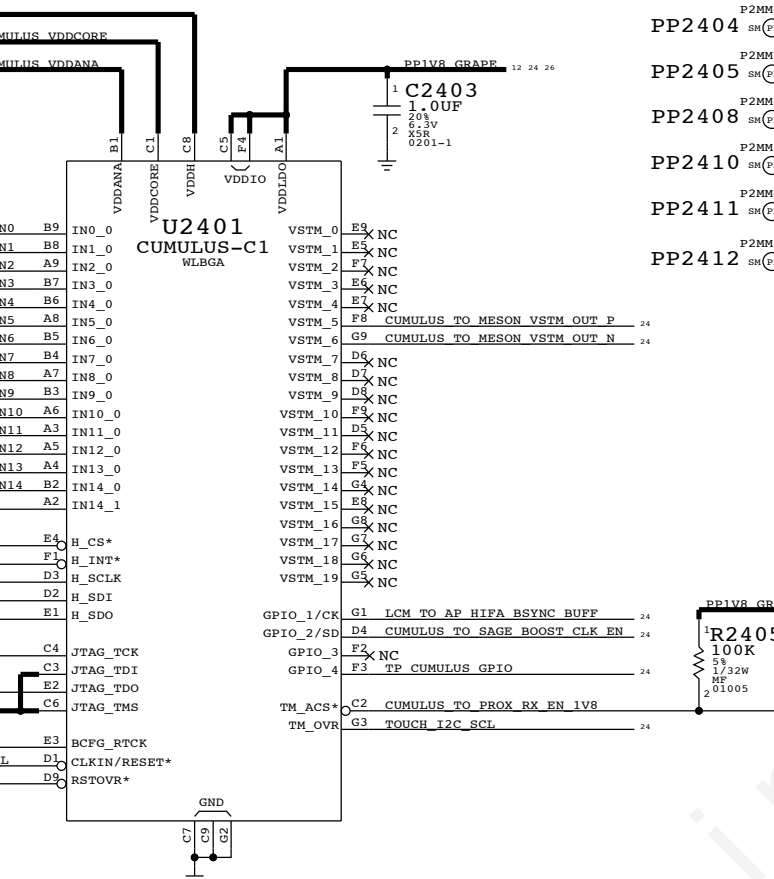
Touch B2B

MLB APN : 516S1183 (Receptacle)
Flex APN: 516S1182 (Plug)

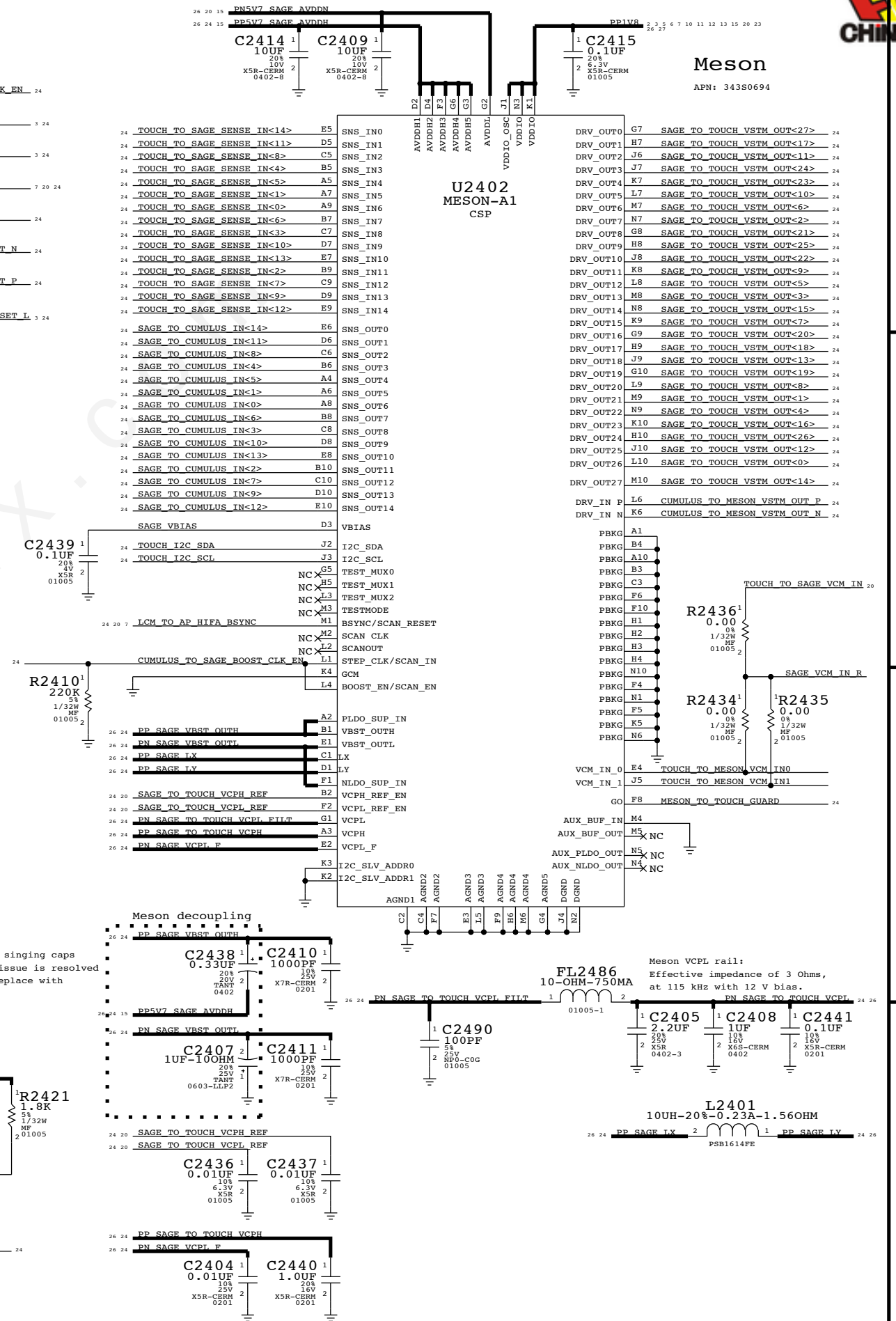
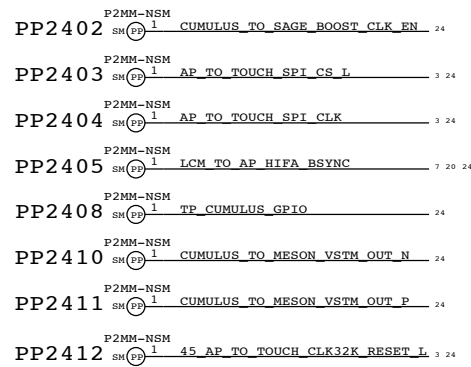


Cumulus

APN: 343S0638



Touch probe points

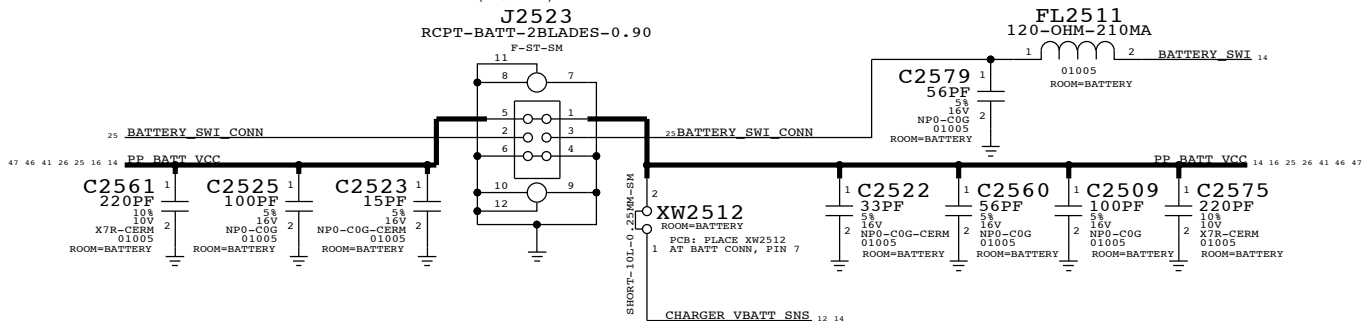




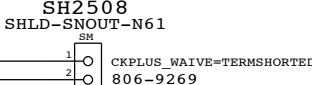
BATT CONN, TPS, STANDOFFS/SHIELDS/FIDUCIALS

BATTERY CONN

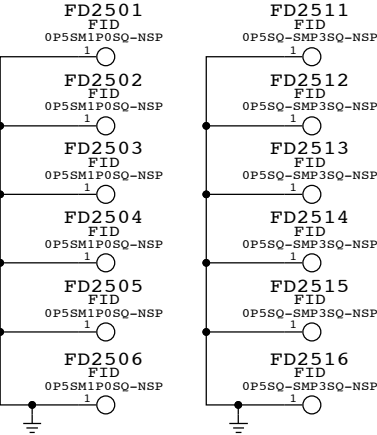
THIS ONE ON MLB ---> 516S1080 RCPT
SCHEMATIC SYMBOL(PIN ORDER) IS WIERD !!



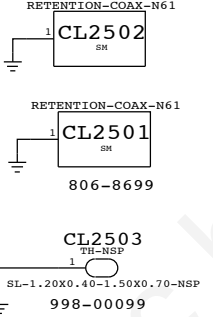
SHIELDS



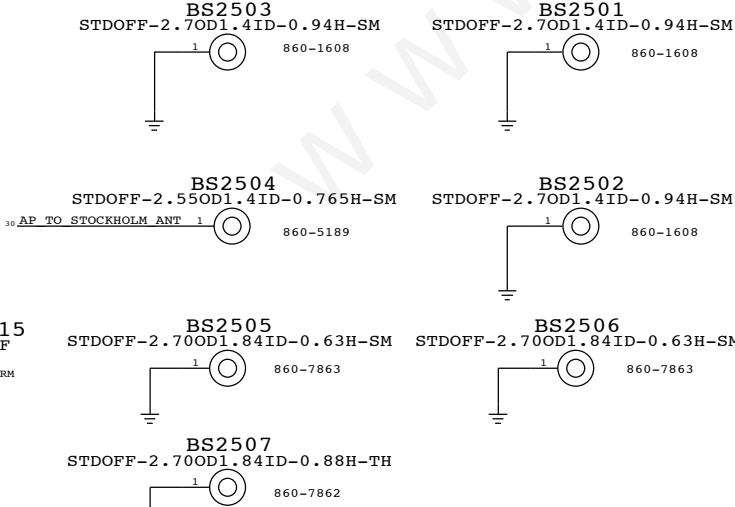
FIDUCIALS



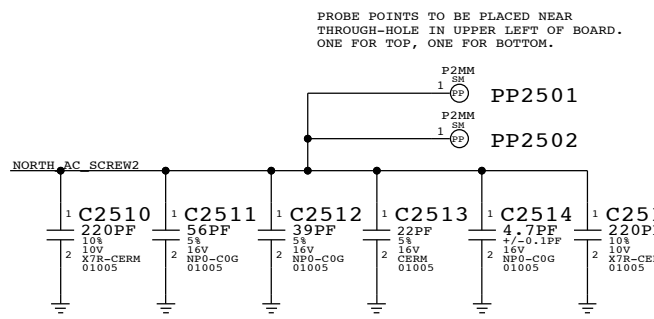
RF CLIPS



STANDOFFS

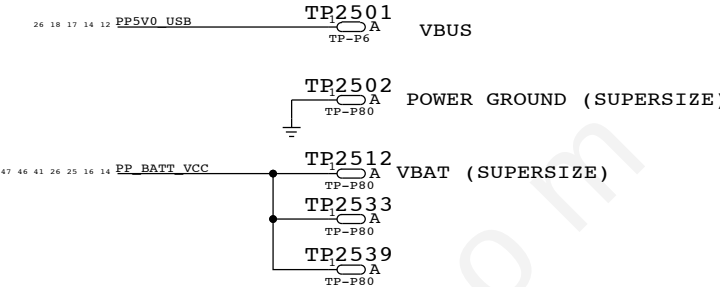


SCREW HOLES + STANDOFFS

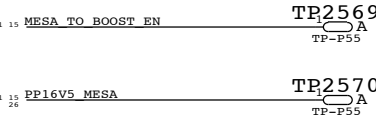


TESTPOINTS

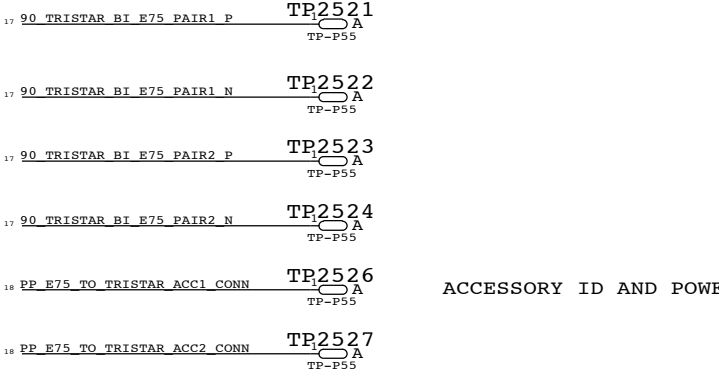
POWER TP



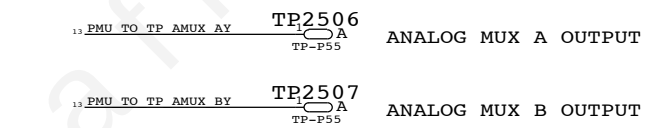
MOJAVE TP



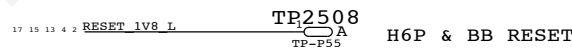
E75 - USB/UART/ID/POWER



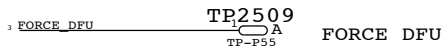
SUPER TP



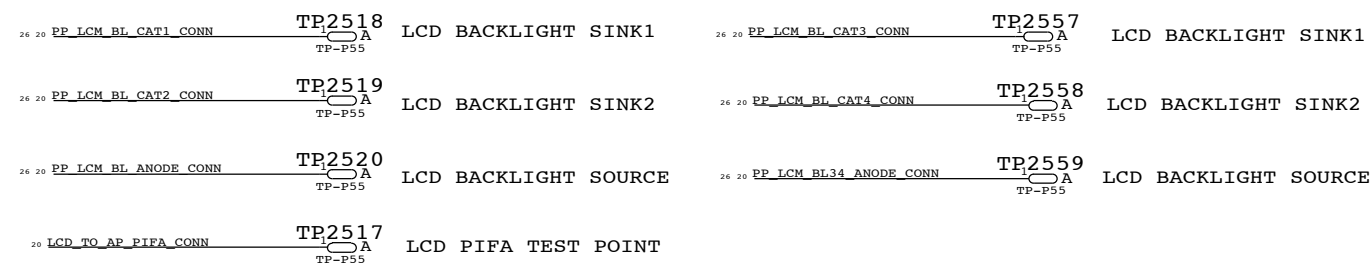
RESET



DFU



LCM BACKLIGHT



VOLTAGE PROPERTIES

E65	VOLTAGE=3.3V	PP3V3 USB	2 12
E66	VOLTAGE=1.8V	PP1V8 VA I19 I67	10 12 16
E67	VOLTAGE=3.0V	PP3V0 TRISTAR	12 15 17 30
E68	VOLTAGE=3.0V	PP3V0 IMU	12 22
E69	VOLTAGE=3.0V	PP3V0 NAND	6 12
E70	VOLTAGE=3.0V	PP3V3 ACC	12 17
E71	VOLTAGE=3.0V	PP3V0 PROX ALS	11 12
E72	VOLTAGE=3.0V	PP2V9 LDO9	12
E73	VOLTAGE=4.6V	PP VCC MAIN	10 12 14 15 16 17 23 30 32
E74	VOLTAGE=1.0V	PP1V0	7 12
E75	VOLTAGE=3.0V	PP3V0 PROX TRIED	11 12
E76	VOLTAGE=1.8V	PP1V8 ALWAYS	3 5 12 14
E77	VOLTAGE=3.0V	PP3V0 MESA	12 21
E78	VOLTAGE=1.1V	PP CPU	4 12
E79	VOLTAGE=1.1V	PP GPU	4 12
E80	VOLTAGE=1.2V	PP1V2 SDRAM	2 4 12 23
E81	VOLTAGE=1.8V	PP1V8 SDRAM	3 4 10 12 13 14 15 17 30
E82	VOLTAGE=1.8V	PP1V8	2 4 12 23
E83	VOLTAGE=1.8V	PP1V8 GRAPE	12 24
E84	VOLTAGE=1.8V	PP1V8 OSCAR	12 22
E85	VOLTAGE=1.2V	PP1V2 NAND VDDT	6
E86	VOLTAGE=1.8V	PP EXTMIC BIAS FILT IN	10
E87	VOLTAGE=1.8V	BOARD ID2	3 27
E88	VOLTAGE=1.2V	PP1V2	2 4 5 11 12
E89	VOLTAGE=5.0V	PP E75 TO TRISTAR ACC1 CONN	18 25
E90	VOLTAGE=5.0V	PP E75 TO TRISTAR ACC1	17 18
E91	VOLTAGE=22.0V	PP LCM BL ANODE	15 20
E92	VOLTAGE=0.2V	PP LCM BL CAT2	15 20
E93	VOLTAGE=0.2V	PP LCM BL CAT1	15 20
E94	VOLTAGE=0.2V	PP LCM BL CAT2 CONN	20 25
E95	VOLTAGE=0.2V	PP LCM BL CAT1 CONN	20 25
E96	VOLTAGE=-5.7V	PN5V7 SAGE AVDDN	15 20 24
E97	VOLTAGE=1.2V	PP1V2 OSCAR	12 22
E98	VOLTAGE=3.0V	PP3V0 MESA CONN	21
E99	VOLTAGE=6V	PP6V0 LCM BOOST	15
E100	VOLTAGE=5.0V	PP STRB DRIVER TO LED WARM	9 16
E101	VOLTAGE=5.0V	PP STRB DRIVER TO LED COOL	8 16
E102	VOLTAGE=1.8V	PP CODEC TO MIC1 BIAS	10 18
E103	VOLTAGE=1.8V	PP EXTMIC BIAS FILT IN	10
E104	VOLTAGE=1.8V	PP EXTMIC BIAS FILT	10
E105	VOLTAGE=1.8V	PP CODEC TO FRONTMIC3 BIAS	10 11
E106	VOLTAGE=1.8V	PP CODEC TO REARMIC2 BIAS	8 10
E107	VOLTAGE=1.8V	PP CODEC FILT4	10
E108	VOLTAGE=2.2V	PP CODEC SPKR VO	10
E109	VOLTAGE=2.5V	PP CODEC VCPFILT	10
E110	VOLTAGE=2.5V	PP CODEC VCPFILT4	10
E111	VOLTAGE=2.5V	PP CODEC VHP ELYN	10
E112	VOLTAGE=0.2V	PP CODEC VHP ELYC	10
E113	VOLTAGE=2.5V	PP CODEC VHP ELYP	10
E114	VOLTAGE=1.8V	PP1V8 FCAM CONN	11
E115	VOLTAGE=3.0V	PP2V85 FCAM AVDD CONN	11
E116	VOLTAGE=1.8V	PP CODEC TO FRONTMIC1 BIAS CONN	11
E117	VOLTAGE=3.0V	PP3V0 ALS CONN	11
E118	VOLTAGE=1.2V	PP1V2 FCAM VDDIO CONN	11
E119	VOLTAGE=5.0V	PP5V0 USB	12 14 17 18 25
E120	VOLTAGE=5.0V	PP5V0 USB TO PMU	12
E121	VOLTAGE=4.6V	PP BUCK5 LX0	12
E122	VOLTAGE=4.6V	PP BUCK3 LX	12
E123	VOLTAGE=4.6V	PP BUCK4 LX	12
E124	VOLTAGE=4.6V	PP BUCK2 LX	12
E125	VOLTAGE=4.6V	PP BUCK1 LX1	12
E126	VOLTAGE=4.6V	PP BUCK1 LX0	12
E127	VOLTAGE=4.6V	PP BUCK0 LX3	12
E128	VOLTAGE=4.6V	PP BUCK0 LX2	12
E129	VOLTAGE=4.6V	PP BUCK0 LX1	12
E130	VOLTAGE=4.6V	PP BUCK0 LX0	12
E131	VOLTAGE=6.0V	PP CHESTNUT LXP	15
E132	VOLTAGE=6.0V	PP CHESTNUT CP	15
E133	VOLTAGE=6.0V	PP CHESTNUT CN	15
E134	VOLTAGE=5.7V	PP5V7 SAGE AVDDH	15 24
E135	VOLTAGE=5.7V	PP5V7 LCM AVDDH	15 20
E136	VOLTAGE=5.1V	PP5V1 GRAPE VDDH	15 24
E137	VOLTAGE=22.0V	PP WLED LX	15
E138	VOLTAGE=18.0V	PP18V0 MESA SW	15
E139	VOLTAGE=17.0V	PP17V0 MESA JDRIN	15
E140	VOLTAGE=16.5V	PP16V5 MESA	15 21 25
E141	VOLTAGE=8.0V	PP SPKAMP SW	16
E142	VOLTAGE=8.0V	PP I19 VBOOST	16
E143	VOLTAGE=1.8V	PP SPKAMP FILT	16
E144	VOLTAGE=1.8V	PP SPKAMP LDO FILT	16
E145	VOLTAGE=5.0V	PP LED DRV LX	16
E146	VOLTAGE=5.0V	PP LED BOOST OUT	16
E147	VOLTAGE=2.7V	PP BB VDD 2V7 CONN	18
E148	VOLTAGE=1.8V	PP CODEC TO MIC1 BIAS CONN	18
E149	VOLTAGE=4.6V	PP E75 TO TRISTAR ACC2	17 18
E150	VOLTAGE=4.6V	PP E75 TO TRISTAR ACC2 CONN	18 25
E151	VOLTAGE=1.8V	PP1V8 LCM CONN	20
E152	VOLTAGE=22.0V	PP LCM BL ANODE CONN	20 25
E153	VOLTAGE=-5.7V	PN5V7 LCM AVDDN CONN	20
E154	VOLTAGE=5.7V	PP5V7 LCM AVDDH CONN	20
E155	VOLTAGE=2.95V	PP IDO13 GPS	51
E156	VOLTAGE=1.8V	PP1V8 MESA	21
E157	VOLTAGE=16.5V	PP16V5 MESA CONN	21
E158	VOLTAGE=5.0V	PP TRISTAR PIN	17
E159	VOLTAGE=1.2V	PP1V2 RCAM CONN	23
E160	VOLTAGE=1.8V	PP1V8 RCAM CONN	23
E161	VOLTAGE=3.0V	PP2V85 CAM VDD	23 25
E162	VOLTAGE=1.8V	PP2V85 RCAM AVDD CONN	23
E163	VOLTAGE=1.8V	PP CUMULUS VDDCORE	24
E164	VOLTAGE=1.2V	PP CUMULUS VPDANA	24
E165	VOLTAGE=13.5V	PP SAGE TO TOUCH VCPH CONN	24
E166	VOLTAGE=-12V	PN SAGE TO TOUCH VCPH CONN	24
E167	VOLTAGE=13.5V	PP SAGE TO TOUCH VCPH	24
E168	VOLTAGE=-12V	PN SAGE TO TOUCH VCPH	24
E169	VOLTAGE=-12V	PN SAGE VCPH F	24
E170	VOLTAGE=5.7V	PP SAGE LX	24
E171	VOLTAGE=17.0V	PP SAGE LX	24
E172	VOLTAGE=1.8V	PP PMU VREF	13
E173	VOLTAGE=14V	PP SAGE VBST OUTH	24
E174	VOLTAGE=5.0V	PP TIGRIS VBUS DET	14
E175	VOLTAGE=2.5V	PP PMU VDD REF	13
E176	VOLTAGE=1.8V	PP EXTMIC BIAS	10
E177	VOLTAGE=1.8V	PP1V8 XTAL	2
E178	VOLTAGE=1.8V	PP PMU VDD RTC	13
E179	VOLTAGE=3.80V	PP BATT VCC	14 16 25 41 46 47
E180	VOLTAGE=1.8V	PP1V8 MESA CONN	21
E181	VOLTAGE=3.0V	PP3V0 PROX CONN	11
E182	VOLTAGE=1.0V	PP0V95 FIXED SOC	4 7 12
E183	VOLTAGE=1.0V	PP0V95 FIXED SOC PCIE	7
E184	VOLTAGE=1.2V	PP1V2 PLL	2
E185	VOLTAGE=1.0V	PP BUCK5 LX1	12
E186	VOLTAGE=1.0V	PP VAR SOC	5 12
E187	VOLTAGE=3.00V	PP PN65 STM PMU	53 55
E188	VOLTAGE=1.8V	PP1V8 HALL CONN	
E189	VOLTAGE=1.8V	PP1V8 MESA GND	21
E190	VOLTAGE=5.0V	CHARGER LDO	14
E191	VOLTAGE=5.0V	PMID_CAP	14

N56 SPECIFIC VOLTAGE PROPERTIES

E192	VOLTAGE=22.0V	PP WLED14 LX	15
E193	VOLTAGE=22.0V	PP LCM BL34 ANODE	15 20
E194	VOLTAGE=22.0V	PP LCM BL34 ANODE CONN	20 25
E195	VOLTAGE=0.2V	PP LCM BL CAT3	15 20
E196	VOLTAGE=0.2V	PP LCM BL CAT4	15 20
E197	VOLTAGE=0.2V	PP LCM BL CAT3 CONN	20 25
E198	VOLTAGE=0.2V	PP LCM BL CAT4 CONN	20 25
E199	VOLTAGE=-12V	PN SAGE TO TOUCH VCPH FILT	24
E200	VOLTAGE=2.0V	PP RCAM AF	12 23
E201	VOLTAGE=2.0V	PP RCAM AF CONN	23
E202	VOLTAGE=-14.0V	PN SAGE VBST OUTL	24
E203	VOLTAGE=-5.7V	PN5V7 SAGE AVDDN FILT	
E204	VOLTAGE=2.0V	PP BUCK6 LX	12



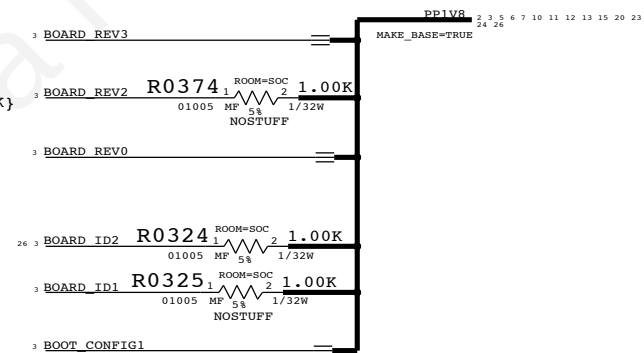
N56 SPECIFIC

BOOTSTRAPPING (BOARD_REV, BOARD_ID, BOOT_CFG, DISPLAY ID)

```
BOARD_REV[3:0]={GPIO34, GPIO35, GPIO36, GPIO37}  
FLOAT=LOW, PULLUP=HIGH  
1111  PROTO1  
1110  PROTO1, ALTERNATE  
1100  PROTO2  
1011  EVT  
1001  CARRIER BUILD <--- SELECTED
```

```
BOARD_ID[4:0]={GPIO29, GPIO16, SPI00_MISO, SPI0_MOSI, SPI0_SCLK}  
FLOAT=LOW, PULLUP=HIGH  
00100  N56, T133 MLB <--- SELECTED  
00101  N56 DEV  
00110  FIJI N61 MLB
```

```
BOOT_CONFIG[2:0]={GPIO28, GPIO25, GPIO18}  
FLOAT=LOW, PULLUP=HIGH  
000  SPI0  
001  SPI0 TEST MODE  
010  NAND <--- SELECTED  
011  NAND TEST MODE  
100  NVME  
101  NVME TEST MODE  
111  FAST SPI
```





87654321																			
D																		D	
C																		C	
B																		B	
A																		A	

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87654321																
D																D
C																C
B																B
A																A

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RADIO_MLB HIERARCHICAL SYMBOL

POWER

32	30	26	23	17	16	15	14	12	10	PP VCC MAIN	MAKE_BASE=TRUE	I42	PP VCC MAIN	10	12	14	15	16	17	23	26	30	32
												I44	PP VCC MAIN WLAN										

CELLULAR HOUSE KEEPING

										I44	AP TO RADIO ON L	MAKE_BASE=TRUE	I44	RADIO ON L	31	33
3										I45	BB TO AP RESET DET L	MAKE_BASE=TRUE	I45	BB RESET DET L	31	36
13										I47	PMU TO BB RST L	MAKE_BASE=TRUE	I47	RF PMIC RESET L	31	33
3											AP TO BB RST L	MAKE_BASE=TRUE		BB RST L	31	33
										I190	AP TO BB WAKE MODEM	MAKE_BASE=TRUE	I50	AP WAKE MODEM	36	
3										I51	BB TO PMU HOST WAKE L	MAKE_BASE=TRUE	I51	BB WAKE HOST L	31	36
13										I52	BB TO AP IPC GPIO	MAKE_BASE=TRUE	I52	BB IPC GPIO	31	36
16										I53	BB TO LEDDRV GSM BLANK	MAKE_BASE=TRUE	I53	GSM TXBURST IND	36	
3											BB TO AP IPC GPIO1	MAKE_BASE=TRUE		BB IPC GPIO1	36	

HSIC IPC

										I55	50 AP BI BB HSIC1 DATA	MAKE_BASE=TRUE	I54	50 BB HSIC1 DATA	31	35
2										I56	50 AP BI BB HSIC1 STB	MAKE_BASE=TRUE	I56	50 BB HSIC1 STROBE	31	35
2										I57	AP TO BB HOST RDY	MAKE_BASE=TRUE	I57	BB HOST RDY	31	36
3										I58	BB TO AP DEVICE RDY	MAKE_BASE=TRUE	I58	BB DEVICE RDY	31	36
3											BB TO AP GPS SYNC	MAKE_BASE=TRUE		BB GPS SYNC	31	36

UART IPC

										I60	AP TO BB UART2 RTS L	MAKE_BASE=TRUE	I59	BB UART CTS L	31	36
3										I62	BB TO AP UART2 CTS L	MAKE_BASE=TRUE	I62	BB UART RTS L	31	36
17										I63	AP TO BB UART2 TXD	MAKE_BASE=TRUE	I63	BB UART RXD	31	36
17											BB TO AP UART2 RXD	MAKE_BASE=TRUE		BB UART TXD	31	36

AUDIO I2S

										I64	45 AP TO BB I2S3_BCLK	MAKE_BASE=TRUE	I64	BB I2S CLK	36	
3										I65	AP TO BB I2S3_DOUT	MAKE_BASE=TRUE	I65	BB I2S RXD	31	36
3										I66	BB TO AP I2S3 DIN	MAKE_BASE=TRUE	I66	BB I2S TXD	31	36
3											AP TO BB I2S3_LRCLK	MAKE_BASE=TRUE		BB I2S WS	31	36

OSCAR UART

										I67	OSCAR TO BB UART TXD	MAKE_BASE=TRUE	I68	BB OTHER RXD	31	36
22											BB TO OSCAR UART RXD	MAKE_BASE=TRUE		BB OTHER TXD	31	36

BB DEBUG INTERFACES

										I70	AP TO BB COREDUMP	MAKE_BASE=TRUE	I72	BB CORE DUMP	31	36
13										I74	PMU TO BB VBUS DET	MAKE_BASE=TRUE	I74	BB USB VBUS	31	35
17										I73	90 TRISTAR BI BB USB N	MAKE_BASE=TRUE	I73	90 BB USB N	31	35
17											90 TRISTAR BI BB USB P	MAKE_BASE=TRUE		90 BB USB P	31	35

RADIO ANTENNA CONTROL

										I75	PP BB VDD 2V7	MAKE_BASE=TRUE	I75	PP LDO14 RF5W	32	42	43	51
											BB GPIO0	MAKE_BASE=TRUE		BB LAT GPIO0	36			
										I74	BB GPIO2	MAKE_BASE=TRUE	I74	BB LAT GPIO2	36			
										I80	BB GPIO3	MAKE_BASE=TRUE	I80	BB LAT GPIO3	36			
											BB GPIO4	MAKE_BASE=TRUE		BB LAT GPIO4	36			

FCT TESTING

										I83	RADIO TO PMU ADC SMPS1	MAKE_BASE=TRUE	I82	ADC SMPS1	31	
13										I84	RADIO TO PMU ADC PP LDO11 VDDIO	MAKE_BASE=TRUE	I84	ADC PP LDO11	31	
13										I85	RADIO TO PMU ADC PP LDO5 SIM	MAKE_BASE=TRUE	I85	ADC PP LDO5	31	
13											RADIO TO PMU ADC SMPS4	MAKE_BASE=TRUE		ADC SMPS4	31	

UPPER RADIO ANTENNA CONTROL

											PP3V0 TRISTAR	MAKE_BASE=TRUE		PAC VDD 3V0	54	
--	--	--	--	--	--	--	--	--	--	--	---------------	----------------	--	-------------	----	--

POWER

										I2	PP WL BT VDDIO AP	MAKE_BASE=TRUE	I2	PP WL BT VDDIO AP	52	
										I1	PP STOCKHOLM 1V8 S2R	MAKE_BASE=TRUE	I1	PP STOCKHOLM 1V8 S2R	53	55
											BFEF VIO S2R	MAKE_BASE=TRUE		BFEF VIO S2R	54	

WLAN/BT HOUSE KEEPING

										I3	45 PMU TO WLAN CLK32K	MAKE_BASE=TRUE	I5	CLK32K AP	31	52
13										I6	PMU TO WLAN REG ON	MAKE_BASE=TRUE	I6	WLAN REG ON	31	52
13										I7	WLAN TO PMU HOST WAKE	MAKE_BASE=TRUE	I7	HOST WAKE WLAN	31	52
13										I8	PMU TO BT REG ON	MAKE_BASE=TRUE	I8	BT REG ON	31	52
3										I9	AP TO BT WAKE	MAKE_BASE=TRUE	I9	WAKE BT	31	52
13											BT TO PMU HOST WAKE	MAKE_BASE=TRUE		HOST WAKE BT	52	

										I9	AP TO WLAN JTAG SWCLK	MAKE_BASE=TRUE	I11	WLAN JTAG SWCLK	31	52
3										I10	AP TO WLAN JTAG SWDIO	MAKE_BASE=TRUE	I10	WLAN JTAG SWDIO	31	52
13										I14	WLAN TO PMU PCIE WAKE L	MAKE_BASE=TRUE	I14	WLAN PCIE WAKE L	31	52
3										I13	AP TO WLAN DEVICE WAKE	MAKE_BASE=TRUE	I13	PCIE DEV WAKE	31	52
7										I12	90 WLAN TO AP PCIE1 RXDP P	MAKE_BASE=TRUE	I12	90 WLAN PCIE TDP	31	52
7										I16	90 WLAN TO AP PCIE1 RXDP N	MAKE_BASE=TRUE	I16	90 WLAN PCIE TDN	31	52
7										I15	90 AP TO WLAN PCIE1 TXDP P	MAKE_BASE=TRUE	I15	90 WLAN PCIE RDP	31	52
7										I17	90 AP TO WLAN PCIE1 TXDP N	MAKE_BASE=TRUE	I17	90 WLAN PCIE RDN	31	52
7										I19	90 AP TO WLAN PCIE1 REFCLK1 P	MAKE_BASE=TRUE	I19	90 WLAN PCIE REFCLK P	52	
7										I18	90 AP TO WLAN PCIE1 REFCLK1 N	MAKE_BASE=TRUE	I18	90 WLAN PCIE REFCLK N	52	
7										I20	WLAN TO AP PCIE1 CLKREQ L	MAKE_BASE=TRUE	I20	WLAN PCIE CLKREQ L	31	52
7											AP TO WLAN PCIE1 RST L	MAKE_BASE=TRUE		WLAN PCIE PERST L	31	52

WLAN HSIC IPC

										I21	WLAN TO AP UART4 RXD	MAKE_BASE=TRUE	I21	WLAN UART TXD	31	52
3										I22	AP TO WLAN UART4 TXD	MAKE_BASE=TRUE	I22	WLAN UART RXD	31	52
3										I23	WLAN TO AP UART4 CTS L	MAKE_BASE=TRUE	I23	WLAN UART RTS L	31	52
3											AP TO WLAN UART4 RTS L	MAKE_BASE=TRUE		WLAN UART CTS L	31	52

BT UART IPC

										I26	AP TO BT UART1 RTS L	MAKE_BASE=TRUE	I25	BT UART CTS L	52	
3										I27	BT TO AP UART1 CTS L	MAKE_BASE=TRUE	I27	BT UART RTS L	52	
3										I28	AP TO BT UART1 TXD	MAKE_BASE=TRUE	I28	BT UART RXD	31	52
3											BT TO AP UART1 RXD	MAKE_BASE=TRUE		BT UART TXD	31	52

BT AUDIO PCM

										I29	45 AP TO BT I2S1_BCLK	MAKE_BASE=TRUE	I30	BT PCM CLK	52	
3										I31	AP TO BT I2S1_DOUT	MAKE_BASE=TRUE	I31	BT PCM IN	52	
3										I32	BT TO AP I2S1 DIN	MAKE_BASE=TRUE	I32	BT PCM OUT	52	
3											AP TO BT I2S1_LRCLK	MAKE_BASE=TRUE		BT PCM SYNC	52	

OSCAR STATES

										I34	OSCAR TO RADIO_CONTEXT_A	MAKE_BASE=TRUE	I33	OSCAR_CONTEXT_A	52	
22											OSCAR TO RADIO_CONTEXT_B	MAKE_BASE=TRUE		OSCAR_CONTEXT_B	52	

STOCKHOLM

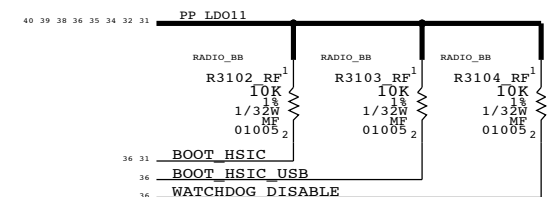
										I35	STOCKHOLM TO AP UART3_CTS_L	MAKE_BASE=TRUE	I36	STOCKHOLM_RTS_L	31	53
3										I37	AP TO STOCKHOLM_UART3_RTS_L	MAKE_BASE=TRUE	I37	STOCKHOLM_CTS_L	31	53
3										I38	STOCKHOLM_TO_AP_UART3_RXD	MAKE_BASE=TRUE	I38	STOCKHOLM_UART_TXD	31	53
3										I39	AP_TO_STOCKHOLM_UART3_TXD	MAKE_BASE=TRUE	I39	STOCKHOLM_UART_RXD	31	53
3										I41	AP TO STOCKHOLM_DWLD_REQ	MAKE_BASE=TRUE	I41	STOCKHOLM_FW_DWLD_REQ	53	
13										I40	STOCKHOLM TO PMU HOST WAKE	MAKE_BASE=TRUE	I40	STOCKHOLM_HOST_WAKE	31	53
3											AP TO STOCKHOLM_EN	MAKE_BASE=TRUE		STOCKHOLM_ENABLE	53	
30	26	17	15	12							PP3V0 TRISTAR	MAKE_BASE=TRUE		STOCKHOLM_VDD_MUX_3V0	55	
3											AP TO STOCKHOLM_SIM_SEL	MAKE_BASE=TRUE		STOCKHOLM_SIM_SEL	55	
25											AP TO STOCKHOLM_ANT	MAKE_BASE=TRUE		STOCKHOLM_ANT	53	

AP INTERFACE & DEBUG CONNECTORS

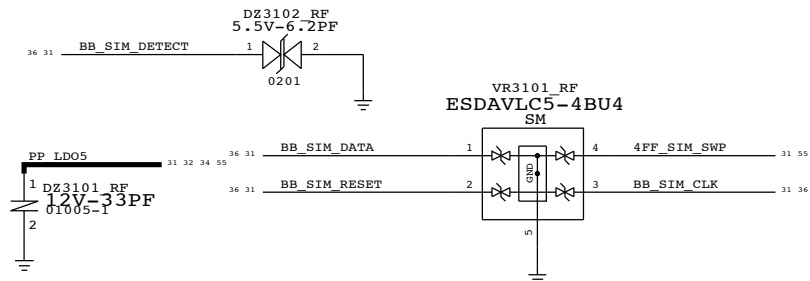
PROBE POINTS

PP3105 RF P2MM-NSM SM 1 CLK32K_AP 30 52	PP3121 RF P2MM-NSM SM 1 STOCKHOLM_HOST_WAKE 30 53	PP3115 RF P4MM-NSM SM 1 50_BB_HSIC_STROBE 30 35	PP3130 RF P4MM-NSM SM 1 BB_JTAG_RST_L 30 35	PP3141 RF P4MM-NSM SM 1 BB_UART_TXD 30 36	PP3170 RF P4MM-NSM SM 1 RFFE1_CLK 36 40 41 42 43 44 45
PP3113 RF P4MM-NSM SM 1 BB_COEX_UART_RXD 36 52	PP3122 RF P4MM-NSM SM 1 BB_REQUEST_XO_CLK 33 53	PP3116 RF P4MM-NSM SM 1 50_BB_HSIC_DATA 30 35	PP3131 RF P4MM-NSM SM 1 BB_JTAG_TCK 30 35	PP3142 RF P4MM-NSM SM 1 BB_UART_RXD 30 36	PP3171 RF P4MM-NSM SM 1 RFFE1_DATA 36 40 41 42 43 44 45
PP3114 RF P4MM-NSM SM 1 BB_COEX_UART_TXD 36 52	PP3123 RF P2MM-NSM SM 1 STOCKHOLM_UART_RXD 30 53	PP3101 RF P4MM-NSM SM 1 BB_DEBUG_ERROR 36	PP3132 RF P4MM-NSM SM 1 BB_JTAG_TMS 30 35	PP3143 RF P4MM-NSM SM 1 BB_UART_RTS_L 30 36	PP3172 RF P4MM-NSM SM 1 RFFE2_CLK 36 46 47 49
PP3119 RF P2MM-NSM SM 1 BT_UART_TXD 30 52	PP3124 RF P2MM-NSM SM 1 STOCKHOLM_UART_TXD 30 53	PP3102 RF P4MM-NSM SM 1 RF_PMIC_RESET_L 30 33	PP3133 RF P4MM-NSM SM 1 BB_JTAG_TDO 30 35	PP3144 RF P4MM-NSM SM 1 BB_UART_CTS_L 30 36	PP3173 RF P4MM-NSM SM 1 RFFE2_DATA 36 46 47 49
PP3120 RF P2MM-NSM SM 1 BT_UART_RXD 30 52	PP3125 RF P2MM-NSM SM 1 STOCKHOLM_CTS_L 30 53	PP3103 RF P4MM-NSM SM 1 PS_HOLD_PMIC 33	PP3134 RF P4MM-NSM SM 1 BB_JTAG_TDI 30 35	PP3145 RF P4MM-NSM SM 1 BB_HOST_RDY 30 36	PP3175 RF P4MM-NSM SM 1 BB_I2S_WS 30 36
PP3152 RF P2MM-NSM SM 1 WAKE_BT 30 52	PP3126 RF P2MM-NSM SM 1 STOCKHOLM_RTS_L 30 53	PP3127 RF P4MM-NSM SM 1 PMIC_RESOUT_L 33 35	PP3135 RF P4MM-NSM SM 1 BB_JTAG_TRST_L 30 35	PP3146 RF P4MM-NSM SM 1 BB_DEVICE_RDY 30 36	PP3176 RF P4MM-NSM SM 1 BB_I2S_RXD 30 36
PP3153 RF P4MM-NSM SM 1 WLAN_REG_ON 30 52	PP3128 RF P4MM-NSM SM 1 PP_PN65_VCC_SIM 53	PP3104 RF P4MM-NSM SM 1 MDM_CLK 33 35	PP3136 RF P4MM-NSM SM 1 BB_DEBUG_STATUS 36	PP3147 RF P4MM-NSM SM 1 BB_GPS_SYNC 30 36	PP3177 RF P4MM-NSM SM 1 BB_I2S_TXD 30 36
PP3154 RF P4MM-NSM SM 1 BT_REG_ON 30 52	PP3144 RF P4MM-NSM SM 1 STOCKHOLM_SIM_SWP 53 55	PP3109 RF P4MM-NSM SM 1 PP_LDO11 31 32 34 35 36 38 39	PP3137 RF P4MM-NSM SM 1 BB_CORE_DUMP 30 36	PP3148 RF P4MM-NSM SM 1 BB_WAKE_HOST_L 30 36	PP3178 RF P4MM-NSM SM 1 BB_OTHER_TXD 30 36
PP3155 RF P2MM-NSM SM 1 HOST_WAKE_WLAN 30 52	PP3129 RF P4MM-NSM SM 1 REF_CLK_FROM_BB 33 53	PP3110 RF P4MM-NSM SM 1 RADIO_ON_L 30 33	PP3138 RF P4MM-NSM SM 1 BB_USB_VBUS 30 35	PP3149 RF P4MM-NSM SM 1 BB_RESET_DET_L 30 36	PP3179 RF P4MM-NSM SM 1 BB_OTHER_RXD 30 36
PP3156 RF P2MM-NSM SM 1 WLAN_PCIE_WAKE_L 30 52	PP3160 RF P4MM-NSM SM 1 DSDS_SIM_CLK 35 55	PP3111 RF P4MM-NSM SM 1 SPMI_DATA 33 35	PP3139 RF P4MM-NSM SM 1 90_BB_USB_N 30 35	PP3150 RF P4MM-NSM SM 1 BB_RST_L 30 33	PP3182 RF P4MM-NSM SM 1 RFFE2_CLK_BUFFER 36 54
PP3157 RF P2MM-NSM SM 1 WLAN_PCIE_PERST_L 30 52	PP3183 RF P4MM-NSM SM 1 DSDS_SIM_RESET 35 55	PP3112 RF P4MM-NSM SM 1 SPMI_CLK 33 35	PP3140 RF P4MM-NSM SM 1 90_BB_USB_P 30 35	PP3151 RF P4MM-NSM SM 1 BOOT_HSIC 31 36	PP3185 RF P4MM-NSM SM 1 RFFE2_DATA_BUFFER 36 54
PP3158 RF P2MM-NSM SM 1 WLAN_PCIE_CLKREQ_L 30 52	PP3184 RF P4MM-NSM SM 1 DSDS_SIM_DATA 35 55				
PP3159 RF P4MM-NSM SM 1 PCIE_DEV_WAKE 30 52	PP3186 RF P4MM-NSM SM 1 DSDS_SIM_DETECT 35				
PP3160 RF P2MM-NSM SM 1 WLAN_UART_RTS_L 30 52	PP3187 RF P4MM-NSM SM 1 PP_LDO6 32 34 55				
PP3161 RF P2MM-NSM SM 1 WLAN_UART_CTS_L 30 52	PP3188 RF P4MM-NSM SM 1 DSDS_SIM_SWP 55				
PP3162 RF P2MM-NSM SM 1 WLAN_UART_RXD 30 52	PP3189 RF P4MM-NSM SM 1 DSDS_SIM_DATA_R 55				
PP3163 RF P2MM-NSM SM 1 WLAN_UART_TXD 30 52					
PP3190 RF P2MM-NSM SM 1 WLAN_JTAG_SWCLK 30 52	PP 3178 RF P2MM-NSM SM 1 BB_SIM_RESET 31 36				
PP3191 RF P2MM-NSM SM 1 WLAN_JTAG_SWDIO 30 52	PP 3179 RF P2MM-NSM SM 1 BB_SIM_CLK 31 36				
	PP 3180 RF P2MM-NSM SM 1 BB_SIM_DATA 31 36				
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	PP 3184 RF P2MM-NSM SM 1 PP_LDO5 31 32 34 55				

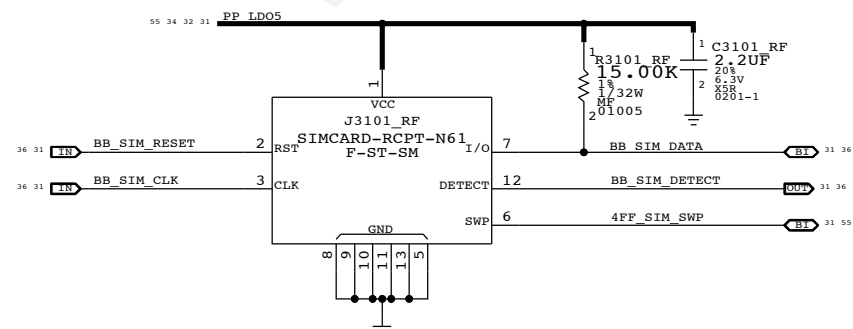
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
197S0565	197S0593	ALTERNATE	Y3301_RF	KDS 19.2MHZ XTAL
197S0598	197S0593	ALTERNATE	Y3301_RF	AVX 19.2MHZ XTAL
138S00005	138S00003	ALTERNATE	C3216_RF	150F CAPACITOR
138S0739	138S0706	ALTERNATE	C4207_RF	1.00UF CAPACITOR
138S0945	138S0706	ALTERNATE	C4207_RF	1.00UF CAPACITOR
138S1103	138S0719	ALTERNATE	C4007_RF	4.7UF CAPACITOR
339S0231	339S0228	ALTERNATE	U5201_RF	CORONA MODULE USI
339S0242	339S0228	ALTERNATE	U5201_RF	CORONA MODULE TDK
155S00024	155S0950	ALTERNATE	F_TRI_RF	TRIPLEXER BIN2



SIM CARD ESD PROTECTION



SIM CARD CONNECTOR



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D



C



A

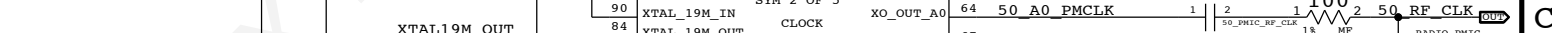
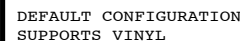
A |



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U404

1.70V	N61/N56	PVT
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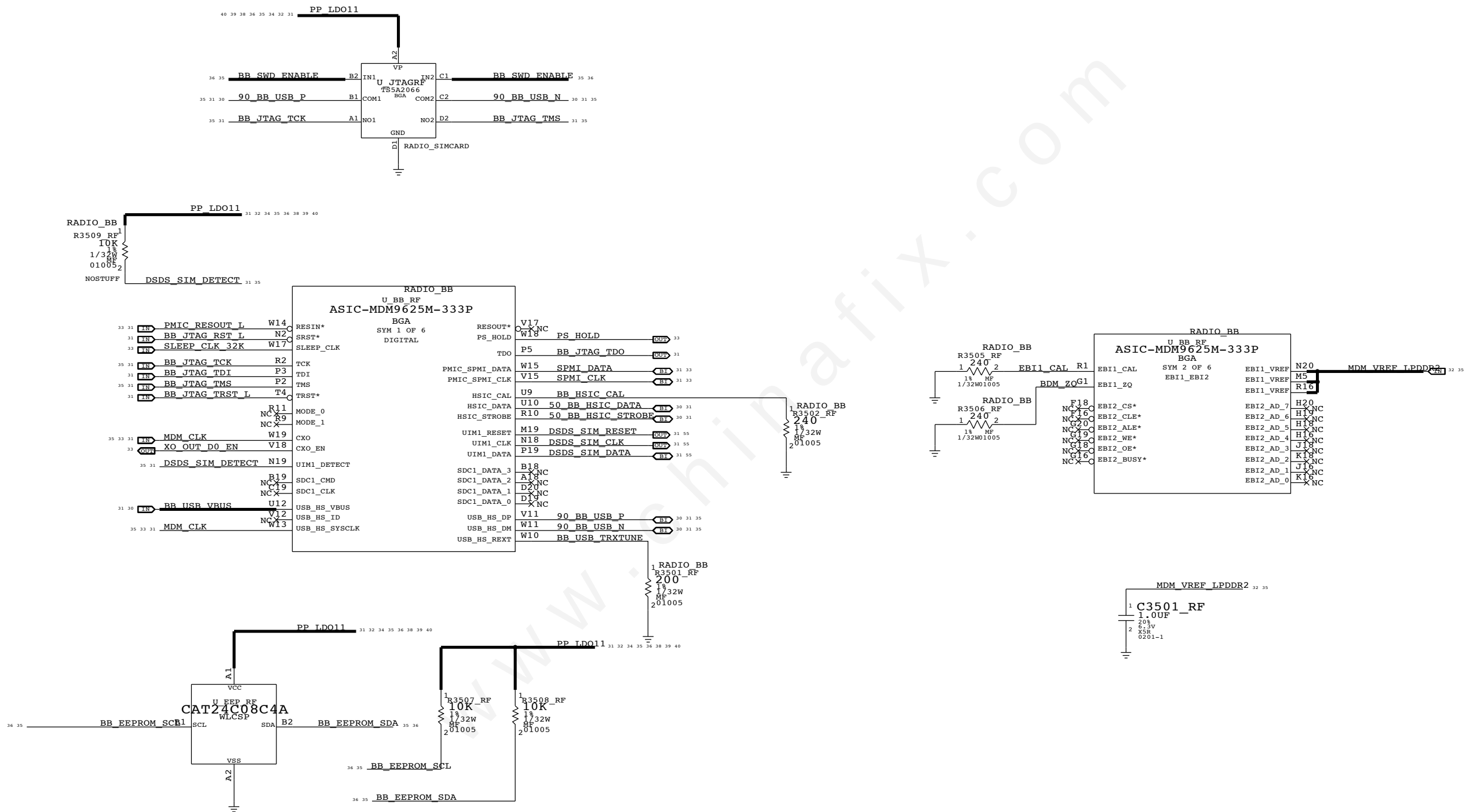
U502





CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

C600
R606
L600
U602

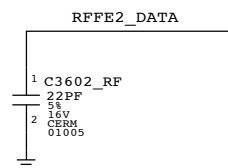
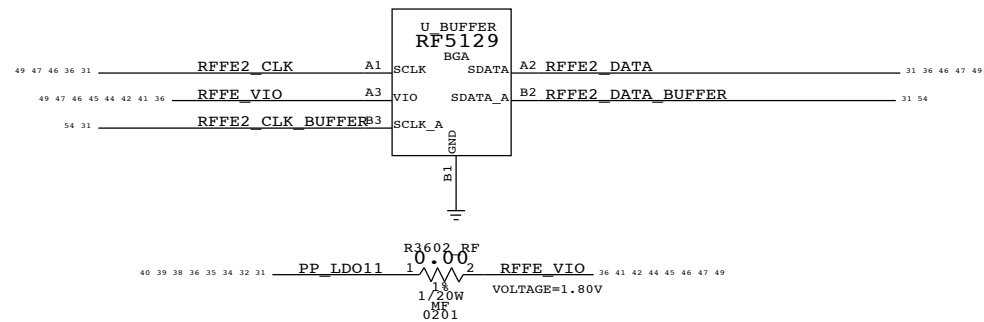
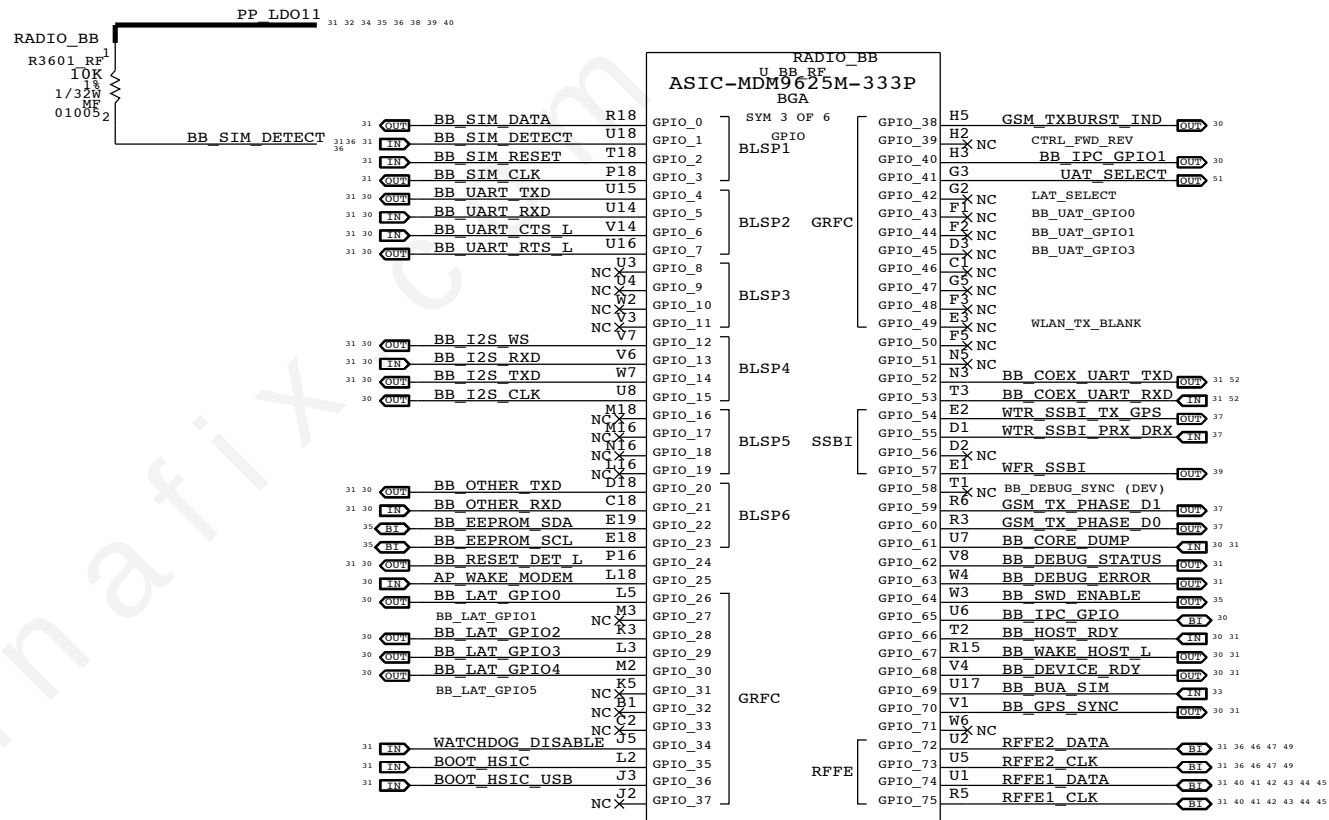
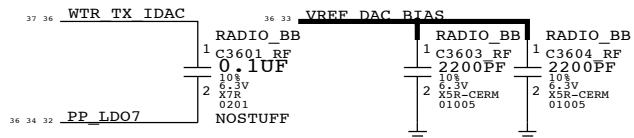
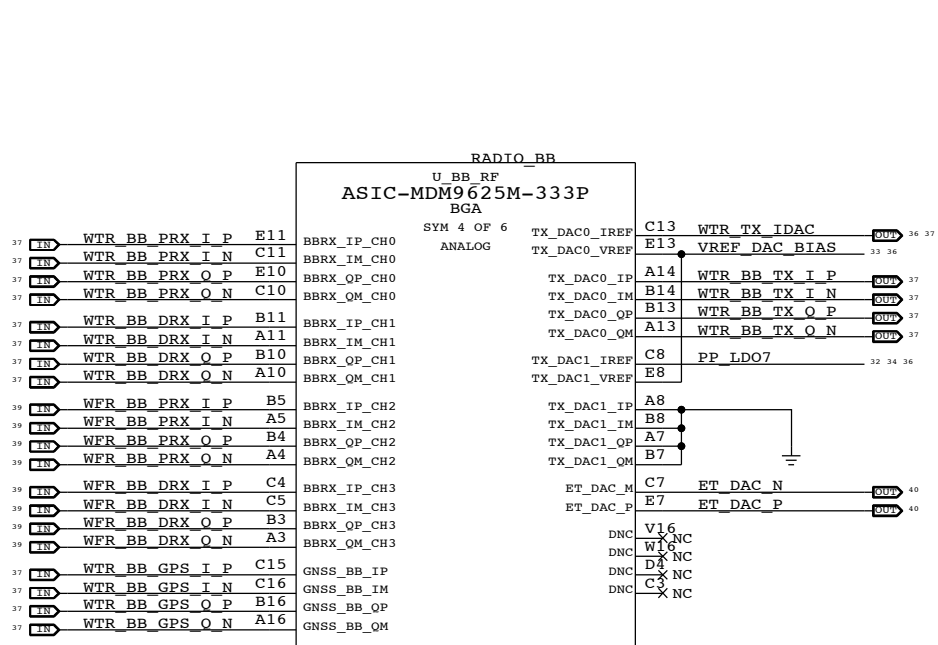


BASEBAND (3 OF 3)

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C704
R700
L700
U702



MOBILE DATA MODEM (2 OF 2)

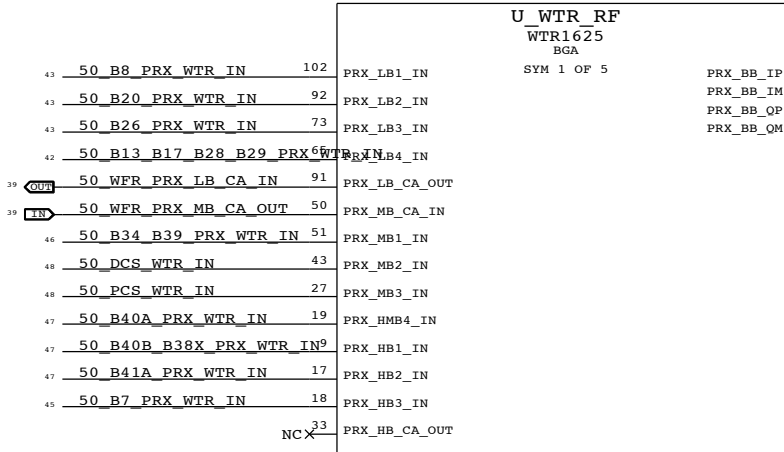
WTR TRANSCEIVER (1 OF 2)

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C802
R802
L800
U803

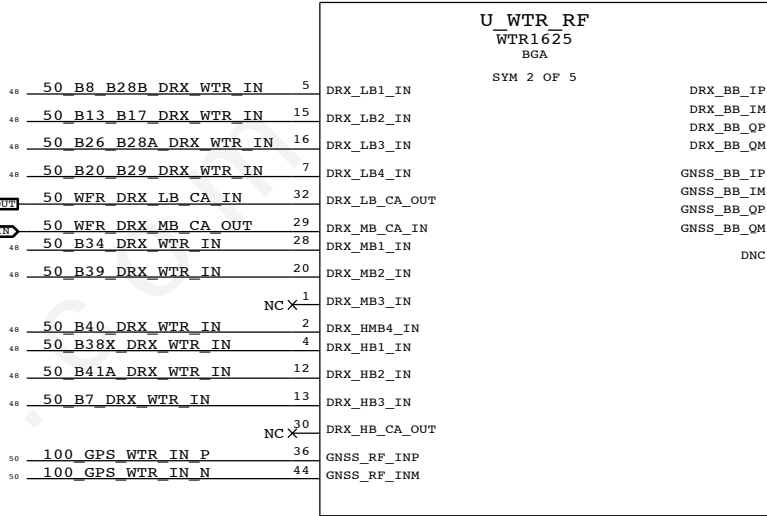


LB1	DC
LB2	DC
LB3	DC
LB4	DC
MB1	NO DC
MB2	DC
MB3	DC
HB1	NO DC
HB2	DC
HB3	DC
HMB4	NO DC



PRX_BB_IP	99	WTR_BB_PRX_I_P	36
PRX_BB_IM	108	WTR_BB_PRX_I_N	36
PRX_BB_QP	107	WTR_BB_PRX_O_P	36
PRX_BB_QM	97	WTR_BB_PRX_O_N	36

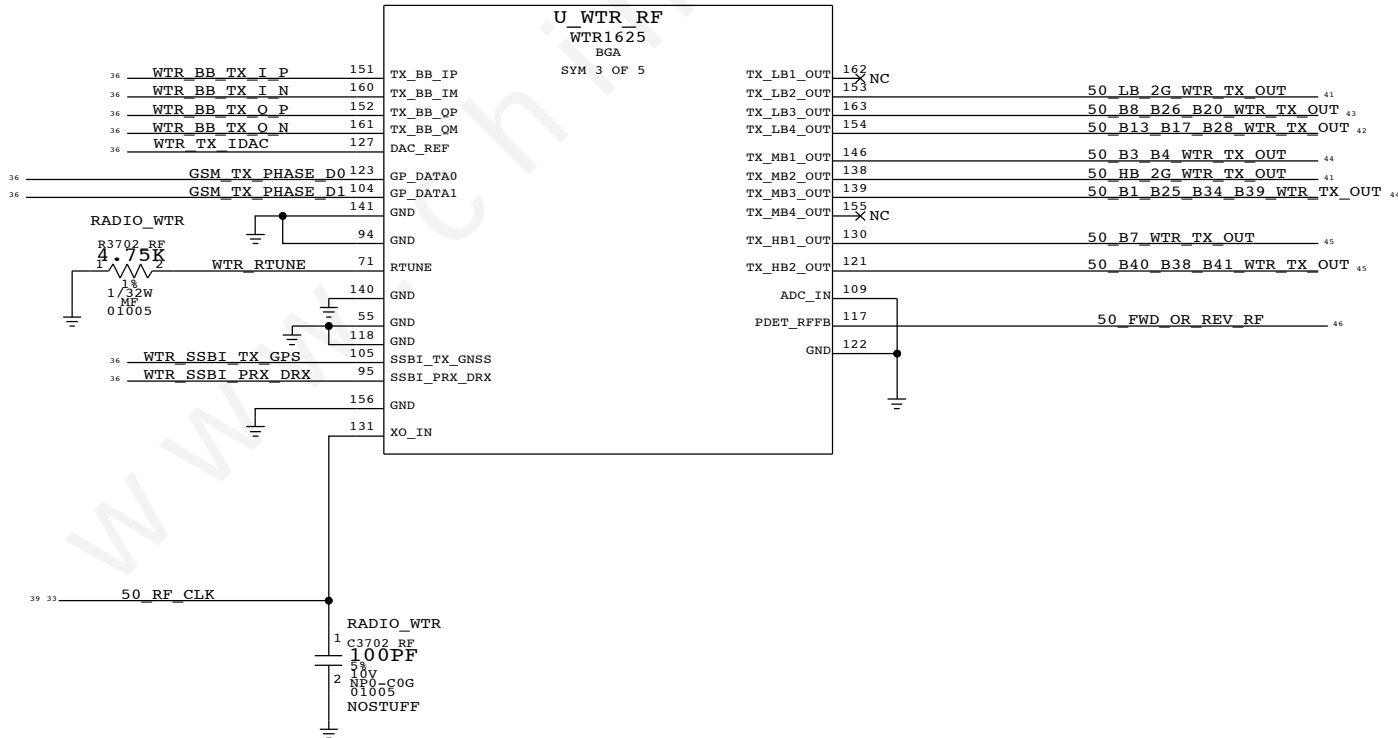
LB1	DC
LB2	DC
LB3	DC
LB4	DC
MB1	NO DC
MB2	DC
MB3	DC
HB1	NO DC
HB2	DC
HB3	DC
HMB4	NO DC



DRX_BB_IP	76	WTR_BB_DRX_I_P	36	RADIO_WTR
DRX_BB_IM	86	WTR_BB_DRX_I_N	36	RADIO_WTR
DRX_BB_QP	61	WTR_BB_DRX_O_P	36	RADIO_WTR
DRX_BB_QM	68	WTR_BB_DRX_O_N	36	RADIO_WTR
GNSS_BB_IP	60	WTR_BB_GPS_I_P	36	RADIO_WTR
GNSS_BB_IM	53	WTR_BB_GPS_I_N	36	RADIO_WTR
GNSS_BB_QP	67	WTR_BB_GPS_O_P	36	RADIO_WTR
GNSS_BB_QM	85	WTR_BB_GPS_O_N	36	RADIO_WTR

DNC

37 X NC



RF_CLK IS SHARED BETWEEN WTR AND WFR. LENGTH DIFFERENCE BETWEEN THE TWO SHOULD BE < 5MM.



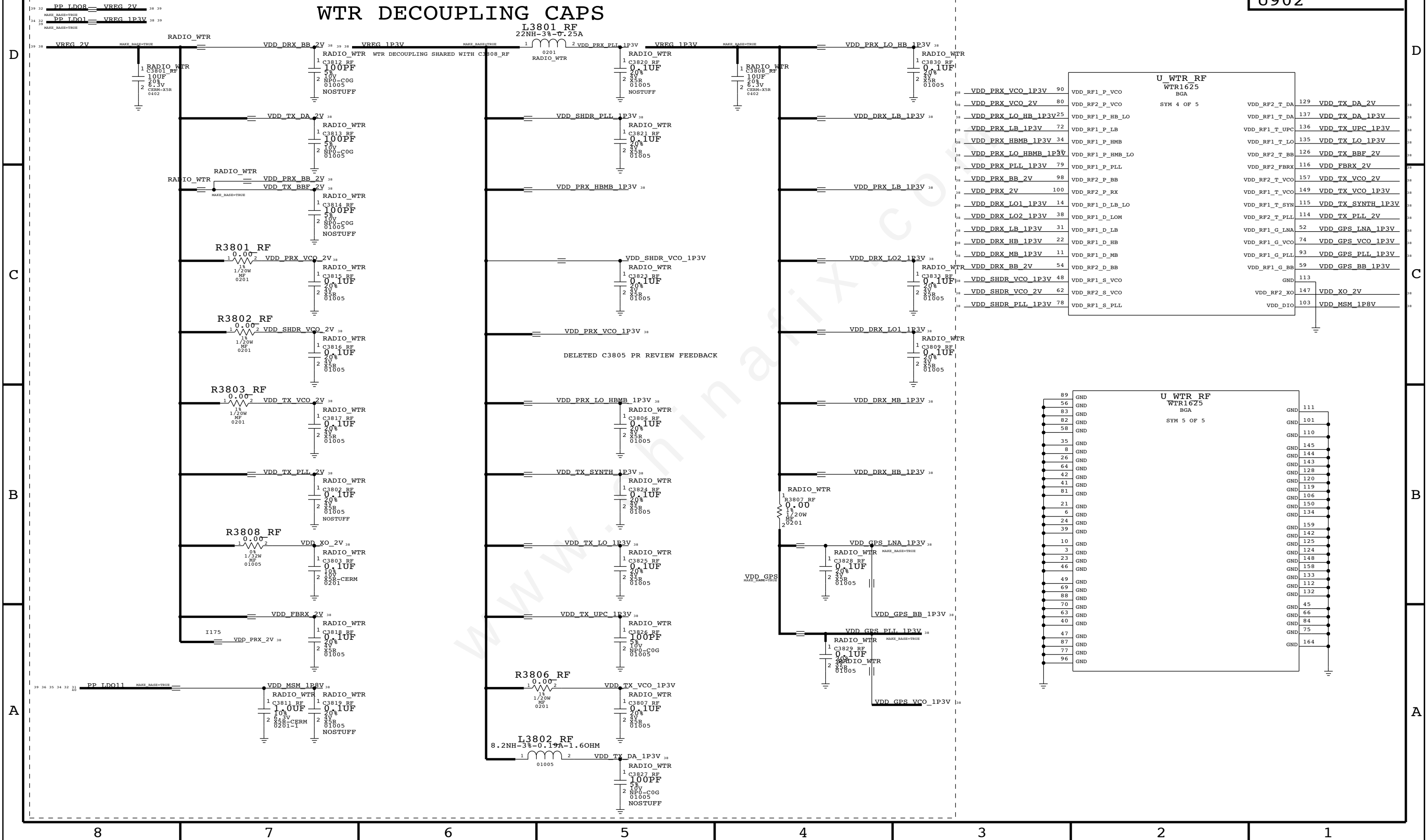
WTR TRANSCEIVER (2 OF 2)

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C934	
R926	
L3802	RF
U902	

WTR DECOUPLING CAPS

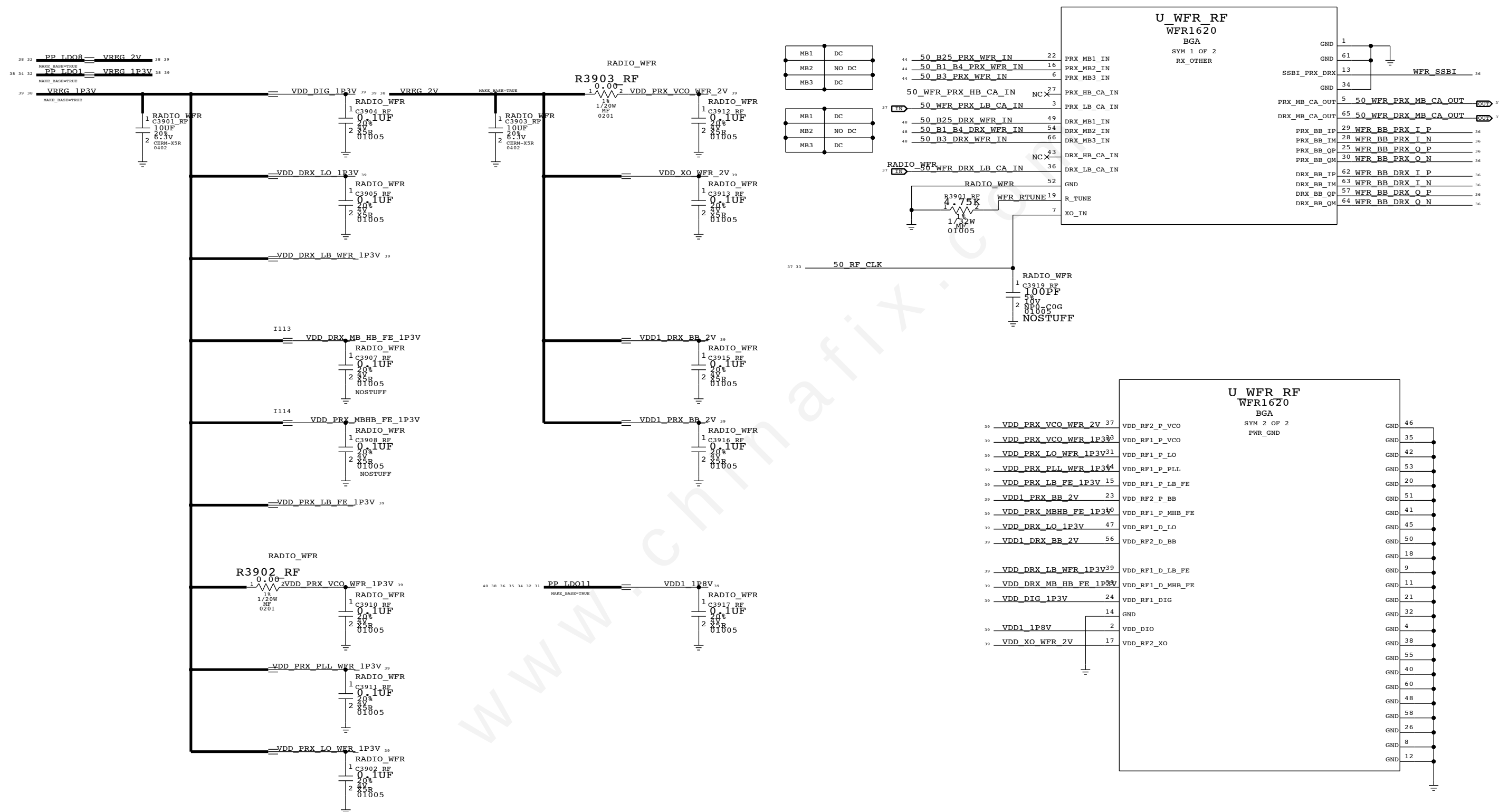
L3801 RF
22NH-3%-0.25A





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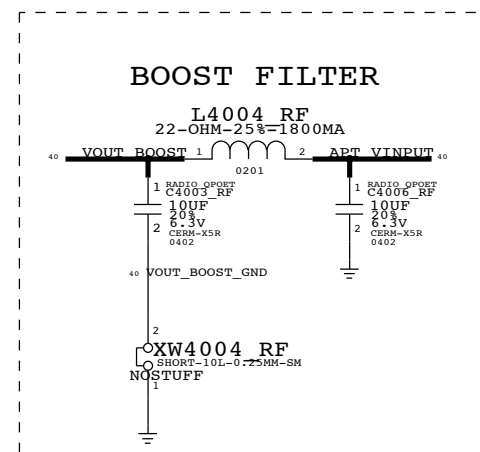
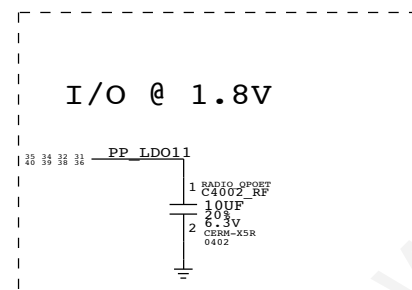
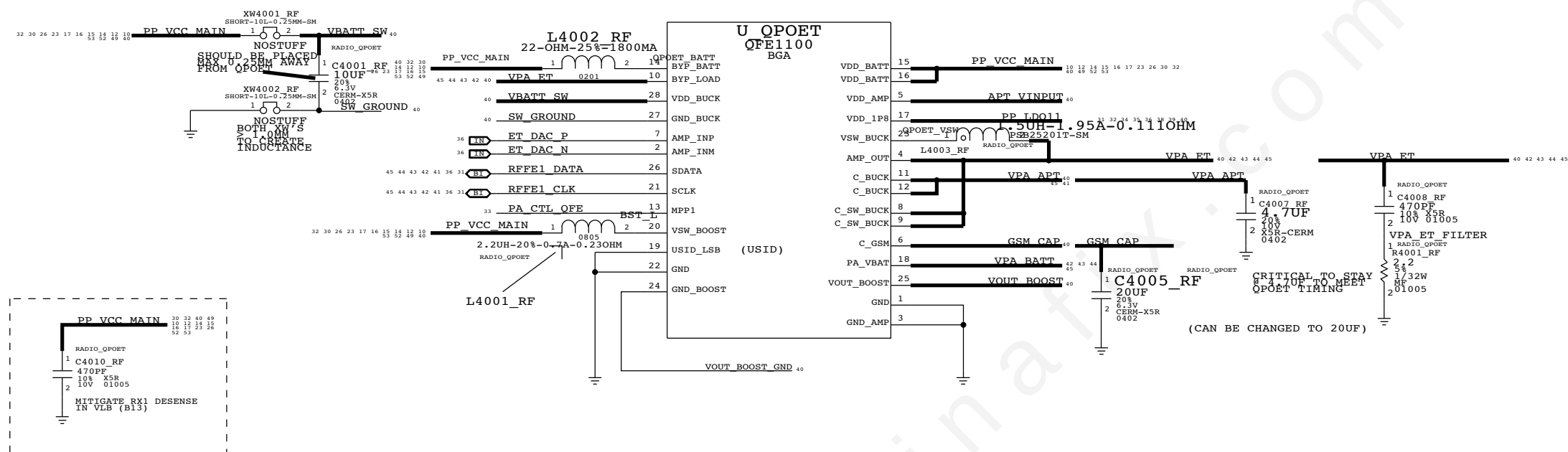
C1019
R1016
L1000
U1002





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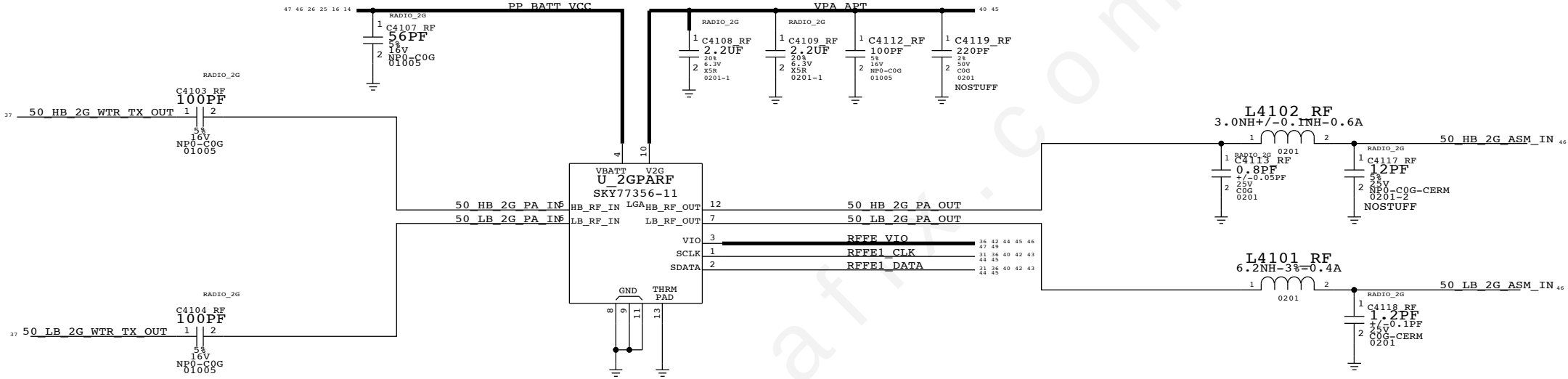
C1110
R1102
L1104
U1101



2G PA

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C1208
R1200
L1204
U1201



VERY LOW BAND PAD (B13, B17, B28)

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C1332
R1300
L4215 RF
U1304



D

C

B

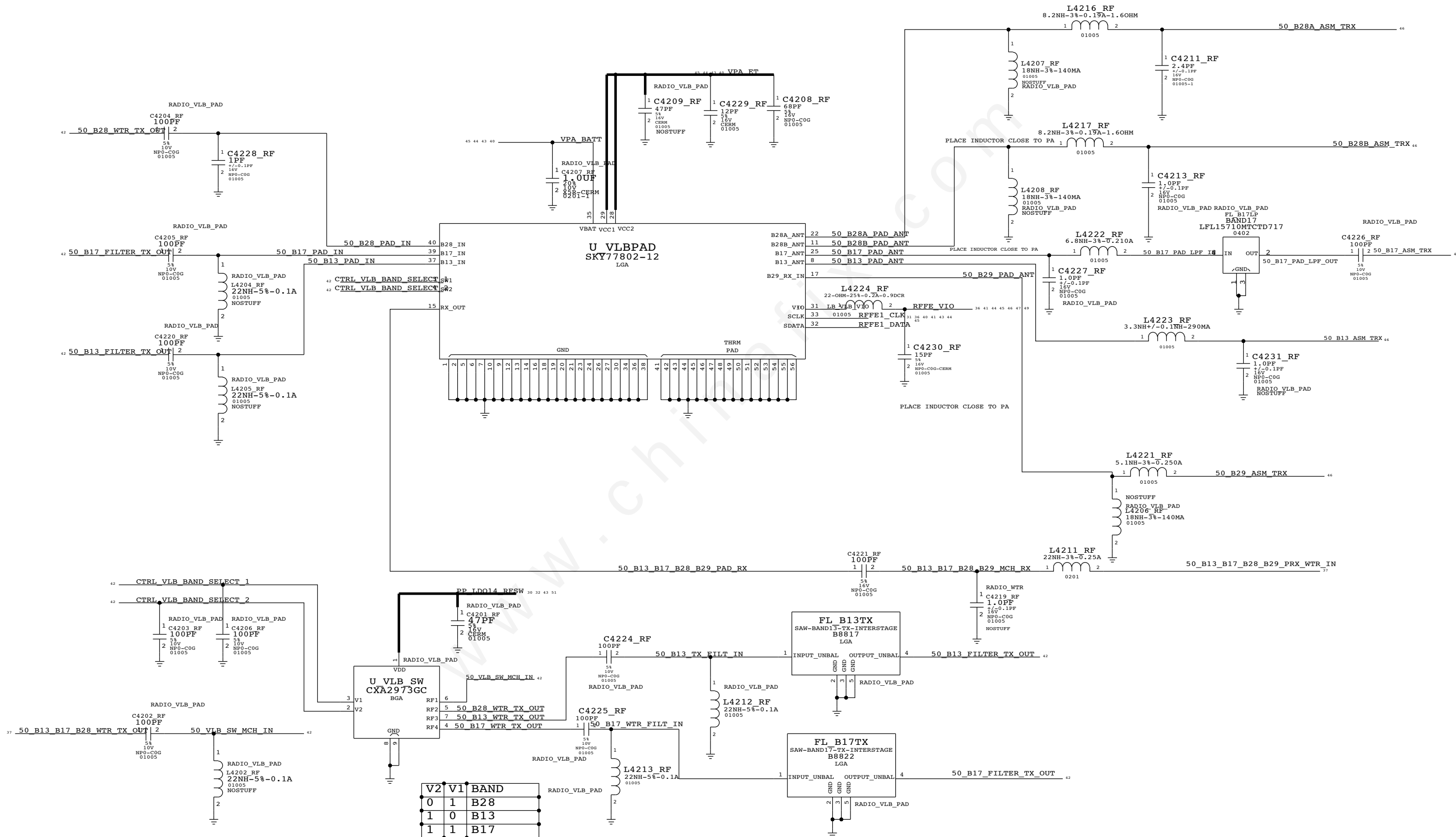
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D

C

B

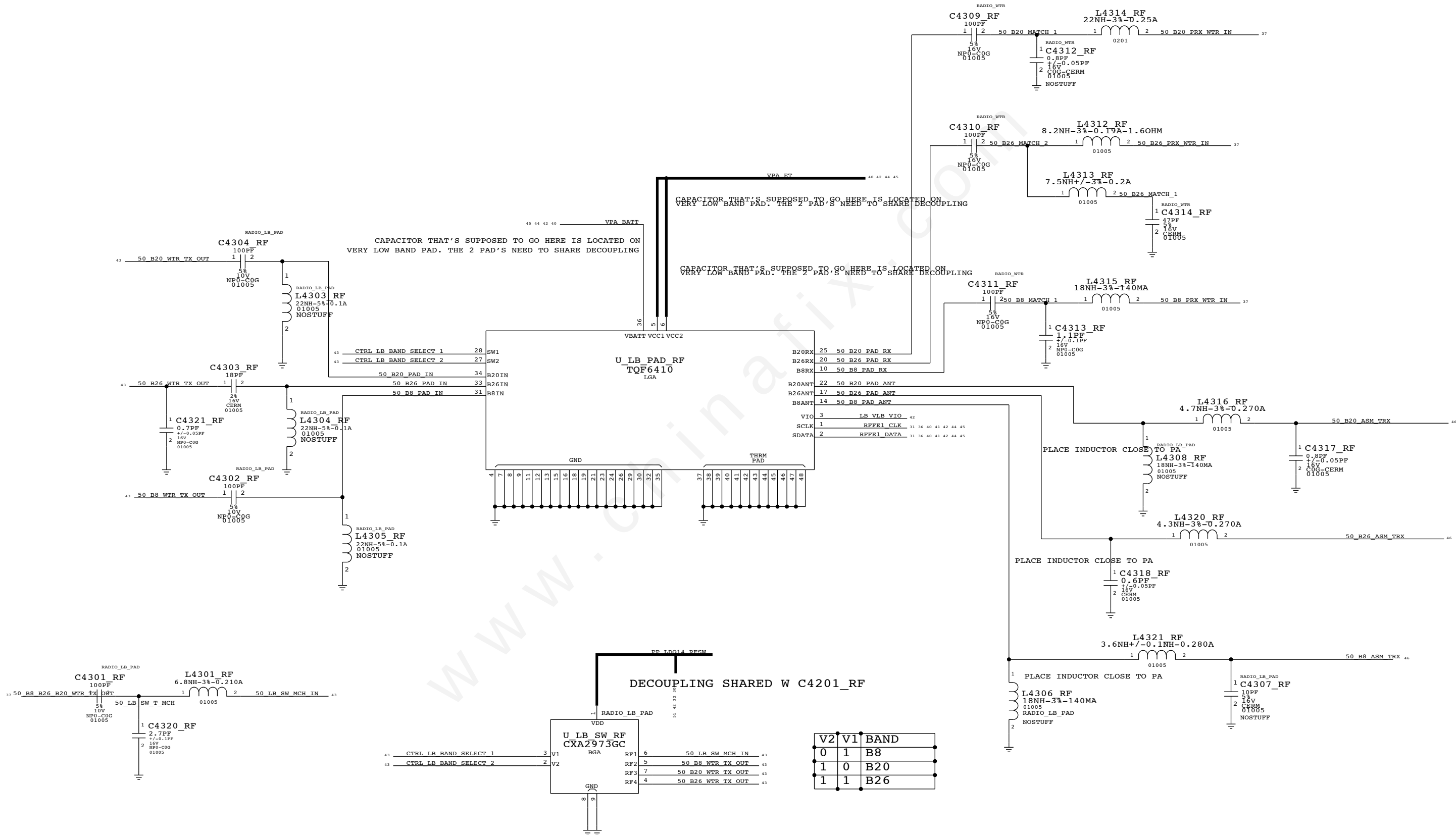
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LOW BAND PAD (B8, B26, B20)

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST

C4318 RF
R1400
L4322 RF
U1402





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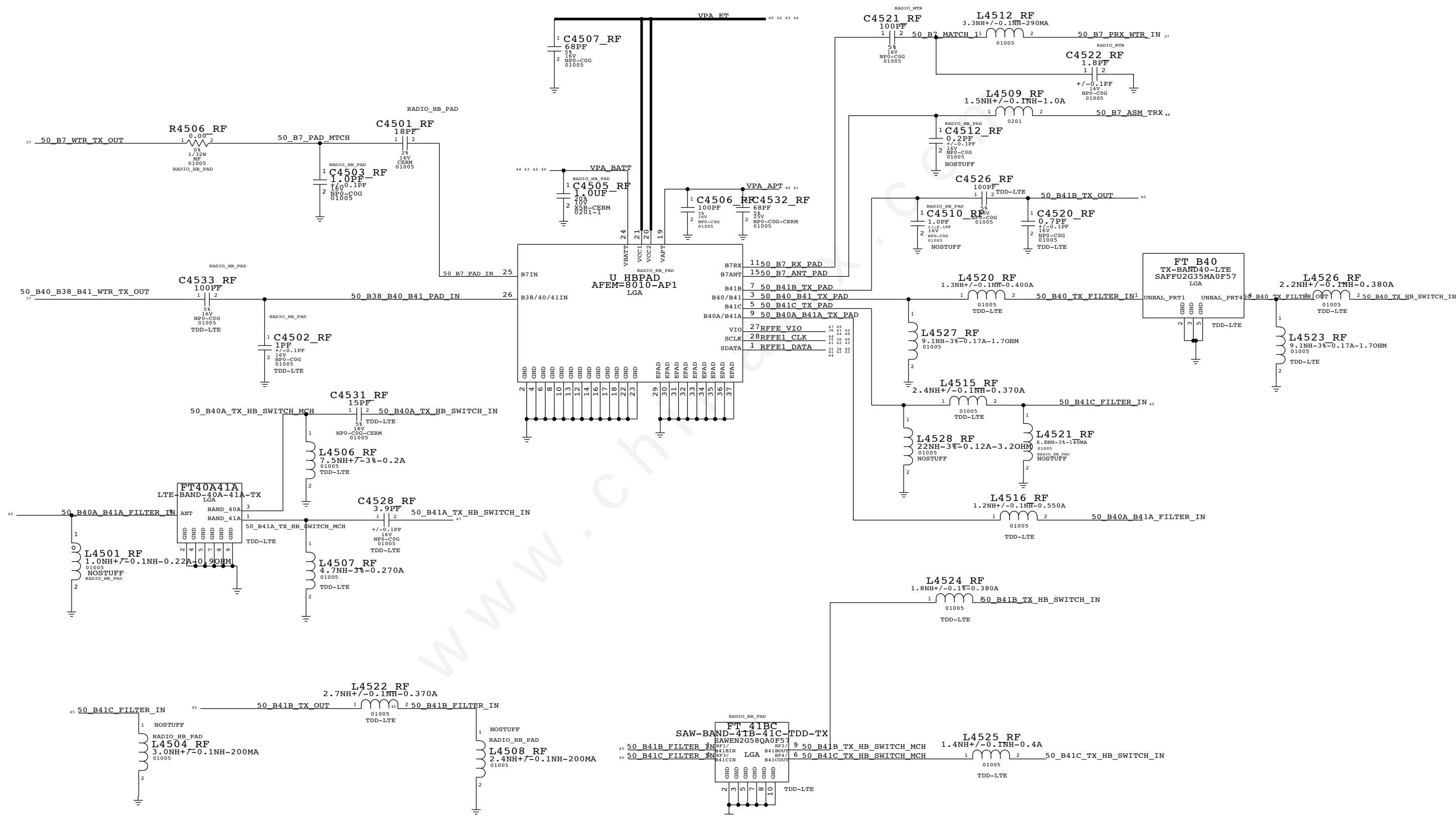
C4426	RF
R1500	
L4409	RF
U1501	





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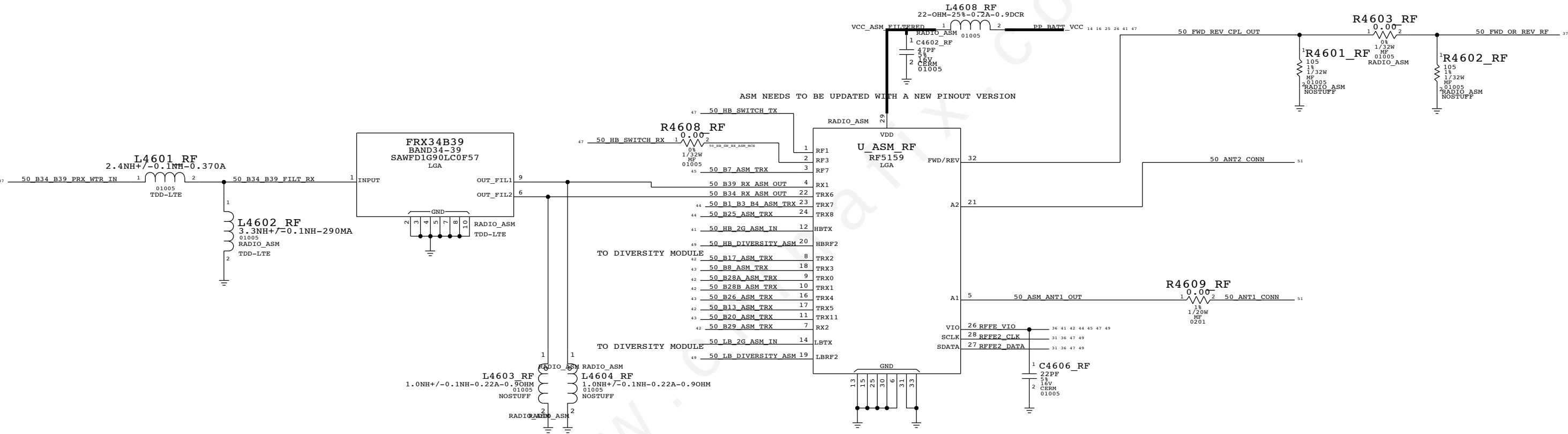
C4533	RF
R1600	
L1616	
U1601	



ANTENNA SWITCH

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

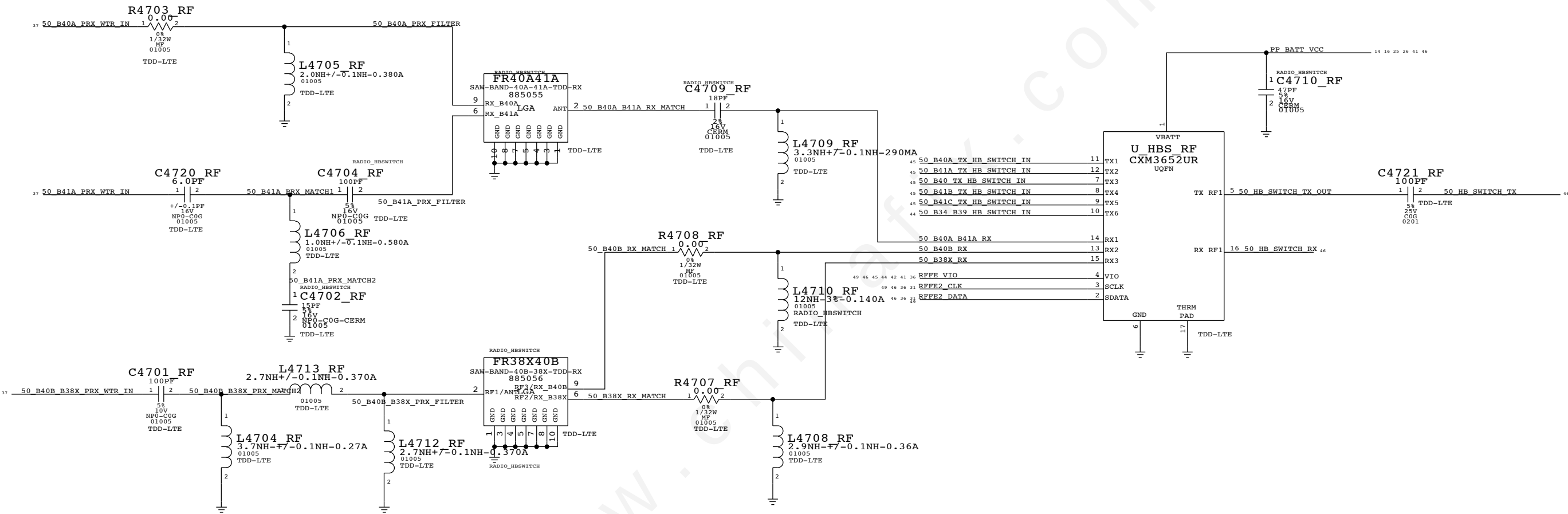
C1702
R1700
L4608 RF
U1702





HIGH BAND SWITCH

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.



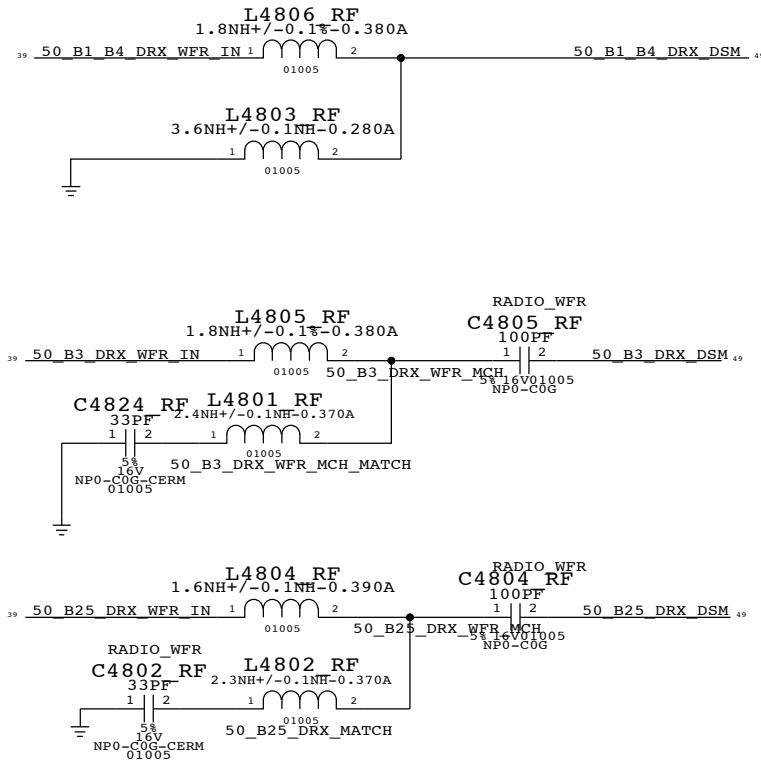
RX DIVERSITY (1)

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSES ONLY - NOT A CHANGE REQUEST.

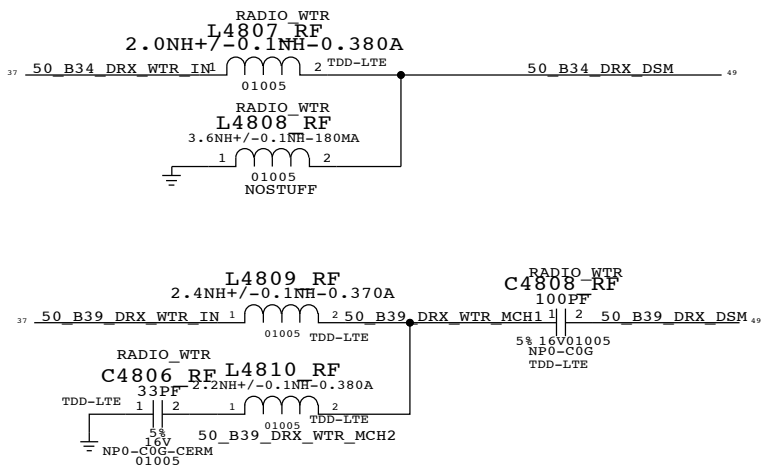
C4826 RF
R1800
L1829
U1801



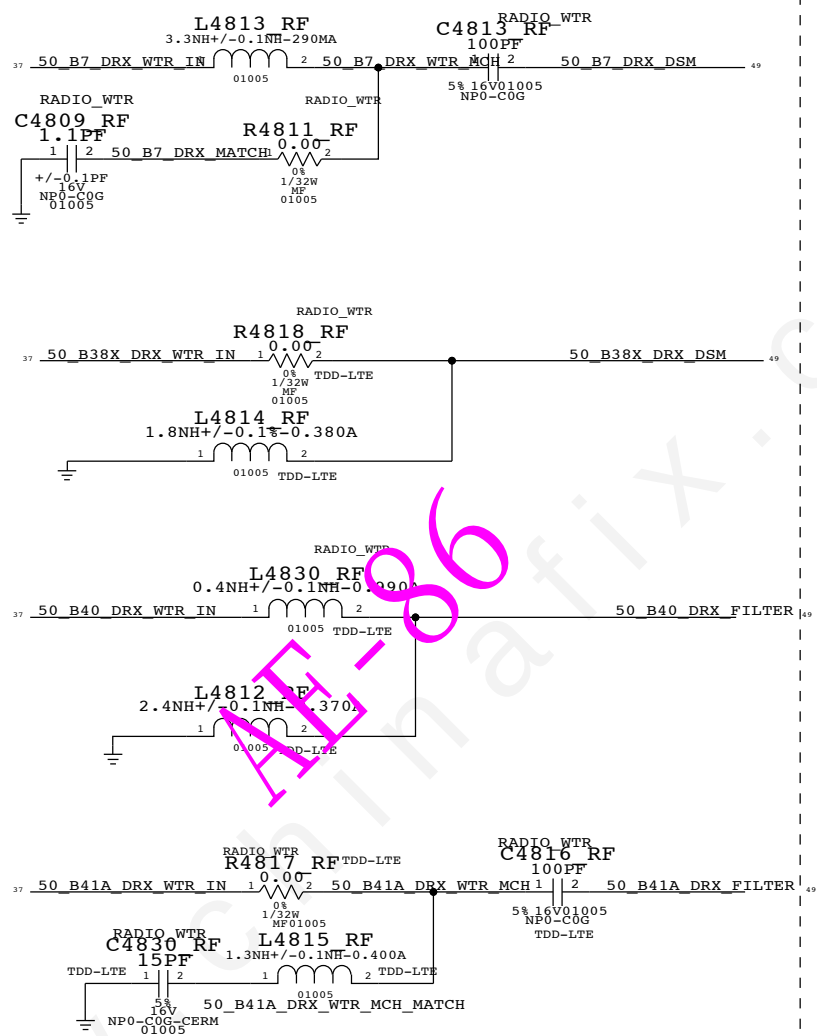
MIDBAND MIDBAND DIVERSITY - WFR



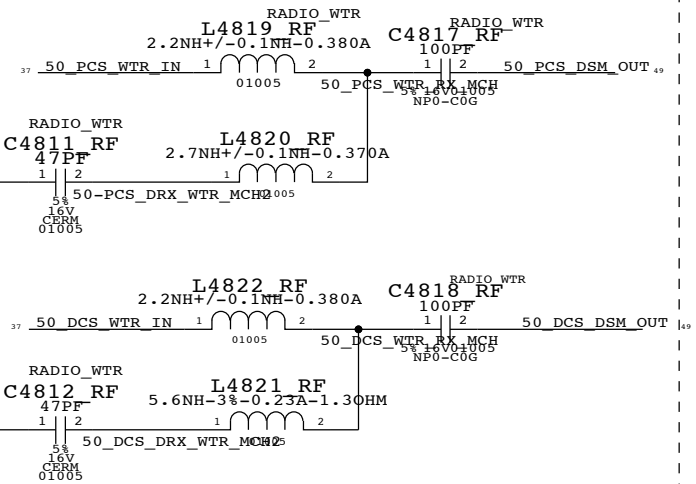
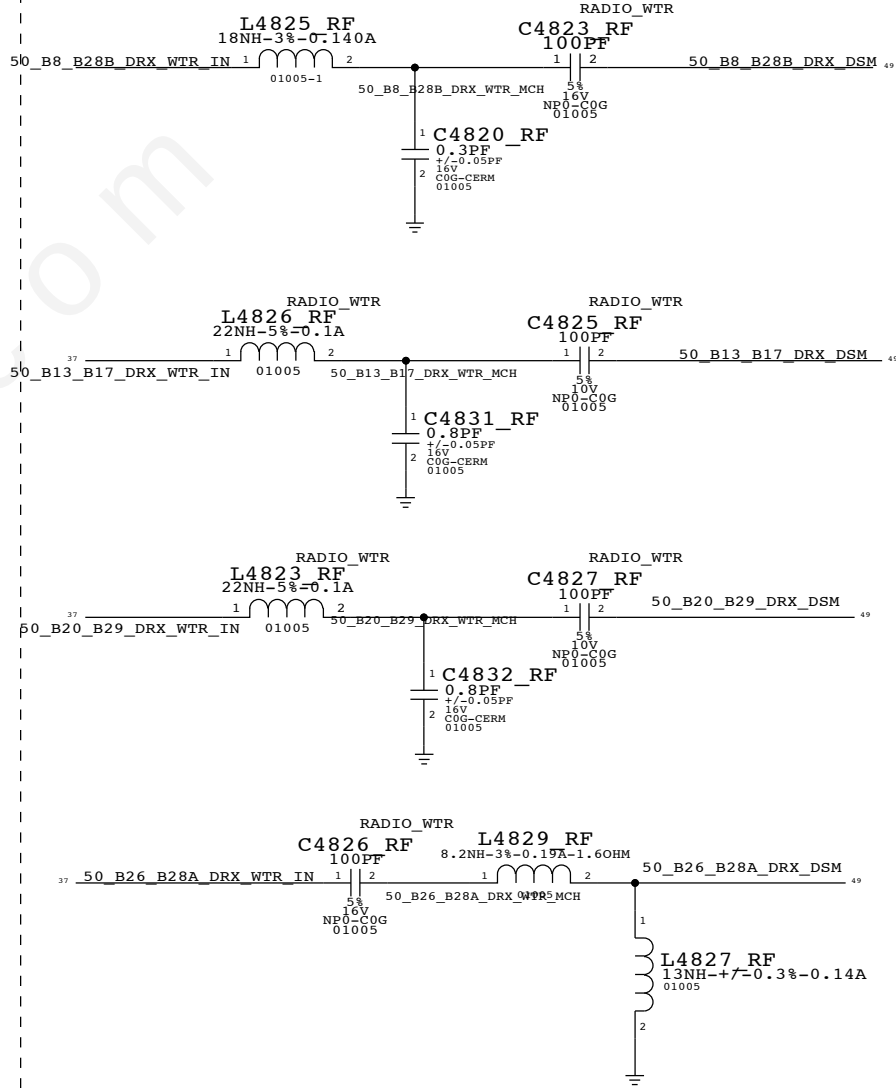
MIDBAND DIVERSITY - WTR



HIGHBAND DIVERSITY - WTR



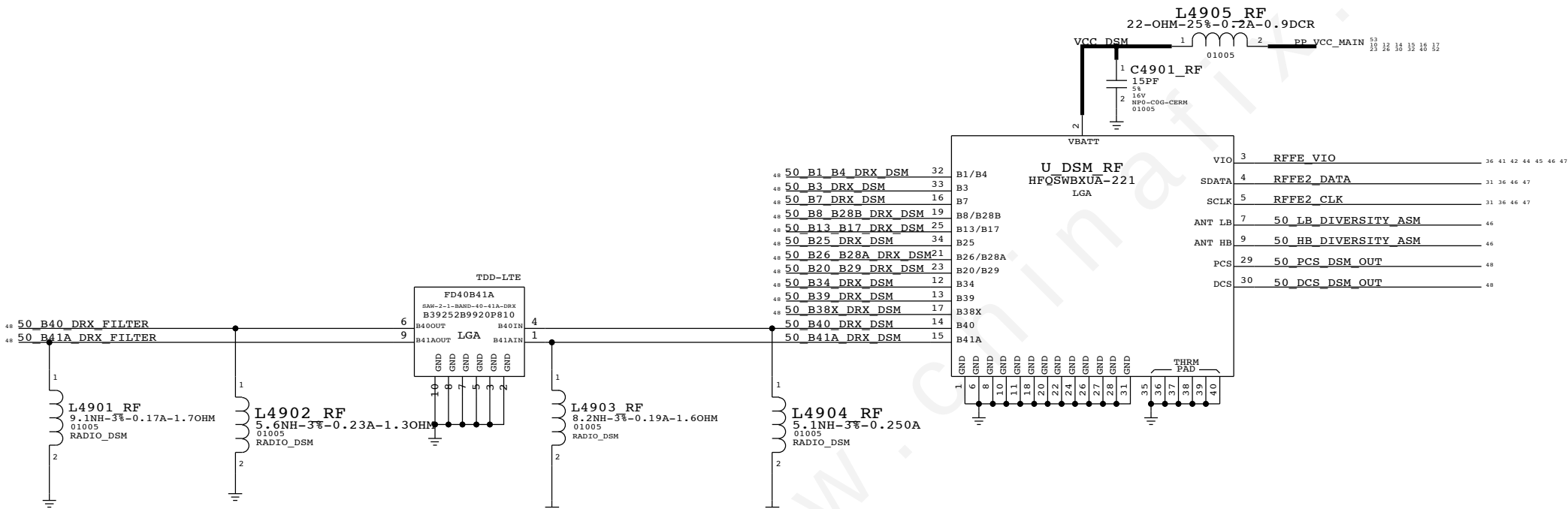
LOWBAND DIVERSITY - WTR

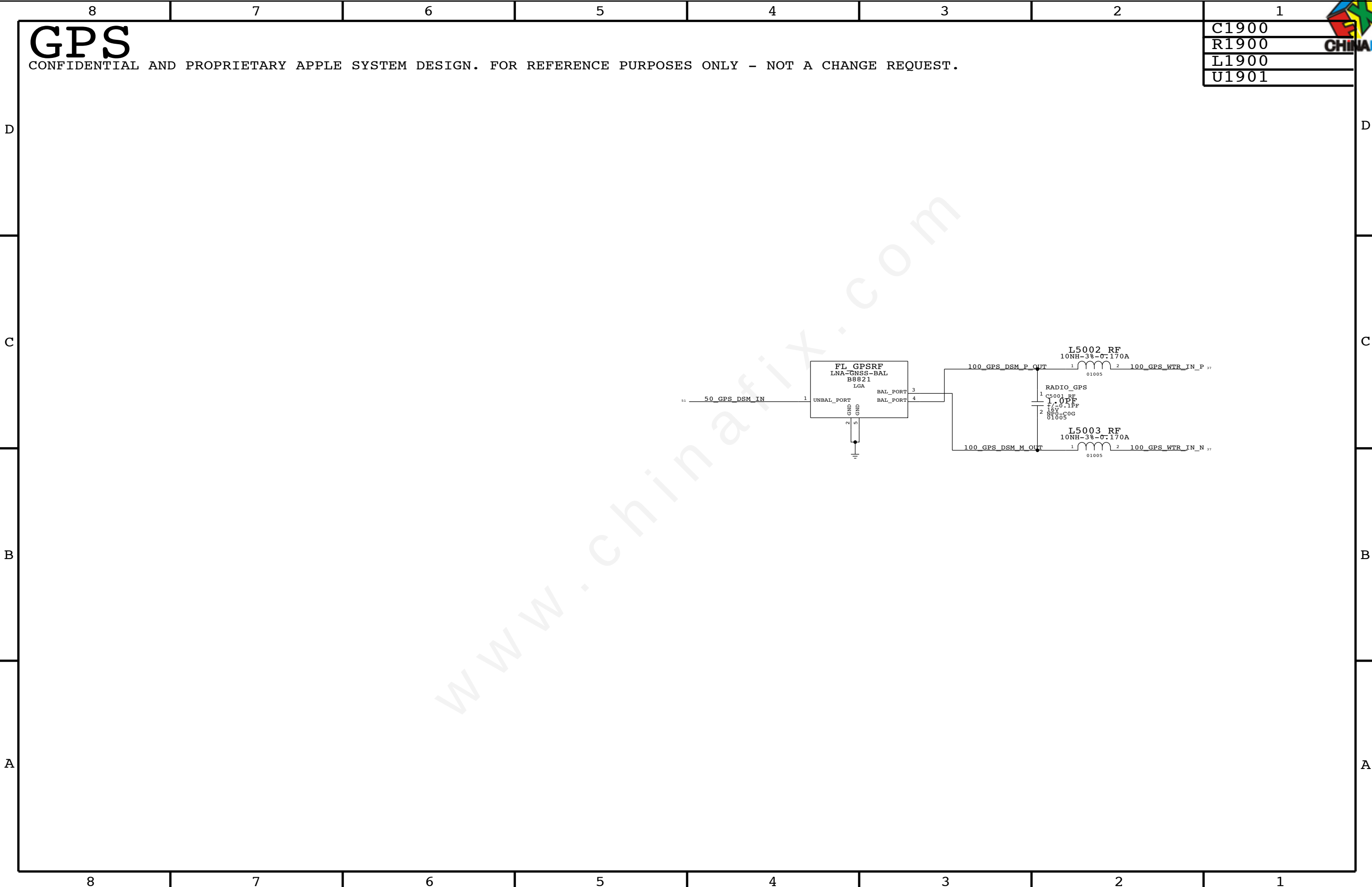


RX DIVERSITY (2)

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C1900
R1900
L1900
U1901



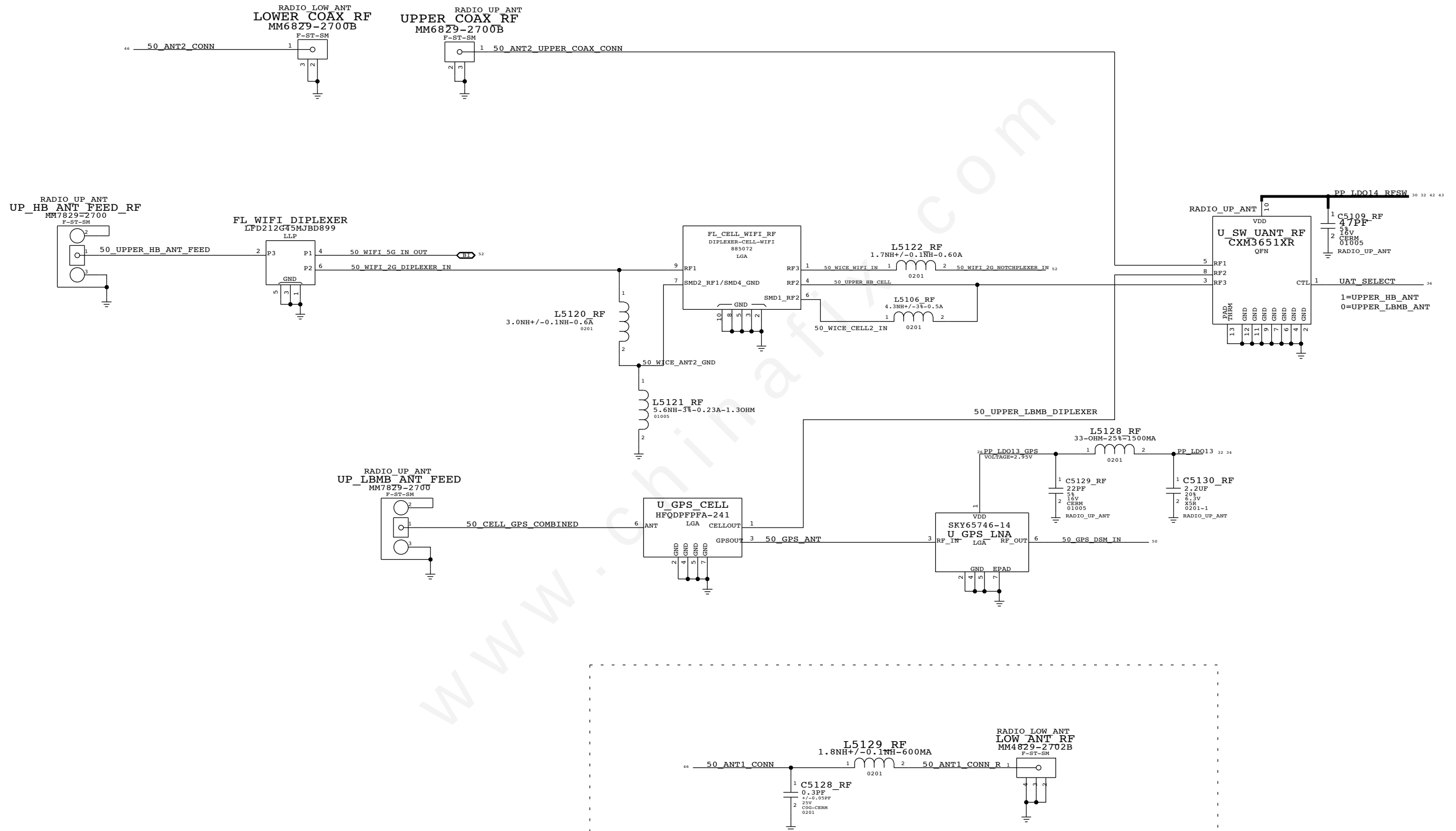




ANTENNA FEED'S

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TEST & COAX CONNECTOR FOR LOWER SECTION OF MLB





The image shows a PCB layout with two antenna connectors. The first connector, labeled 'PP VCC MAIN', is connected to a 'WIFI_BT' pad. It has two pins: Pin 1 is connected to a capacitor (C5202 RF, 10UF, 20V, CERW-X5R, 0402-1) and Pin 2 is connected to ground. The second connector, labeled 'PP WLAN VDDIO_1V8', is connected to a 'WIFI_BT' pad. It has two pins: Pin 1 is connected to a capacitor (C5203 RF, 27PF, 5V, NP0-COG, 01005) and Pin 2 is connected to ground. The layout also shows a 'WIFI_BT' pad connected to a resistor (R5208 RF, 0.00, 0.8, 032W, 01005) and a 'WIFI_BT' pad connected to a capacitor (C5204 RF, 0.01UF, 0V, NP0-COG, 01005). The layout is labeled with '22', '23', '24', '25', '26', '27', '28', '29', '30', '31', '32', '33', '34', '35', '36', '37', '38', '39', '40', '41', '42', '43', '44', '45', '46', '47', '48', '49', '50', '51', '52', '53', '54', '55', '56', '57', '58', '59', '60', '61', '62', '63', '64', '65', '66', '67', '68', '69', '70', '71', '72', '73', '74', '75', '76', '77', '78', '79', '80', '81', '82', '83', '84', '85', '86', '87', '88', '89', '90', '91', '92', '93', '94', '95', '96', '97', '98', '99', '100'.

WIFI_BT		CLK32K_AP		CLK32K_K	
31	30	31	30	31	30
WLAN_BUCK_OUT		26		VIN_LDO	
WLAN_SR_VLX		28		SR_VLX	
WLAN_REG_ON		9		WL_REG_ON	
BT_REG_ON		10		BT_REG_ON	

WIFI_BT		L5201_RF		L5201_RF	
31	30	31	30	31	30
WLAN_SR_LC		0603		WLAN_SR_LC	
HOST_WAKE_WLAN		30		GPIO_0	
WLAN_PCIE_WAKE_L		12		PCIE_WAKE*	
WLAN_PCIE_PERSLT_L		14		PCIE_PRST*	
WLAN_PCIE_CLKREQ_L		13		PCIE_CLKREQ*	
90_WLAN_PCIE_REFCLK_N		16		PCIE_REFCLK_N	
90_WLAN_PCIE_REFCLK_P		17		PCIE_REFCLK_P	
90_WLAN_PCIE_RDN		20		PCIE_RDN	
90_WLAN_PCIE_RDP		21		PCIE_RDP	
90_WLAN_PCIE_TDN		18		PCIE_TDN	
90_WLAN_PCIE_TDP		19		PCIE_TDP	

Pin	Signal	Function
32	JTAG_SEL	11 JTAG_SEL
31	WLAN_JTAG_SWCLK	31 JTAG_TCK(GPIO_2)
30	WLAN_JTAG_SWDIO	34 JTAG_TMS(GPIO_3)
30	OSCAR_CONTEXT_A	32 JTAG_TDI(GPIO_4)
		NC X ³⁵ JTAG_TDO(GPIO_5)
30	OSCAR_CONTEXT_B	33 JTAG_TRST(GPIO_6)

WLAN_PCIE_PERST_L

0

1

2

C5215_RF

100PF

5V

16V

NP0-C0G

01005

	2G_ANT	45	50_WLAN_G_ANT		2	C6
	5G_ANT	58	50_WLAN_A_ANT		02	NC
GPIO_1	8	PCIE_DEV_WAKE				
BT_HOST_WAKE	43	HOST_WAKE_BT			30	
BT_WAKE	42	WAKE_BT			30	
BT_UART_CTS	38	BT_UART_CTS_L			30	
BT_UART_RTS	39	BT_UART_RTS_L			30	
BT_UART_RXD	41	BT_UART_RXD			30	
BT_UART_TXD	40	BT_UART_TXD			30	
BT_PCM_CLK	49	BT_PCM_CLK			30	
BT_PCM_SYNC	50	BT_PCM_SYNC			30	
BT_PCM_IN	48	BT_PCM_IN			30	
BT_PCM_OUT	47	BT_PCM_OUT			30	

UART_RTS(GPIO_7)	56	WLAN_UART_RTS_L	OUT	30 31
UART_CTS(GPIO_8)	4	WLAN_UART_CTS_L	IN	30 31
UART_RX(GPIO_9)	3	WLAN_UART_RXD	IN	30 31
UART_TX(GPIO_10)	2	WLAN_UART_TXD	OUT	30 31

```
SECI_TX(GPIO_13)
SECI_RX(GPIO_14)
RF_SW_CTRL 8
```

PP WLAN VDDIO 1V8

WIFI BT
RS201_RF
10K
51
32W
8F
NOSTUFF

JTAG_SEL

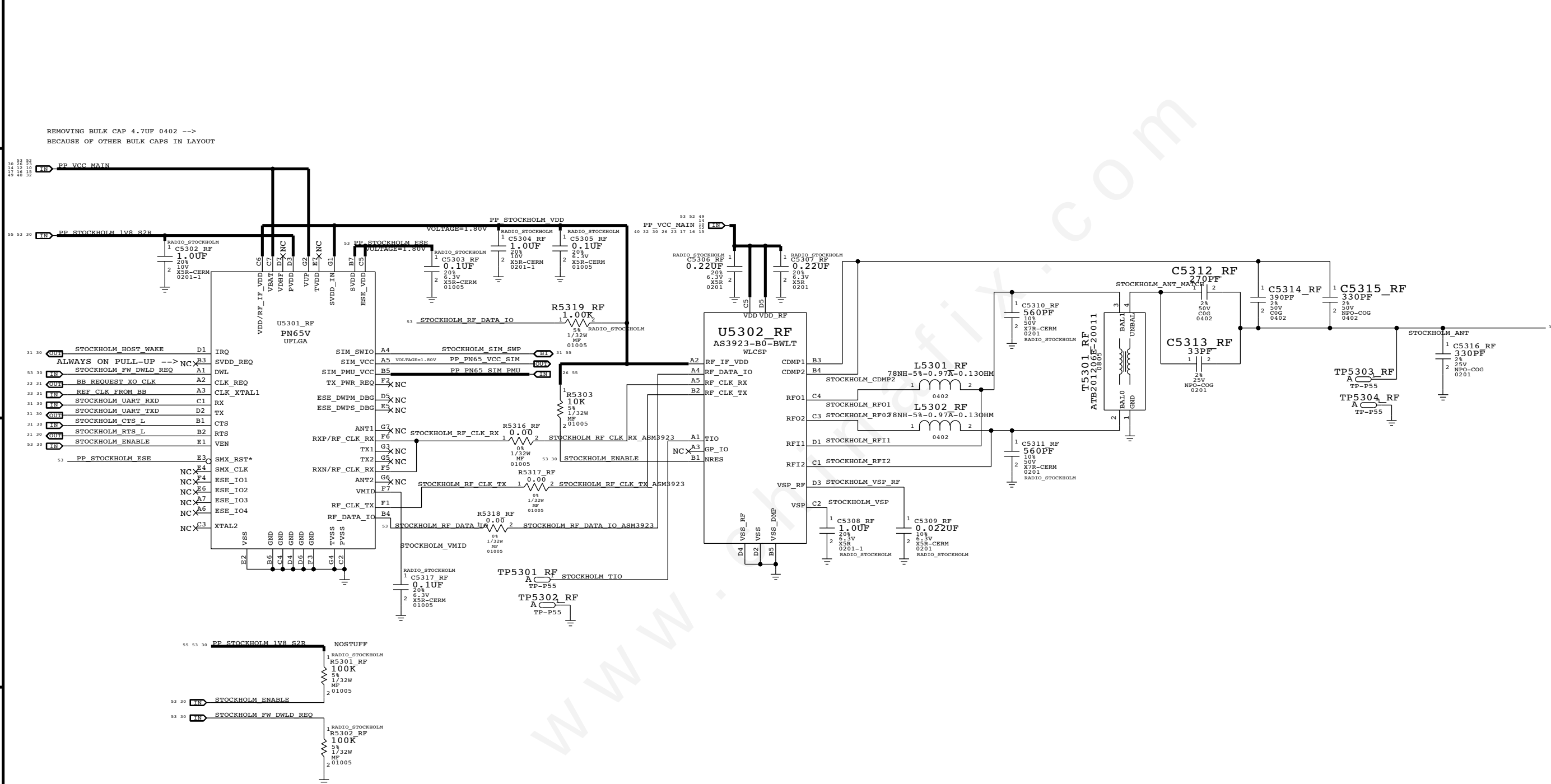
WIFI BT
RS202_RF
10K
51
32W
8F
NOSTUFF

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C2101
R2100
L2102
U2100

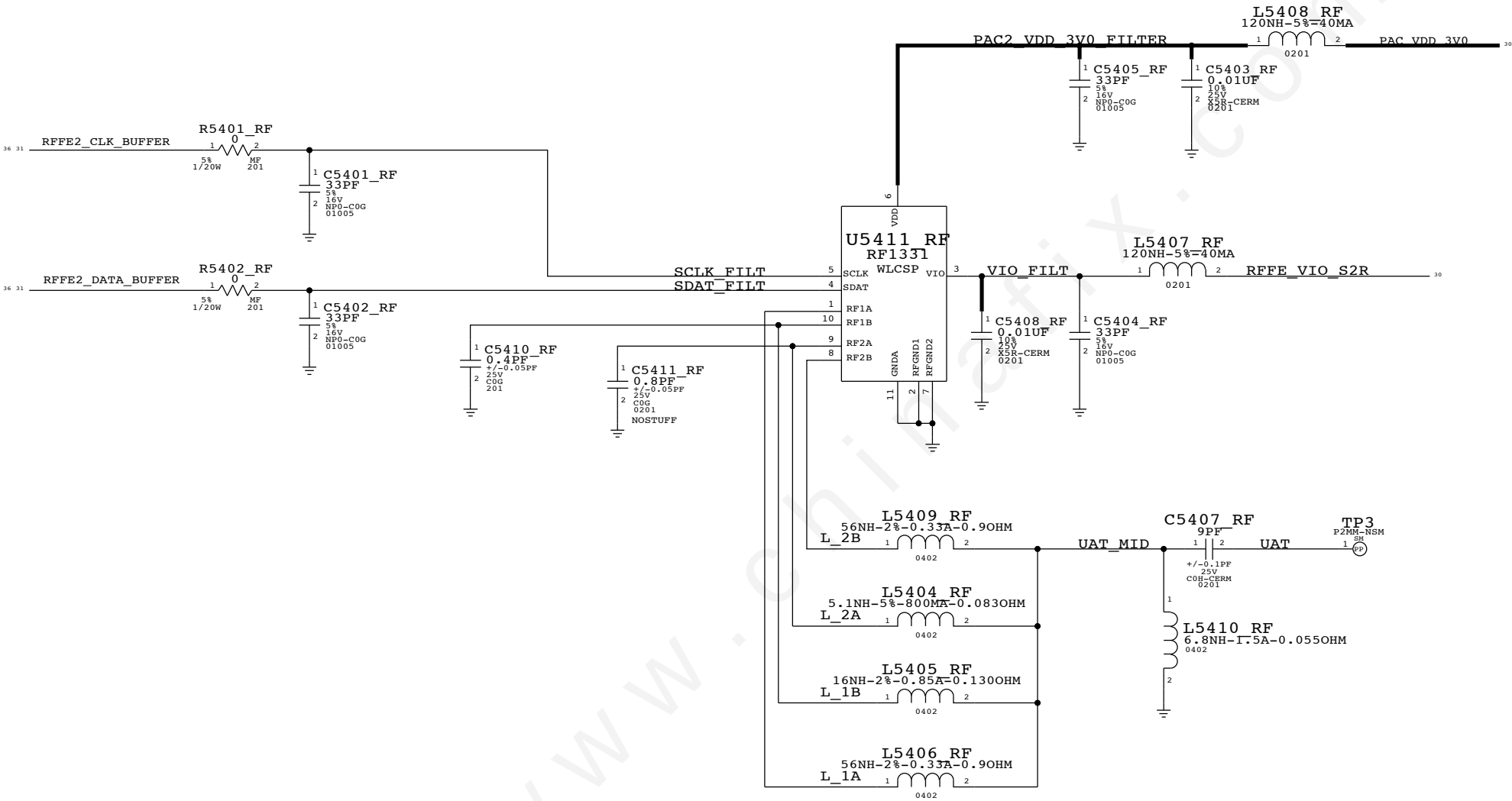




ON-BOARD JUMPER FLEX

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UAT JUMPER



DSDS

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